-->

(https://www.aetna.com/)

Infliximab

Clinical Policy Bulletins | Medical Clinical Policy Bulletins

Number: 0341

Table Of Contents

Policy
Applicable CPT / HCPCS / ICD-10 Codes
Background
References

Policy History

Last Review
01/07/2026
Effective: 08/03/1999
Next Review: 10/22/2026

Review History

Definitions

Additional Information

Clinical Policy Bulletin Notes

Brand Selection for Medically Necessary Indications for Commercial Medical Plans

According to Aetna commercial benefit plans, health care services are considered not medically necessary if they are more costly than an alternative service or sequence of services at least as likely to produce equivalent therapeutic or diagnostic results as to the diagnosis or treatment of that member's illness, injury or disease. Aetna commercial plans may require a trial of a lower-cost drug (preferred medication) that is at least as likely to produce equivalent therapeutic results before approving coverage for a higher-cost drug within the same therapeutic class. For a list of preferred drugs, refer to the Aetna Commercial

Clinical Program Summary (https://www.aetna.com/health-care-professionals/precertification/precertification-lists.html#tab_content_section_responsivegrid_970135637_responsivegrid_tabs_link_tabs_2).

Policy

Scope of Policy

This Clinical Policy Bulletin addresses the following infliximab products for commercial medical plans. For Medicare criteria, see Medicare Part B Criteria (https://www.aetna.com/health-care-professionals/medicare/part-b-step.html).

- infliximab (Remicade or unbranded Remicade)
- infliximab-abda (Renflexis)
- infliximab-axxq (Avsola)
- infliximab-dyyb (Inflectra for intravenous infusion)
- infliximab-dyyb (Zymfentra for subcutaneous injection)

Note: Requires Precertification:

Precertification of an infliximab product is required of all Aetna participating providers and members in applicable plan designs. For precertification of an infliximab product, call (866) 752-7021 or fax (888) 267-3277. For Statement of Medical Necessity (SMN) precertification forms, see Specialty Pharmacy Precertification (https://www.aetna.com/health-care-professionals/health-care-professional-forms.html).

Note: For commercial plans, Site of Care Utilization Management Policy applies. For information on site of service for infliximab products (Avsola, Inflectra, Remicade, and Renflexis), see Utilization Management Policy

on Site of Care for Specialty Drug Infusions (https://www.aetna.com/health-care-professionals/utilization-management/drug-infusion-site-of-care-policy.html).

I. **Prescriber Specialties**

This medication must be prescribed by or in consultation with one of the following:

A. Crohn's disease and ulcerative colitis: gastroenterologist;
B. Rheumatoid arthritis, ankylosing spondylitis, non-radiographic axial spondyloarthritis, Behcet's disease, Takayasu's arteritis, and reactive arthritis: rheumatologist;
C. Psoriatic arthritis and hidradenitis suppurativa: rheumatologist or dermatologist;
D. Plaque psoriasis and pyoderma gangrenosum: dermatologist;
E. Sarcoidosis: dermatologist, pulmonologist, rheumatologist, cardiologist, neurologist, or ophthalmologist;
F. Uveitis: ophthalmologist or rheumatologist;
G. Immune checkpoint inhibitor-related inflammatory arthritis: oncologist, hematologist, or rheumatologist;
H. Immune checkpoint inhibitor-related toxicity: gastroenterologist, oncologist or hematologist;
I. Acute graft versus host disease: oncologist or hematologist.

II. **Criteria for Initial Approval**

Aetna considers infliximab (Remicade), infliximab-abda (Renflexis), infliximab-axxq (Avsola), and inflixmab-dyyb (Inflectra, Zymfentra) medically necessary, where indicated, for members with *any* of the following conditions when criteria are met:

A. *Crohn's disease (CD)*

For treatment of moderately to severely active CD;

B. *Ulcerative colitis (UC)*

For treatment of moderately to severely active UC;

C. *Rheumatoid arthritis (RA)*
(Avsola/Inflectra/infliximab/Remicade/Renflexis only)

1. For adult members who have previously received a biologic or targeted synthetic drug (e.g., Rinvoq, Xeljanz) indicated for moderately to severely active rheumatoid arthritis (RA) within the past 120 days. The requested medication must be prescribed in combination with methotrexate or leflunomide unless the member has a clinical reason not to use methotrexate or leflunomide (see Appendix A); *or*
2. For adult members who have moderately to severely active RA when *all* of the following criteria are met:

   a. Member meets *either* of the following criteria:

      i. Member has been tested for *either* of the following biomarkers and the test was positive:

         - Rheumatoid factor (RF); *or*
         - Anti-cyclic citrullinated peptide (anti-CCP); *or*

      ii. Member has been tested for *all* of the following biomarkers:

         - RF; *and*
         - Anti-CCP; *and*
         - C-reactive protein (CRP) and/or erythrocyte sedimentation rate (ESR); *and*

   b. Member meets *one* of the following criteria:

      i. Member has failed to achieve a low disease activity after a 3-month trial of methotrexate (MTX) monotherapy at a maximum titrated dose of at least 15 mg per week and meets *any* of the following conditions:

         - Member has had a documented inadequate response to MTX in combination with at least one

other conventional synthetic drug (i.e., hydroxychloroquine and/or sulfasalazine) after a 3-month trial at a maximum tolerated dose(s); *or*
- Member has experienced a documented intolerable adverse event to hydroxychloroquine or sulfasalazine; *or*
- Member has a documented contraindication to hydroxychloroquine (see Appendix A) and sulfasalazine (e.g., porphyria, intestinal or urinary obstruction); *or*
- Member has moderate to high disease activity; *or*

ii. Member was unable to tolerate a 3-month trial of MTX monotherapy at a maximum titrated dose of at least 15 mg per week and meets *any* of the following conditions:

- Member has had a documented inadequate response to MTX in combination with at least one other conventional synthetic drug (i.e., hydroxychloroquine and/or sulfasalazine) after a 3-month trial at a maximum tolerated dose(s); *or*
- Member has stopped taking MTX and has had a documented inadequate response to another conventional synthetic drug (i.e., leflunomide, hydroxychloroquine, and/or sulfasalazine) alone or in combination after a 3-month trial at a maximum tolerated dose(s); *or*
- Member has experienced a documented intolerable adverse event to leflunomide, hydroxychloroquine, or sulfasalazine; *or*
- Member has a documented contraindication to leflunomide, hydroxychloroquine (see Appendix A), and sulfasalazine (e.g., porphyria, intestinal or urinary obstruction); *or*
- Member has moderate to high disease activity; *or*

iii. Member has experienced a documented intolerable adverse event or has a documented contraindication to MTX (see Appendix A), discontinues MTX, and meets *any* of the following conditions:

- Member has had a documented inadequate response to another conventional synthetic drug (i.e., leflunomide, hydroxychloroquine, and/or sulfasalazine) alone or in combination after a 3-month trial at a maximum tolerated dose(s); *or*
- Member has experienced a documented intolerable adverse event to leflunomide, hydroxychloroquine, or sulfasalazine; *or*
- Member has a documented contraindication to leflunomide, hydroxychloroquine (see Appendix A), and sulfasalazine (e.g., porphyria, intestinal or urinary obstruction); *or*
- Member has moderate to high disease activity; *and*

c. Member is prescribed the requested medication in combination with methotrexate or leflunomide, or has a clinical reason not to use methotrexate or leflunomide (see Appendix);

D. *Ankylosing spondylitis (AS) and non-radiographic axial spondyloarthritis (nr-axSpA) (Avsola/Inflectra/infliximab/Remicade/Renflexis only)*

1. For adult members who have previously received a biologic or targeted synthetic drug (e.g., Rinvoq, Xeljanz) indicated for active ankylosing spondylitis or active non-radiographic axial spondyloarthritis; *or*
2. For adult members for treatment of active ankylosing spondylitis or active non-radiographic axial spondyloarthritis when *either* of the following criteria is met:

a. Member has had an inadequate response to at least two non-steroidal anti-inflammatory drugs (NSAIDs); *or*
b. Member has an intolerance or contraindication to two or more NSAIDs;

E. *Psoriatic arthritis (PsA)*
(Avsola/Inflectra/infliximab/Remicade/Renflexis only)

1. For adult members who have previously received a biologic or targeted synthetic drug (e.g., Rinvoq, Otezla) indicated for active psoriatic arthritis; *or*
2. For adult members for treatment of active psoriatic arthritis when *either* of the following criteria is met:

   a. Member has mild to moderate disease and meets *one* of the following criteria:

      i. Member has had an inadequate response to methotrexate, leflunomide, or another conventional synthetic drug (e.g., sulfasalazine) administered at an adequate dose and duration; *or*
      ii. Member has an intolerance or contraindication to methotrexate or leflunomide (see Appendix), or another conventional synthetic drug (e.g., sulfasalazine); *or*
      iii. Member has enthesitis or predominantly axial disease; *or*

   b. Member has severe disease;

F. *Plaque psoriasis (PsO)*
(Avsola/Inflectra/infliximab/Remicade/Renflexis only)

1. For adult members who have previously received a biologic or targeted synthetic drug (e.g., Sotyktu, Otezla) indicated for the treatment of moderate to severe plaque psoriasis; *or*
2. For adult members for treatment of moderate to severe plaque psoriasis when *any* of the following criteria is met:

a. Crucial body areas (e.g., hands, feet, face, neck, scalp, genitals/groin, intertriginous areas) are affected; *or*
b. At least 10% of the body surface area (BSA) is affected; *or*
c. At least 3% of body surface area (BSA) is affected and the member meets *either* of the following criteria:

i. Member has had an inadequate response or intolerance to either phototherapy (e.g., UVB, PUVA) or pharmacologic treatment with methotrexate, cyclosporine, or acitretin; *or*
ii. Member has a clinical reason to avoid pharmacologic treatment with methotrexate, cyclosporine, and acitretin (see Appendix);

G. *Behçet's disease (Avsola/Inflectra/infliximab/Remicade/Renflexis only)*

1. For members who have previously received apremilast (Otezla) or a biologic indicated for the treatment of Behcet's disease; *or*
2. For the treatment of Behçet's disease when the member has had an inadequate response to at least one non-biologic medication for Behçet's disease (e.g., azathioprine, colchicine, cyclosporine, systemic corticosteroids);

H. *Hidradenitis suppurativa (Avsola/Inflectra/infliximab/Remicade/Renflexis only)*

1. For members who have previously received a biologic indicated for the treatment of severe, refractory hidradenitis suppurativa; *or*
2. For treatment of severe, refractory hidradenitis suppurativa when *either* of the following is met:

a. Member has had an inadequate response to an oral antibiotic used for the treatment of hidradenitis suppurative (e.g., clindamycin, metronidazole,

moxifloxacin, rifampin, tetracyclines) for at least 90 days; *or*

b. Member has an intolerance or contraindication to oral antibiotics used for the treatment of hidradenitis suppurative;

I. *Pyoderma gangrenosum (Avsola/Inflectra/infliximab/Remicade/Renflexis only)*

1. For members who have previously received a biologic indicated for pyoderma gangrenosum; *or*
2. For treatment of pyoderma gangrenosum when *either* of the following is met:

   a. Member has had an inadequate response to corticosteroids or immunosuppressive therapy (e.g., cyclosporine, mycophenolate mofetil); *or*
   b. Member has an intolerance or contraindication to corticosteroids and immunosuppressive therapy (e.g. cyclosporine, mycophenolate mofetil);

J. *Sarcoidosis (Avsola/Inflectra/infliximab/Remicade/Renflexis only)*

For treatment of sarcoidosis in members when *either* of the following criteria is met:

1. Member has had an inadequate response to corticosteroids or immunosuppressive therapy; *or*
2. Member has an intolerance or contraindication to corticosteroids and immunosuppressive therapy;

K. *Takayasu's arteritis (Avsola/Inflectra/infliximab/Remicade/Renflexis only)*

For treatment of refractory Takayasu's arteritis when *either* of the follow criteria is met:

1. Member has had an inadequate response to corticosteroids or immunosuppressive therapy (e.g., methotrexate, azathioprine, mycophenolate mofetil); *or*
2. Member has an intolerance or contraindication to corticosteroids and immunosuppressive therapy (e.g., methotrexate, azathioprine, mycophenolate mofetil);

L. *Uveitis (Avsola/Inflectra/infliximab/Remicade/Renflexis only)*

1. For members who have previously received a biologic indicated for uveitis; *or*
2. For treatment of uveitis when *either* of the follow criteria is met:

    a. Member has had an inadequate response to corticosteroids or immunosuppressive therapy (e.g., azathioprine, cyclosporine, methotrexate, mycophenolate mofetil); *or*
    b. Member has an intolerance or contraindication to corticosteroids and immunosuppressive therapy (e.g., azathioprine, cyclosporine, methotrexate, mycophenolate mofetil);

M. *Reactive arthritis (Avsola/Inflectra/infliximab/Remicade/Renflexis only)*

1. For members who have previously received a biologic indicated for reactive arthritis; *or*
2. For treatment of reactive arthritis when *either* of the following criteria is met:

    a. Member has had an inadequate response to methotrexate or sulfasalazine; *or*
    b. Member has an intolerance or contraindication to methotrexate (see Appendix) and sulfasalazine (e.g., porphyria, intestinal or urinary obstruction);

N. *Immune checkpoint inhibitor-related toxicity (Avsola/Inflectra/infliximab/Remicade/Renflexis only)*

1. For treatment of immune checkpoint inhibitor-related toxicity when the member has moderate or severe immunotherapy-related inflammatory arthritis and *either* of the following is met:

    a. Member has had an inadequate response to corticosteroids or a conventional synthetic drug (e.g., methotrexate, sulfasalazine, leflunomide, hydroxychloroquine); *or*
    b. Member has an intolerance or contraindication to corticosteroids and a conventional synthetic drug (e.g., methotrexate, sulfasalazine, leflunomide, hydroxychloroquine; *or*

2. For treatment of immune checkpoint inhibitor-related toxicity when *any* of the following criteria is met:

    a. Member has had an inadequate response to systemic corticosteroids; *or*
    b. Member has an intolerance or contraindication to corticosteroids; *or*
    c. Member has immune checkpoint inhibitor-related diarrhea or colitis;

O. *Acute graft versus host disease (aGVHD) (Avsola/Inflectra/infliximab/Remicade/Renflexis only)*

For treatment of acute graft versus host disease when *either* of the following criteria is met:

1. Member has had an inadequate response to systemic corticosteroids; *or*
2. Member has an intolerance or contraindication to corticosteroids.

Aetna considers all other indications as experimental, investigational, or unproven.

III. **Continuation of Therapy**

Aetna considers continuation of therapy medically necessary for the following conditions when criteria are met:

A. *Crohn's disease (CD)*

1. For all members (including new members) who are using the requested medication for moderately to severely active Crohn's disease and who achieve or maintain remission; *or*
2. For all members (including new members) who are using the requested medication for moderately to severely active Crohn's disease and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when there is improvement in *any* of the following from baseline:

    a. Abdominal pain or tenderness; *or*
    b. Diarrhea; *or*
    c. Body weight; *or*
    d. Abdominal mass; *or*
    e. Hematocrit; *or*
    f. Appearance of the mucosa on endoscopy, computed tomography enterography (CTE), magnetic resonance enterography (MRE), or intestinal ultrasound; *or*
    g. Improvement on a disease activity scoring tool (e.g., Crohn's Disease Activity Index [CDAI] score);

B. *Ulcerative colitis (UC)*

1. For all members (including new members) who are using the requested medication for moderately to severely active ulcerative colitis and who achieve or maintain remission; *or*
2. For all members (including new members) who are using the requested medication for moderately to severely active ulcerative colitis and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when

there is improvement in *any* of the following from baseline:

a. Stool frequency; *or*
b. Rectal bleeding; *or*
c. Urgency of defecation; *or*
d. C-reactive protein (CRP); *or*
e. Fecal calprotectin (FC); *or*
f. Appearance of the mucosa on endoscopy, computed tomography enterography (CTE), magnetic resonance enterography (MRE), or intestinal ultrasound; *or*
g. Improvement on a disease activity scoring tool (e.g., Ulcerative Colitis Endoscopic Index of Severity [UCEIS], Mayo score);

C. *Rheumatoid arthritis (RA)*
(Avsola/Inflectra/infliximab/Remicade/Renflexis only)

For all adult members (including new members) who are using the requested medication for moderately to severely active rheumatoid arthritis and who achieve or maintain a positive clinical response as evidenced by disease activity improvement of at least 20% from baseline in tender joint count, swollen joint count, pain, or disability;

D. *Ankylosing spondylitis (AS) and non-radiographic axial spondyloarthritis (nr-axSpA)*
(Avsola/Inflectra/infliximab/Remicade/Renflexis only)

For all adult members (including new members) who are using the requested medication for ankylosing spondylitis or non-radiographic axial spondyloarthritis and who achieve or maintain a positive clinical response with the requested medication as evidenced by low disease activity or improvement in signs and symptoms of the condition when there is improvement in *any* of the following from baseline:

1. Functional status; *or*
2. Total spinal pain; *or*
3. Inflammation (e.g., morning stiffness); *or*

4. Swollen joints; *or*
5. Tender joints; *or*
6. C-reactive protein (CRP);

E. *Psoriatic arthritis (PsA)*
(Avsola/Inflectra/infliximab/Remicade/Renflexis only)

For all adult members (including new members) who are using the requested medication for psoriatic arthritis and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when there is improvement in *any* of the following from baseline:

1. Number of swollen joints; *or*
2. Number of tender joints; *or*
3. Dactylitis; *or*
4. Enthesitis; *or*
5. Axial disease; *or*
6. Skin and/or nail involvement; *or*
7. Functional status; *or*
8. C-reactive protein (CRP);

F. *Plaque psoriasis (PsO)*
(Avsola/Inflectra/infliximab/Remicade/Renflexis only)

For all adult members (including new members) who are using the requested medication for moderate to severe plaque psoriasis and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when *either* of the following is met:

1. Reduction in body surface area (BSA) affected from baseline; *or*
2. Improvement in signs and symptoms from baseline (e.g., itching, redness, flaking, scaling, burning, cracking, pain);

G. *Hidradenitis suppurativa (Avsola/Inflectra/infliximab/Remicade/Renflexis only)*

For all members (including new members) who are using the requested medication for severe, refractory hidradenitis suppurativa and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when *any* of the following is met:

1. Reduction in abscess and inflammatory nodule count from baseline; *or*
2. Reduced formation of new sinus tracts and scarring; *or*
3. Decrease in frequency of inflammatory lesions from baseline; *or*
4. Reduction in pain from baseline; *or*
5. Reduction in suppuration from baseline; *or*
6. Improvement frequency of relapses from baseline; *or*
7. Improvement in quality of life from baseline; *or*
8. Improvement on a disease severity assessment tool from baseline;

H. *Uveitis (Avsola/Inflectra/infliximab/Remicade/Renflexis only)*

For all members (including new members) who are using the requested medication for uveitis and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when the patient meets *any* of the following:

1. Reduced frequency of recurrence compared to baseline; *or*
2. Zero anterior chamber inflammation or reduction in anterior chamber inflammation compared to baseline; *or*
3. Decreased reliance on topical corticosteroids;

I. *Reactive arthritis (Avsola/Inflectra/infliximab/Remicade/Renflexis only)*

For all members (including new members) who are using the requested medication for reactive arthritis and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition (e.g., tender joint count, swollen joint count, or pain);

J. *Immune checkpoint inhibitor-related inflammatory arthritis (Avsola/Inflectra/infliximab/Remicade/ Renflexis only)*

For all members (including new members) who are using the requested medication for immunotherapy-related inflammatory arthritis and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition.

K. *Immune checkpoint inhibitor-related toxicity and acute graft versus host disease (Avsola/Inflectra/ infliximab/Remicade/Renflexis only)*

For all members (including new members) requesting authorization for continuation of therapy must meet all initial authorization criteria;

L. *All other indications (Avsola/Inflectra/infliximab/Remicade/Renflexis only)*

For all members (including new members) who are using the requested medication for an indication outlined in Section II and who achieve or maintain positive clinical response with the requested medication as evidenced by low disease activity or improvement in signs and symptoms of the condition.

IV. **Other**

For all indications: Member has had a documented negative tuberculosis (TB) test (which can include a tuberculosis skin test [TST] or an interferon-release assay [IGRA]* within 12 months of initiating therapy for persons who are naïve to biologic drugs or targeted synthetic drugs associated with an increased risk of TB.

* If the screening testing for TB is positive, there must be further testing to confirm there is no active disease (e.g., chest x-ray). Do not administer the requested medication to members with active TB infection. If there is latent disease, TB treatment must be started before initiation of the requested medication.

For all indications: Member cannot use the requested medication concomitantly with any other biologic drug or targeted synthetic drug for the same indication.

V. **Related Policies**

Refer to pharmacy benefit plans for apremilast (Otezla), brodulumab (Siliq), Ixekizumab (Taltz), infliximab-dyyb (Zymfentra), and tofacitinib (Xeljanz and Xeljanz XR).

For measurement of serum levels of infliximab and/or antibodies to infliximab, see CPB 1091 - Therapeutic Drug Monitoring of Anti-Tumor Necrosis Factor Blocking Agents (../1000_1099/1091.html).

See also:

- CPB 0205 - Phototherapy and Photochemotherapy (PUVA) for Skin Conditions (../200_299/0205.html)
- CPB 0249 - Inflammatory Bowel Disease: Serologic Markers and Pharmacogenomic and Metabolic Assessment of Thiopurine Therapy (../200_299/0249.html)
- CPB 0314 - Rituximab (../300_399/0314.html)
- CPB 0315 - Etanercept (0315.html)
- CPB 0577 - Laser Treatment for Psoriasis and Other Selected Skin Conditions (../500_599/0577.html)
- CPB 0655 - Adalimumab (../600_699/0655.html)
- CPB 0720 - Abatacept (Orencia) (../700_799/0720.html)
- CPB 0751 - Natalizumab (../700_799/0751.html)
- CPB 0761 - Certolizumab Pegol (Cimzia) (../700_799/0761.html)
- CPB 0790 - Golimumab (Simponi and Simponi Aria) (../700_799/0790.html)
- CPB 0885 - Vedolizumab (Entyvio) (../800_899/0885.html)
- CPB 0905 - Secukinumab (Cosentyx) (../900_999/0905.html)

- CPB 0912 - Ustekinumab (../900_999/0912.html)
- CPB 1009 - Risankizumab-rzaa (Skyrizi) (../1000_1099/1009.html)
- CPB 1011 - Guselkumab (Tremfya) (../1000_1099/1011.html)
- CPB 1091 - Therapeutic Drug Monitoring of Anti-Tumor Necrosis Factor Blocking Agents (../1000_1099/1091.html)

Dosage and Administration

Note: Approvals may be subject to dosing limits in accordance with FDA-approved labeling, accepted compendia, and/or evidence-based practice guidelines. See Medical Specialty Medication Quantity Limits (https://aetna.com/content/dam/aetna/pdfs/aetnacom/pdf/medical-specialty-medication-quantity-limits.pdf) for more information.

Below includes dosing recommendations as per the FDA-approved prescribing information.

Avsola, Inflectra, infliximab, Remicade, Renflexis

Infliximab products (Avsola, Inflectra, Remicade, and Renflexis) are available for injection in a 20 mL single-dose vial containing 100 mg of lyophilized infliximab or its biosimilars. These products are reconstituted and diluted before being administered as an intravenous infusion.

Table: Dosing for Infliximab, Infliximab-axxq, Infliximab-dyyb, and Infliximab-abda

Indication	Dose
Ankylosing spondylitis (adults)	Induction: 5 mg/kg IV at weeks 0, 2, and 6 Maintenance: 5 mg/kg IV every 6 weeks

Crohn's disease, moderate-to-severe or fistulizing (adults)	Induction: 5 mg/kg IV at weeks 0, 2, and 6 Maintenance: 5 mg/kg IV every 8 weeks For persons who respond and then lose their response, consideration may be given to treatment with 10 mg/kg IV every 8 weeks. Persons who do not respond by week 14 are unlikely to respond and consideration should be given to discontinue infliximab in these persons.
Crohn's disease, moderate-to-severe (pediatrics 6 years of age and older)	Induction: 5 mg/kg IV at weeks 0, 2, and 6 Maintenance: 5 mg/kg IV every 8 weeks
Plaque psoriasis, chronic (severe) (adults)	Induction: 5 mg/kg IV at weeks 0, 2, and 6 Maintenance: 5 mg/kg IV every 8 weeks
Psoriasis arthritis (adults)	Induction: 5 mg/kg at weeks 0, 2, and 6 Maintenance: 5 mg/kg IV every 8 weeks
Rheumatoid arthritis, moderate-to-severe (adults)	Induction: 3 mg/kg IV at weeks 0, 2 and 6 Maintenance: 3 mg/kg IV every 8 weeks Infliximab should be given in combination with methotrexate. Incomplete response: consideration may be given to adjusting the dose up to 10 mg/kg every 8 weeks or treating as often as every 4 weeks bearing in mind that risk of serious infections is increased at higher doses.
Ulcerative colitis (6 years of age and older)	Induction: 5 mg/kg IV at weeks 0, 2 and 6 Maintenance: 5 mg/kg IV every 8 weeks

Sources: Prescribing Information (Amgen, 2025, Janssen Biotech, 2025; Organon, 2023, Pfizer, 2025)

Zymfentra

Zymfentra (infliximab-dyyb) is available as a subcutaneous injection that can be self-administered, if appropriate, after proper training. It is supplied in the following dosage forms and strengths:

- 120 mg/mL in a single-dose prefilled syringe
- 120 mg/mL in a single-dose prefilled syringe with needle shield
- 120 mg/mL in a single-dose prefilled pen.

Zymfentra is indicated for adults as maintenance treatment only, starting at Week 10 and thereafter. All individuals must complete an intravenous induction regimen with an infliximab product before starting Zymfentra.

Recommended dosage for maintenance treatment in ulcerative colitis and Crohn's disease

Starting at Week 10 and thereafter: 120 mg subcutaneously once every two weeks. To switch persons who are responding to maintenance therapy with an infliximab product administered intravenously, the first subcutaneous dose of Zymfentra is administered in place of the next scheduled intravenous infusion and every two weeks thereafter.

Source: Celltrion USA, 2024

Experimental, Investigational, or Unproven

I. Combination with Other Biologics

Aetna considers concomitant use of infliximab, or its biosimilar, with any other biologic drug (e.g., abatacept, adalimumab, anakinra, etanercept, tocilizumab) or targeted synthetic drug (e.g. tofacitinib) experimental, investigational, or unproven for the same indication because the effectiveness of this approach has not been established.

II. Other Experimental, Investigational, or Unproven Indications

Aetna considers infliximab, infliximab-abda, infliximab-axxq, and infliximab-dyyb experimental, investigational, or unproven for all other indications (not an all-inclusive list) because its effectiveness for these indications has not been established:

- Acne fulminans
- Amyloid angiopathy
- Asthma
- Autoimmune cholangiopathy
- Birdshot retinochoroidopathy
- Bronchiolitis obliterans
- Celiac disease
- Central nervous system amyloidosis
- Chemotherapy induced enterocolitis (not due to Yervoy or Opdivo)
- Chronic immune-mediated myelitis
- Chronic obstructive pulmonary disease
- Cogan's syndrome
- Corneal ulcer
- Coronavirus disease 2019 (COVID-19) pneumonia
- Cranial nerve palsy
- Cystoid macular degeneration
- Disc herniation-induced sciatica
- Discoid lupus erythematosus
- Eczema
- Eosinophilic fasciitis
- Granuloma annulare
- Granulomatous angiitis
- Granulomatous mastitis
- Hepatitis C genotype 1
- IgG4-related disease
- Iritis
- Kawasaki disease
- Localized scleroderma/morphea
- Membranous glomerulopathy
- Microscopic colitis

- Multifocal osteomyelitis (e.g., (chronic recurrent multifocal osteomyelitis (CRMO))
- Necrobiosis lipoidica diabeticorum
- Nodular scleritis
- Panniculitis
- Polyarteritis nodosa
- Polymyositis
- Prevention of post-operative recurrence of Crohn's disease
- Rejection following small bowel transplantation
- Relapsing polychondritis
- Renal cell carcinoma
- Retinal vasculitis
- Scleroderma
- Sjogren's syndrome
- Still's disease
- Systemic lupus erythematosus
- Tolosa-Hunt syndrome
- Tubulo-interstitial nephritis with uveitis (TINU) syndrome.

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
CPT codes not covered for indications listed in the CPB:	
0350U	Drug metabolism or processing (multiple conditions), whole blood or buccal specimen, DNA analysis, 27 gene report, with variant analysis and reported phenotypes
0514U	Gastroenterology (irritable bowel disease [IBD]), immunoassay for quantitative determination of adalimumab (ADL) levels in venous serum in patients undergoing adalimumab therapy, results reported as a numerical value as micrograms per milliliter (µg/mL)

Code	Code Description
0515U	Gastroenterology (irritable bowel disease [IBD]), immunoassay for quantitative determination of infliximab (IFX) levels in venous serum in patients undergoing infliximab therapy, results reported as a numerical value as micrograms per milliliter (µg/mL)
80230	Infliximab
83516	Immunoassay for analyte other than infectious agent antibody or infectious agent antigen, qualitative or semiquantitative; multiple step method [measurement of anti-histone antibodies for monitoring infliximab therapy]
83520	Immunoassay, analyte, quantitative, not otherwise specified [human anti-climeric antibody (HACA), infliximab-specific ELISA] or [serum infliximab level] [InformTx therapeutic drug monitoring]
86235	Extractable nuclear antigen, antibody to, any method (e.g., nRNP, SS-A, SS-B, Sm, RNP, Sc170, J01), each antibody [measurement of anti-histone antibodies for monitoring infliximab therapy]
Other CPT codes related to the CPB:	
71045 - 71048	Radiologic examination, chest
80220	Hydroxychloroquine
85651	Sedimentation rate, erythrocyte; non-automated
85652	Sedimentation rate, erythrocyte; automated
86140	C-reactive protein
86141	C-reactive protein; high sensitivity (hsCRP)
86200	Cyclic citrullinated peptide (CCP), antibody
86430	Rheumatoid factor; qualitative
86431	Rheumatoid factor; quantitative
86480	Tuberculosis test, cell mediated immunity antigen response measurement; gamma interferon
86481	Tuberculosis test, cell mediated immunity antigen response measurement; enumeration of gamma interferon - producing T cells in cell suspension

Code	Code Description
86580	Skin test; tuberculosis, intradermal
96360 - 96361	Intravenous infusion, hydration
96365 - 96368	Intravenous infusion, for therapy, prophylaxis, or diagnosis (specify substance or drug)
96372	Therapeutic, prophylactic, or diagnostic injection (specify substance or drug); subcutaneous or intramuscular
96374	intravenous push, single or initial substance/drug
96375	each additional sequential intravenous push of a new substance/drug
96379	Unlisted therapeutic, prophylactic, or diagnostic intravenous or intra-arterial injection of infusion
96401 - 96450	Chemotherapy administration
96910 - 96913	Photochemotherapy
99601 - 99602	Home infusion/specialty drug administration
HCPCS codes covered if selection criteria are met:	
J1745	Injection, infliximab, 10 mg
J1748	Injection, infliximab-dyyb (zymfentra), 10 mg
Q5103	Injection, infliximab-dyyb, biosimilar, (inflectra), 10 mg
Q5104	Injection, infliximab-abda, biosimilar, (renflexis), 10 mg
Q5109	Injection, infliximab-qbtx, biosimilar, (ixifi), 10 mg
Q5121	Injection, infliximab-axxq, biosimilar, (AVSOLA), 10 mg
S9359	Home infusion therapy, anti-tumor necrosis factor intravenous therapy; (e.g., Infliximab); administrative services, professional pharmacy services, care coordination, and all necessary supplies and equipment (drugs and nursing visits coded separately), per diem
Other HCPCS codes related to the CPB (examples of other drug therapies):	
Rinvoq, Xeljanz, Otezla, Sulfasalazine, Hydroxychloroquine, Leflunomide - no specific code	
G0069	Professional services for the administration of subcutaneous immunotherapy for each infusion drug administration calendar day in the individual's home, each 15 minutes

Code	Code Description
J0129	Injection, abatacept, 10 mg
J0139	Injection, adalimumab, 1 mg
J0702	Injection, betamethasone acetate 3 mg and betamethasone sodium phosphate 3 mg
J1010	Injection, methylprednisolone acetate, 1 mg
J1020	Injection, methylprednisolone acetate, 20 mg
J1030	Injection, methylprednisolone acetate, 40 mg
J1040	Injection, methylprednisolone acetate, 80 mg
J1100	Injection, dexamethasone sodium phosphate, 1 mg
J1438	Injection, etanercept, 25 mg
J1600	Injection, gold sodium thiomalate, up to 50 mg
J1628	Injection, guselkumab, 1 mg
J1720	Injection, hydrocortisone sodium succinate, up to 100 mg
J2650	Injection, prednisolone acetate, up to 1 ml
J2804	Injection, rifampin, 1 mg
J2919	Injection, methylprednisolone sodium succinate, 5 mg
J2920	Injection, methylprednisolone sodium succinate, up to 40 mg [Solu-Medrol]
J2930	Injection, methylprednisolone sodium succinate, up to 125 mg [Solu-Medrol]
J3245	Injection, tildrakizumab, 1 mg
J3300	Injection, triamcinolone acetonide, preservative free, 1 mg
J3301	Injection, triamcinolone acetonide, not otherwise specified, 10 mg
J3302	Injection, triamcinolone diacetate, per 5 mg
J3303	Injection, triamcinolone hexacetonide, per 5 mg
J3304	Injection, triamcinolone acetonide, preservative-free, extended-release, microsphere formulation, 1 mg
J7500	Azathioprine, oral, 50 mg
J7501	Azathioprine, parenteral, 100 mg
J7502	Cyclosporine, oral, 100 mg

Code	Code Description
J7509	Methylprednisolone, oral, per 4 mg
J7510	Prednisolone, oral, per 5 mg
J7512	Prednisone, immediate release or delayed release, oral, 1 mg
J7515	Cyclosporine, oral, 25 mg
J7516	Cyclosporine, parenteral, 250 mg
J7637	Dexamethasone, inhalation solution, compounded product, administered through dme, concentrated form, per milligram
J7638	Dexamethasone, inhalation solution, compounded product, administered through dme, unit dose form, per milligram
J7683	Triamcinolone, inhalation solution, compounded product, administered through dme, concentrated form, per milligram
J7684	Triamcinolone, inhalation solution, compounded product, administered through dme, unit dose form, per milligram
J8530	Cyclophosphamide, oral, 25 mg
J8540	Dexamethasone, oral, 0. 25 mg
J8541	Dexamethasone (hemady), oral, 0.25 mg
J8610	Methotrexate, oral, 2.5 mg
J8611	Methotrexate (jylamvo), oral, 2.5 mg
J8612	Methotrexate (xatmep), oral, 2.5 mg
J9070	Cyclophosphamide, 100 mg
J9073	Injection, cyclophosphamide (ingenus), 5 mg
J9074	Injection, cyclophosphamide (sandoz), 5 mg
J9075	Injection, cyclophosphamide, not otherwise specified, 5mg
J9076	Injection, cyclophosphamide (baxter), 5 mg
J9250	Methotrexate sodium, 5 mg
J9255	Injection, methotrexate (accord) not therapeutically equivalent to j9250 or j9260, 50 mg
J9260	Methotrexate sodium, 50 mg
Q5140	Injection, adalimumab-fkjp, biosimilar, 1 mg
Q5141	Injection, adalimumab-aaty, biosimilar, 1 mg
Q5142	Injection, adalimumab-ryvk biosimilar, 1 mg

Code	Code Description
Q5143	Injection, adalimumab-adbm, biosimilar, 1 mg
Q5144	Injection, adalimumab-aacf (idacio), biosimilar, 1 mg
Q5145	Injection, adalimumab-afzb (abrilada), biosimilar, 1 mg
S0108	Mercaptopurine, oral, 50 mg
S9338	Home infusion therapy, immunotherapy, administrative services, professional pharmacy services, care coordination, and all necessary supplies and equipment (drug and nursing visits coded separately), per diem
S9563	Home injectable therapy, immunotherapy, including administrative services, professional pharmacy services, care coordination, and all necessary supplies and equipment (drugs and nursing visits coded separately), per diem
ICD-10 codes covered if selection criteria are met:	
D86.0 - D86.9	Sarcoidosis
D89.810	Acute graft-versus-host disease
D89.812	Acute on chronic graft-versus-host disease
H20.00 - H20.9	Iridocyclitis [uveitis]
H44.131 - H44.139	Sympathetic uveitis
I30.0 - I30.9	Acute pericarditis [immune checkpoint inhibitor toxicity]
I40.0 - I40.9	Acute myocarditis [immune checkpoint inhibitor toxicity]
K50.00 - K50.919	Crohn's disease
K51.00 - K51.919	Ulcerative colitis
K52.1	Toxic gastroenteritis and colitis
K52.21 - K52.29	Allergic and dietetic gastroenteritis and colitis
K52.3	Indeterminate colitis
K52.82	Eosinophilic colitis
K52.89	Other specified noninfective gastroenteritis and colitis

Code	Code Description
K52.9	Noninfective gastroenteritis and colitis, unspecified
K60.30 - K60.529	Anal fistula
K63.2	Fistula of intestine, excluding rectum and anus
L40.0 - L40.9	Psoriasis
L73.2	Hidradenitis suppurativa
L88	Pyoderma gangrenosum
M00.00 - M01.x9	Arthropathy associated with infections
M02.30 - M02.9	Reiter's disease and other reactive arthropathies
M05.00 - M06.9	Inflammatory polyarthropathies [except bursitis] [age 18 and older]
M05.00 – M06.09, M06.0A, M06.20 – M07.69	Ankylosing spondylitis [age 18 and older]
M13.80 - M13.89	Other specified arthritis [immunotherapy arthritis] [age 18 and older]
M31.4	Aortic arch syndrome [Takayasu]
M35.2	Behcet's syndrome
N17.0 - N17.9	Acute kidney failure [immune checkpoint inhibitor toxicity]
R19.7	Diarrhea, unspecified [moderate or severe][immune checkpoint inhibitor toxicity]
R19.7 T45.AX5A - T45.AX5S	Adverse effect of immune checkpoint inhibitors and immunostimulant drugs [inflammatory arthritis]
ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):	
B17.10 - B17.11	Acute hepatitis C with or without hepatic coma
B18.2	Chronic viral hepatitis C

Code	Code Description
B19.20 - B19.21	Unspecified viral hepatitis C with or without hepatic coma
B58.01	Toxoplasma chorioretinitis
B97.29	Other coronavirus as the cause of diseases classified elsewhere [COVID-19 pneumonia]
C64.1 - C65.9	Malignant neoplasm of kidney and of renal pelvis [renal cell carcinoma]
D80.3	Selective deficiency of immunoglobulin G [IgG] subclasses
D89.811	Chronic graft-versus-host disease
D89.813	Graft versus-host disease, unspecified
E08.620	Diabetes mellitus due to underlying condition with diabetic dermatitis [necrobiosis lipoidica diabeticorum]
E09.620	Drug or chemical induced diabetes mellitus with diabetic dermatitis [necrobiosis lipoidica diabeticorum]
E10.620	Type 1 diabetes mellitus with diabetic dermatitis [necrobiosis lipoidica diabeticorum]
E11.620	Type 2 diabetes mellitus with diabetic dermatitis [necrobiosis lipoidica diabeticorum]
E13.620	Other specified diabetes mellitus with diabetic dermatitis [necrobiosis lipoidica diabeticorum]
E50.3	Vitamin A deficiency with corneal ulceration and xerosis
E85.81 - E85.89	Other amyloidosis [central nervous system amyloidosis]
H15.111 - H15.119	Episcleritis periodica fugax
H16.001 - H16.079	Corneal ulcer
H16.321 - H16.329	Diffuse interstitial keratitis [Cogan's syndrome]
H30.131 - H30.139	Disseminated chorioretinal inflammation, generalized

Code	Code Description
H30.90 - H30.93	Unspecified chorioretinal inflammation [birdshot retinochoroidopathy]
H35.061 - H35.069	Retinal vasculitis
H35.351 - H35.359	Cystoid macular degeneration
J12.89	Other viral pneumonia [COVID-19 pneumonia]
J42	Unspecified chronic bronchitis [bronchiolitis obliterans]
J44.9	Chronic obstructive pulmonary disease, unspecified
J45.20 - J45.998	Asthma
K52.0	Gastroenteritis and colitis due to radiation
K52.831 – K52.839	Microscopic colitis
K65.4	Sclerosing mesenteritis [mesenteric panniculitis]
K83.01 - K83.09	Cholangitis [autoimmune cholangiopathy]
K90.0	Celiac disease
L70.1	Acne conglobate [fulminans]
L92.0	Granuloma annulare
L92.1	Necrobiosis lipoidica, not elsewhere classified
L93.0	Discoid lupus erythematosus
L94.0	Localized scleroderma [morphea]
M06.1	Adult - onset Still's disease
M30.3	Mucocutaneous lymph node syndrome [Kawasaki]
M33.20 - M33.29	Polymyositis
M34.0 - M34.9	Systemic sclerosis [scleroderma]
M35.00 - M35.09	Sicca syndrome (Sjögren)
M35.4	Diffuse (eosinophilic) fasciitis

Code	Code Description
M51.26 - M51.27	Other intervertebral disc displacement lumbar or lumbosacral region [disc-herniation-induced sciatica]
M54.00 - M54.09	Panniculitis affecting regions of neck and back
M79.3	Panniculitis, unspecified
M86.00 - M86.9	Osteomyelitis [multi-focal]
M94.1	Relapsing polychondritis
N02.1 - N02.7	Recurrent and persistent hematuria [membranous glomerulopathy]
N04.1 - N04.7	Nephrotic syndrome [membranous glomerulopathy]
N05.0 - N05.9	Unspecified nephritic syndrome [membranous glomerulopathy]
N61.0 - N61.1	Inflammatory disorders of breast [granulomatous mastitis]
T86.850	Intestine transplant rejection
U07.1	COVID-19 [COVID-19 pneumonia]

Background

U.S. Food and Drug Administration (FDA)-Approved Indications

- infliximab/Avsola/Inflectra/Remicade/Renflexis

  - Adult patients with moderately to severely active Crohn's disease (CD) who have had an inadequate response to conventional therapy and adult patients with fistulizing CD
  - Pediatric patients 6 years of age and older with moderately to severely active Crohn's disease who have had an inadequate response to conventional therapy
  - Moderately to severely active ulcerative colitis (UC) in patients 6 years of age and older who have had an inadequate response to conventional therapy

  - Adult patients with moderately to severely active rheumatoid arthritis (RA) in combination with methotrexate
  - Adult patients with active ankylosing spondylitis (AS)
  - Adult patients with active psoriatic arthritis (PsA)
  - Adult patients with chronic severe plaque psoriasis (PsO) who are candidates for systemic therapy and when other systemic therapies are medically less appropriate

- Zymfentra

  - Maintenance treatment of moderately to severely active ulcerative colitis in adults following treatment with an infliximab product administered intravenously
  - Maintenance treatment of moderately to severely active Crohn's disease in adults following treatment with an infliximab product administered intravenously

Compendial Uses

- Acute graft-versus-host disease
- Behcet's disease
- Hidradenitis suppurativa
- Immune checkpoint inhibitor-related toxicity
- Moderate to severe plaque psoriasis
- Non-radiographic axial spondyloarthritis
- Pyoderma gangrenosum
- Reactive arthritis
- Sarcoidosis
- Takayasu's arteritis
- Uveitis

Infliximab is a monoclonal antibody that works by blocking tumor necrosis factor-alpha (TNF-α), a cytokine that plays a role in various inflammatory processes in the body. Biological activities attributed to TNF-α include: induction of pro-inflammatory cytokines such as interleukins (IL) 1 and 6, enhancement of leukocyte migration by increasing endothelial layer permeability and expression of adhesion molecules by endothelial cells and leukocytes, activation of neutrophil and eosinophil functional activity, induction of acute phase reactants and other liver proteins, as well as

tissue degrading enzymes produced by synoviocytes and/or chondrocytes. Inappropriate or excessive activation of TNF-α signaling is associated with chronic inflammation which can lead to the development of pathological complications such as autoimmune diseases. Infliximab neutralizes the biological activity of TNF-α by binding to the soluble and transmembrane forms of TNF-α, therefore effectively inhibiting the binding of TNF-α with its receptors.

Infliximab is available as an intravenous (IV) injection for infusion (Avsola [Amgen], Inflectra [Pfizer], Remicade [Janssen Biotech, Inc.], Renflexis [Organon LLC], and unbranded infliximab [Janssen Biotech, Inc.]), and as a subcutaneous (SC) injection (Zymfentra [Celltrion USA, Inc.]).

In 1998, the FDA approved Remicade for the treatment of Crohn's disease (CD), which was found to reduce infiltration of inflammatory cells and TNF-α production in inflamed areas of the intestine, and reduce the proportion of mononuclear cells from the lamina propia able to express TNF-α and interferon gamma. In1999, the FDA expanded the approval in combination with methotrexate for treatment of rheumatoid arthritis (RA), which reduced infiltration of inflammatory cells into inflamed areas of the joint as well as expression of molecules mediating cellular adhesion, chemoattraction, and tissue degradation. Over the years, approved labeling included use for ankylosing spondylitis, psoriatic arthritis, ulcerative colitis (UC), plaque psoriasis, and pediatric CD and UC.

The FDA-approved product labeling for Remicade includes a boxed warning that patients treated with infliximab are at increased risk for infections, including progression to serious infections leading to hospitalization or death. These infections have included bacterial sepsis, tuberculosis (TB), invasive fungal and other opportunistic infections. The boxed warning states that patients should be educated about the symptoms of infection, closely monitored for signs and symptoms of infection during and after treatment with infliximab, and should have access to appropriate medical care. Patients who develop an infection should be evaluated for appropriate antimicrobial therapy and for serious infections infliximab should be discontinued. The labeling states that TB (frequently disseminated or extrapulmonary at clinical presentation) has been observed in patients receiving infliximab. Patients should be evaluated for TB risk factors and be tested for latent TB infection prior to

initiating infliximab and during therapy. The labeling recommends that treatment of latent TB infection should be initiated prior to therapy with infliximab. Treatment of latent TB in patients with a reactive tuberculin test reduces the risk of TB reactivation in patients receiving infliximab. Some patients who tested negative for latent TB prior to receiving infliximab have developed active TB. Physicians should monitor patients receiving infliximab for signs and symptoms of active TB, including patients who tested negative for latent TB infection.

According to the CDC's (Centers for Disease Control and Prevention) Quality ID #176 measure, TB screening is recommended within 12 months prior to beginning a first course of biologic or immune response modifier therapy that includes a warning about the risk of reactivating latent infections. This guideline applies to adult patients (18 years and older) who are newly prescribed these therapies and have not undergone similar treatment in the last 15 months. The objective is to ensure that TB testing is recorded in the medical history before treatment begins, thereby reducing the risk of reactivating latent TB infection. The CDC states that both the tuberculin (TB) skin test (also known as the Mantoux test) and the TB blood test (interferon-gamma release assays or IGRAs) can be utilized to detect *M. tuberculosis* infection. Additional testing is necessary to confirm active TB disease. A posterior-anterior chest X-ray can help identify abnormalities in the chest, as lesions may appear in various sizes, shapes, densities, and cavitations throughout the lungs. While these abnormalities may indicate TB, they cannot definitively diagnose the disease. However, a chest X-ray can be used to exclude the possibility of pulmonary TB in individuals who have tested positive on a tuberculosis skin test (TST) or TB blood test but show no symptoms of the disease.

In addition, the boxed warning states that lymphoma and other malignancies, some fatal, have been reported in children and adolescent patients treated with tumor necrosis factor (TNF) blockers, including infliximab. Postmarketing cases of fatal hepatosplenic T-cell lymphoma (HSTCL) have been reported in patients treated with TNF blockers including infliximab. Almost all had received azathioprine or 6-mercaptopurine concomitantly with a TNF blocker at or prior to diagnosis.

The majority of infliximab cases were reported in patients with Crohn's disease or ulcerative colitis, most of whom were adolescent or young adult males (Janssen Biotech, 2021).

The use of Remicade at doses exceeding 5 mg/kg is contraindicated for patients with moderate to severe heart failure. The labeling for Remicade indicates that infliximab has been associated with adverse outcomes in individuals with heart failure and should only be considered after exploring alternative treatment options. A randomized study involving patients with heart failure (NYHA Functional Class III/IV) suggested an increased mortality rate among those receiving 10 mg/kg of infliximab, as well as a higher incidence of cardiovascular adverse events at both 5 mg/kg and 10 mg/kg doses. The labeling also notes that there have been post-marketing reports of worsening heart failure, with or without identifiable triggers, in patients taking infliximab. Additionally, there have been cases of new or worsening heart failure, including in individuals without prior cardiovascular issues, some of whom were under 50 years old. If infliximab is prescribed to patients with heart failure, it is advised that they be closely monitored during treatment, and the medication should be discontinued if any new or worsening heart failure symptoms arise.

In addition to the risks of serious infections, malignancies, and heart failure, other labeled warnings and precautions include:

- Invasive fungal infections
- Malignancies, including cervical cancer and lymphoma
- Reactivation of hepatitis B virus
- Hepatotoxicity
- Cytopenias
- Hypersensitivity reactions, including serious infusion reactions such as anaphylaxis or serum sickness-like reactions
- Cardiovascular and cerebrovascular events, including strokes, myocardial infarctions (some of which may be fatal), and arrhythmias occurring during or within 24 hours of infliximab infusion
- Exacerbation or new onset of demyelinating diseases
- Lupus-like syndrome

- Vaccination considerations: Before initiating infliximab, ensure that both pediatric and adult patients are up to date on all vaccinations. Live vaccines or therapeutic infectious agents should not be given with infliximab. For infants exposed to infliximab in utero, it is recommended to wait at least six months after birth before administering live vaccines.

Most common adverse reactions (greater than 10%) include infections (e.g., upper respiratory, sinusitis, and pharyngitis), infusion-related reactions, headache, and abdominal pain.

The product labeling for Remicade recommends against use with other biologics. The prescribing information states: "The combination of Remicade with other biological therapeutics used to treat the same conditions as Remicade is not recommended."

In April 2016, the FDA approved Inflectra (infliximab-dyyb) (Pfizer, Inc.) as a biosimilar to Remicade (infliximab) (Janssen Biotech, Inc). The approval included treatment of RA, adult UC, plaque psoriasis, psoriatic arthritis, ankylosing spondylitis, and adult and pediatric CD. In June 2019, the FDA approval was expanded to include pediatric UC.

In 2017, the FDA approved Renflexis (infliximab-abda), a biosimilar referencing infliximab (Remicade), across all eligible indications. Renflexis has labeled indications for reducing signs and symptoms in adult and pediatric CD, adult UC, RA, ankylosing spondylitis, psoriatic arthritis, and adult plaque psoriasis. The indication and usage to include pediatric UC was later FDA-approved in June 2019. Renflexis is manufactured by Samsung Bioepis of Incheon, Korea. Renflexis was originally marketed and distributed in the United States by Merck, and now by Organon LLC.

In December 2019, the FDA approved the biosimilar, Avsola (infliximab-axxq) (Amgen), for all approved indications of the reference product, Remicade (infliximab). Thus, Avsola is FDA approved for the treatment of moderate-to-severe RA, moderate-to-severe CD in the adult and pediatric population, moderate-to-severe UC in the adult and pediatric population, chronic severe plaque psoriasis (PsO), psoriatic arthritis (PsA) and ankylosing spondylitis (AS). Avsola, an anti-tumor necrosis factor alpha

(anti-TNF) monoclonal antibody, was found to be highly similar to Remicade with no clinically meaningful differences based on a totality of evidence which included comparative analytical, nonclinical and clinical data (Amgen, 2019b).

FDA approval was based on a randomized, double-blind comparative clinical study which evaluated the efficacy and safety of infliximab-axxq (ABP710; Avsola) compared to infliximab (Remicade) in 558 patients with moderate-to-severe RA. The patients were randomized (1:1) to receive either Avsola or Remicade at a dose of 3 mg/kg administered as an infusion on day 1, at weeks 2 and 6, and every 8 weeks thereafter. The primary endpoint was the response difference (RD) of 20% improvement in American College of Rheumatology core set measurements (ACR20) at week 22. Key secondary endpoints included DAS28-CRP change from baseline, RD of ACR20, ACR50 and ACR70 at weeks 2, 6, 14, 22, 30, 34, 38, 46 and 50. The study also incorporated the evaluation of a single transition in 119 subjects from Remicade to Avsola at week 22, which demonstrated similar safety and immunogenicity in patients who were previously on Remicade (Amgen 2019b, 2019c).

Avsola has the same pharmaceutical dosage form and strength as Remicade.

In October 2023, the FDA approved infliximab-dyyb subcutaneous (SC) injection, branded as Zymfentra (Celltrion USA, Inc.), for maintenance therapy in adults with moderately to severely active UC and CD following treatment with an infliximab product administered intravenously. FDA approval was based on the safety and efficacy data obtained from the LIBERTY-UC and LIBERTY-CD trials, which demonstrated superiority in the primary endpoints of clinical remission (UC and CD) and endoscopic response (CD) compared to placebo when Zymfentra was used as maintenance therapy in patients who had received intravenous infliximab induction therapy. The overall safety profile was found to be similar to that of placebo (Celltrion USA, 2023).

Per the label, Zymfentra is indicated for maintenance treatment only in adults with CD or UC after completing an intravenous (IV) induction regimen with an infliximab product. The recommended maintenance dosage starting at Week 10 and thereafter is 120 mg administered SC

once every two weeks. Persons who are responding to IV infliximab maintenance therapy may switch to SC dosing by administering the first SC dose of Zymfentra in place of the next scheduled IV infusion and every two weeks thereafter. If a healthcare professional determines that it is appropriate, persons may self-inject Zymfentra or caregivers may inject Zymfentra using either the Zymfentra prefilled syringe, Zymfentra pre-filled syringe with needle shield, or Zymfentra pen after proper training in subcutaneous injection technique (Celltrion USA, 2024).

Zymfentra carries boxed warning as other infliximab products. Other warnings and precautions include risk of hepatitis B virus reactivation, hepatotoxicity, congestive heart failure, hematologic reactions, risk of infection with concurrent administration of other biological products, risk of additive immunosuppressive effects from prior biological product use, and formation of autoantibodies and development of lupus-like syndrome. Exacerbation or new onset of CNS demyelinating disorders may occur. The most common adverse reactions (3% or more) include:

- Ulcerative Colitis: COVID-19, anemia, arthralgia, injection site reaction, increased alanine aminotransferase, and abdominal pain
- Crohn's Disease: COVID-19, headache, upper respiratory tract infection, injection site reaction, diarrhea, increased blood creatine phosphokinase, arthralgia, increased alanine aminotransferase, hypertension, urinary tract infection, neutropenia, dizziness, and leukopenia.

Biosimilar and Interchangeable Drugs

Biosimilar drugs are FDA-approved biological products that have demonstrated that they are highly similar without clinically meaningful differences in terms of safety and effectiveness from the biologic reference product. Only minor differences in clinically inactive components are allowable in biosimilar drugs. Some biosimilar drugs may have the 'interchangeable' designation. An interchangeable biological product is a biosimilar that meets additional FDA requirements and may be substituted for the reference product at the pharmacy without the intervention of the prescribing health care provider, similar to how generic drugs are routinely substituted for brand-name drugs (FDA, 2023; FDA, undated). For approval as an interchangeable product, the

biosimilar must be expected to produce the same clinical result as the reference product in any given patient, and for a product administered more than once, the risk of switching between a reference product and an interchangeable product must not be greater than the risk of using the reference product without switching (Herndon et al, 2023).

Despite the adoption of biosimilars in many therapeutic areas, concerns persist regarding switching a patient to a biosimilar whose condition is stable while on the reference biologic. Herndon et al (2023) performed a systematic review and meta-analysis to examine the occurrence of safety events following a switch to or from a biosimilar and its reference biologic in all identified controlled clinical studies that included a biosimilar approved by the FDA. The authors reviewed FDA databases containing publicly available information for all randomized controlled studies and extension studies with a switch treatment period (STP) to or from a biosimilar and its reference biologic. Information not captured in FDA reviews was identified by searching the Embase, MEDLINE, and PubMed databases for the period of January 1, 2000, through December 31, 2022. Forty-four STPs were identified from 31 unique studies for 21 different biosimilars. Data were extracted and synthesized following PRISMA guidelines. Meta-analysis was conducted to estimate the overall risk difference across studies. A total of 5,252 patients who were switched to or from a biosimilar and its reference biologic were identified. Safety data including deaths, serious adverse events (SAEs), and treatment discontinuation showed an overall risk difference (95% CI) of -0.00 (-0.00, 0.00), 0.00 (-0.01, 0.01), -0.00 (-0.01, 0.00) across STPs, respectively. Immunogenicity data showed similar incidence of anti-drug antibodies and neutralizing antibodies in patients within a STP who were switched to or from a biosimilar to its reference biologic and patients who were not switched. Immune related adverse events such as anaphylaxis, hypersensitivity reactions, and injections site reactions were similar in switched and non-switched patients. The authors report that this is the first systematic review using statistical methods to address the risk of switching patients between reference biologics and biosimilars which found no difference in the safety profiles or immunogenicity rates in patients who were switched and those who remained on a reference biologic or a biosimilar. It is of note that endpoints related to pharmacokinetics (PK) or effectiveness of a drug that were collected during the STPs were not included in this systematic review. If efficacy

data were collected periodically during STPs, it was not powered for an efficacy determination. The efficacy data used to determine equivalent efficacy of a biosimilar to a reference biologic is contained in publicly available FDA reviews and in the publications for the individual studies that led to approval of the biosimilars.

Hinshaw et al (2022) state that infliximab biosimilar agents have been recommended for pediatric inflammatory bowel disease (IBD), and although institutions are initiating therapy with the biosimilar agents (IFX-B), few are switching maintenance therapy from the originator Remicade (IFX-O). In a single-center, retrospective chart review, the authors compared biochemical markers of disease activity of 25 children with IBD on maintenance therapy with Inflectra (IFX-B) to their previous markers on Remicade (IFX-O) between March 2018 and June 2018. Analysis included demographics and various biochemical markers of disease control. The nonparametric-related samples Wilcoxon signed-rank test was used to compare mean ranks of these markers (C-reactive protein, erythrocyte sedimentation rate, hemoglobin, platelet count, albumin, body mass index z score) between the last 12 months on IFX-O and the first 12 months on IFX-B. Of the 25 children included, 17 were diagnosed with Crohn disease and 8 with ulcerative colitis. The results of all, except albumin value, supported retention of the null hypothesis that there would not be a statistically significant difference in the biochemical markers of disease activity between the 2 medications. The authors concluded that Inflectra (IFX-B) is as effective as Remicade (IFX-O) for maintenance therapy in pediatric IBD when comparing biochemical markers of disease activity.

In summary, all biological products are approved only after they meet the FDA's rigorous approval standards. This means health care providers and patients can expect the same safety and effectiveness from both a biosimilar and an interchangeable biosimilar, just as they would for a reference product (FDA, 2023; FDA, undated).

Acne Fulminans

Iqbal and Kolodney (2005) stated that acne fulminans is a syndrome of sudden onset hemorrhagic and ulcerative acne involving the back, chest, and face combined with systemic symptoms. It can be the dermatologic

manifestation of the synovitis-acne-pustulosis-hyperostosis-osteitis (SAPHO) syndrome. Current therapy for acne fulminans consists of wound care, topical and systemic corticosteroids, isotretinoin, and non-steroidal anti-inflammatory drugs (NSAIDs). Infliximab has shown efficacy in the treatment of psoriatic arthritis and ankylosing spondylitis both of which share clinical similarities to the SAPHO syndrome. These investigators reported the case of a patient with the SAPHO syndrome and acne fulminans who was treated with infliximab; 10 months after initiating therapy with infliximab, the area of the patient's ulcerative lesions was reduced by 70 %. The authors concluded that infliximab might be considered as a treatment option for patients with acne fulminans unresponsive to conventional therapies.

Furthermore, an UpToDate review on “Treatment of acne vulgaris” (Graber, 2019) states that “Oral glucocorticoids (typically, prednisone 0.5 to 1 mg/kg per day) and oral isotretinoin are the mainstays of treatment for acne fulminans. There is a paucity of high-quality data to guide the approach to treatment”.

Currently, there is insufficient evidence to support the use of infliximab for the treatment of acne fulminans.

Age-Related Macular Degeneration

An UpToDate review on “Age-related macular degeneration: Treatment and prevention” (Arroyo, 2013) states that “Several investigational clinical trials are underway to evaluate treatment options for dry AMD. Potential new treatments include topical antioxidant eye drops, implantation of encapsulated human NTC cells, and fetal cell transplantation. The first report of transplantation of cells derived from human embryonic stem cells (hESCs) into human beings involved subretinal transplantation of cells differentiated to retinal pigment epithelium in a patient with dry AMD. At four months there was no evidence of rejection, tumorigenicity, or ectopic tissue; there was minimal visual improvement although the patient had advanced disease at baseline. Ongoing trials for patients with neovascular AMD include evaluation of the angiogenesis inhibitor pazopanib given as topical eye drops, subconjunctival injection of an antiproliferative polyamine analog (CGC-11047), and various combination treatment protocols that combine photodynamic therapy with intravitreal

anti-VEGF agents. A topical kinase inhibitor with multiple growth factor targets, including VEGF, suppressed choroidal neovascularization in a mouse model. Genetic defects in various loci in the complement factor H gene can be found in over 50 percent of cases of macular degeneration. Gene-environment interactions compound the genetic risk for AMD and have been demonstrated for smokers and for antibodies to C pneumonia. Rapid and affordable genetic screening is being developed commercially, with the intent to identify high risk patients who could modify some risk factors or benefit from targeted therapy. Some genetic mutations associated with AMD are thought to result in increased complement activation and inflammation. The effects of anti-inflammatory drugs or vaccines on the progression of AMD are under investigation. Research trials include study of an intravitreal complement inhibitor (POT-4), vaccination with the immunomodulator copaxone, and treatment with anti-inflammatory agents (infliximab, sirolimus, and daclizumab). Whenever possible, patients should be encouraged to enroll in clinical trials of treatments for AMD".

Ankylosing Spondylitis and Spondyloarthropathies

Two published randomized controlled trials have reported on significant reductions in disease activity in patients with ankylosing spondylitis and other spondyloarthropathies who were treated with infliximab. Spondyloarthropathy (literally arthritis of the spine) may be associated with ankylosing spondylitis, Reiter's syndrome, reactive arthritis, psoriatic arthritis, and inflammatory bowel disease, or may be idiopathic (undifferentiated spondyloarthropathy). Van den Bosch et al (2002) reported on a 12-week long clinical study involving forty patients with active spondyloarthropathy who were randomly assigned to receive an intravenous loading dose (weeks 0, 2, and 6) of 5 mg/kg infliximab or placebo. Both patient and physician global assessments of disease activity on a visual analog scale improved significantly in the infliximab group compared with the baseline value, with no improvement in the placebo group. As early as week 2 and sustained up to week 12, there was a highly statistically significant difference between the values for these 2 endpoints in the infliximab versus the placebo group. In most of the other assessments of disease activity (laboratory measures, assessments of specific peripheral and/or axial disease), significant improvements were observed in the infliximab group compared with the

baseline value and compared with placebo. There was 1 severe drug-related adverse event, in which a patient developed disseminated tuberculosis.

Braun et al (2002) reported the results of a 12-week randomized placebo-controlled clinical trial involving 35 patients with active ankylosing spondylitis treated with intravenous 5 mg/kg infliximab infusion (at weeks 0, 2 and 6) and 35 patients assigned to placebo. Eighteen (53 %) of 34 patients on infliximab had a regression of disease activity at week 12 of at least 50 % compared with 3 (9 %) of 35 on placebo (difference 44 % (95 % confidence interval [CI]: 23 to 61, $p < 0.0001$). Function and quality of life also improved significantly on infliximab but not on placebo ($p < 0.0001$ and $p < 0.0001$, respectively). The investigators reported that treatment with infliximab was generally well-tolerated, but 3 patients had to stop treatment because of systemic tuberculosis, allergic granulomatosis of the lung, or mild leucopenia.

These randomized controlled trials confirm the findings of an open label study of infliximab in 21 patients with active spondyloarthropathy who received a maintenance regimen of 5 mg/kg infliximab every 14 weeks, 19 of whom were followed for 1 year (Kruithof et al, 2002). The investigators reported that, after each re-treatment a sustained significant decrease of all disease manifestations was observed. Before re-treatment, symptoms recurred in 3/19 (16 %) at week 20, in 13/19 (68 %) at week 34, and in 15/19 (79 %) at week 48. Twelve minor infectious episodes were observed in this cohort.

Based on the evidence of efficacy of infliximab in a variety of spondyloarthropathies where TNF plays a role, the U.S. Pharmacopeial Convention (2003) has concluded that reactive arthritis and inflammatory bowel disease arthritis are accepted off-label indications for infliximab.

Atopic Eczema

Schmitt et al (2007) noted that systemic immunosuppressive agents are recommended for patients with atopic eczema in whom disease activity can not be controlled adequately with topical treatments. Guidelines do not give clear advice which agents to prefer. These investigators systematically reviewed clinical trials on systemic treatment for severe

atopic eczema to provide evidence-based treatment recommendations. Standardized literature search, independent standardized assessment of eligibility and data abstraction was performed by 2 reviewers. A total of 27 studies totaling 979 patients were included. Eleven studies consistently showed effectiveness of cyclosporine. Cyclosporine is recommended as first option for patients with atopic eczema refractory to conventional treatment. Evidence from randomized controlled trials also exists for gamma interferon and azathioprine. Although frequently used in clinical practice, systemic glucocorticosteroids have not been assessed adequately in studies. Mycophenolate mofetile showed effectiveness in 2 small uncontrolled studies. Intravenous immunoglobulins and infliximab are not recommended based on published data.

Belloni et al (2008) stated that atopic eczema is a common inflammatory skin disease showing chronically relapsing eczema and high association with elevated serum IgE levels. A subgroup of atopic eczema patients requires systemic immunomodulatory treatment for long time periods. However, beyond cyclosporine A and azathioprine, only limited consent exists on systemic treatment options. Timely published systemic treatment modalities include studies on efalizumab, infliximab, adalimumab, and etanercept, omalizumab, rituximab, specific immunotherapy, leflunomide, and leukotriene receptor antagonists with varying clinical results and with particular safety profiles. The authors concluded that although there is not yet a treatment modality reaching clinical efficacy of cyclosporine A as gold standard of systemic therapy, limitation in its application duration as in its side effect profile as well as the search for alternatives has set a focus on the new alternatives of which especially B-cell-directed therapies might be promising candidates.

Autoimmune Cholangiopathy

Rojas-Feria et al (2013) noted that abnormal liver biochemical tests are present in up to 30% of patients with inflammatory bowel disease (IBD), and therefore become a diagnostic challenge. Liver and biliary tract diseases are common extra-intestinal manifestations for both Crohn's disease and ulcerative colitis (UC), and typically do not correlate with intestinal activity. Primary sclerosing cholangitis (PSC) is the most common hepatobiliary manifestation of IBD, and is more prevalent in UC. Approximately 5% of patients with UC develop PSC, with the prevalence

reaching up to 90 %. Cholangiocarcinoma and colon cancer risks are increased in these patients. Less common disorders include autoimmune hepatitis/PSC overlap syndrome, IgG4-associated cholangiopathy, primary biliary cirrhosis, hepatic amyloidosis, granulomatous hepatitis, cholelithiasis, portal vein thrombosis, liver abscess, and non-alcoholic fatty liver disease. Hepatitis B reactivation during immunosuppressive therapy is a major concern, with screening and vaccination being recommended in serologically negative cases for patients with IBD. Reactivation prophylaxis with entecavir or tenofovir for 6 to 12 months after the end of immunosuppressive therapy is mandatory in patients showing as hepatitis B surface antigen (HBsAg) positive, independently from viral load. HBsAg negative and anti-HBc positive patients, with or without anti-HBs, should be closely monitored, measuring alanine aminotransferase and hepatitis B virus DNA within 12 months after the end of therapy, and should be treated if the viral load increases. On the other hand, immunosuppressive therapy did not seem to promote reactivation of hepatitis C, and hepatitis C anti-viral treatment did not influence IBD natural history either. Most of the drugs used for IBD treatment may induce hepatotoxicity, although the incidence of serious adverse events (AEs) is low. Abnormalities in liver biochemical tests associated with aminosalicylates are uncommon and are usually not clinically relevant. Methotrexate-related hepatotoxicity has been described in 14% of patients with IBD, in a dose-dependent manner. Liver biopsy is not routinely recommended. Biologics-related hepatotoxicity is rare, but has been shown most frequently in patients treated with infliximab. Thiopurines have been associated with veno-occlusive disease, regenerative nodular hyperplasia, and liver peliosis. Routine liver biochemical tests are recommended, especially during the 1st month of treatment. All these conditions should be considered in IBD patients with clinical or biochemical features suggestive of hepatobiliary involvement. Diagnosis and management of these disorders usually involve hepatologists and gastroenterologists due to its complexity.

Furthermore, an UpToDate review on “Autoimmune hepatitis variants: Definitions and treatment” (Heneghan, 2018) does not mention infliximab as a therapeutic option.

Behçet's disease

In a review on standard and novel treatments to Behçet's disease, Gul (2007) stated that multi-center, multi-disciplinary and long-term trials aiming to assess the effectiveness of interventions (including infliximab) in both the treatment of acute inflammatory attacks and the prevention of relapses are needed in order to provide more generalizable results that can lead to better management plans. Kobayashi et al (2007) stated that 5-aminosalycylic acid, corticosteroids, immunosuppressants, enteral nutrition, total parenteral nutrition, and surgical therapy were considered standard therapy for intestinal Behçet's disease. Moreover, infliximab, colchicines, thalidomide, other pharmacological therapy, endoscopic therapy, and leukocytapheresis were deemed experimental therapy.

In the 2018 update of the European League Against Rheumatism (EULAR) recommendations for the management of Behcet's syndrome (BS), Hatemi and colleagues (2018) noted that several new therapeutic modalities with different mechanisms of action have been studied in patients with BS. These researchers updated the recommendations in the light of these new data under the auspices of EULAR Standing Committee for Clinical Affairs. The recommendations on the medical management of muco-cutaneous, joint, eye, vascular, neurological and GI involvement of BS were modified; 5 overarching principles and a new recommendation about the surgical management of vascular involvement were added. For BS with eye involvement, among the monoclonal anti-TNF antibodies, although there is more accumulated experience with infliximab, adalimumab also appeared to be an effective alternative. Switching between these agents appeared to be possible in patients with primary or secondary unresponsiveness or AEs. Patients presenting with an initial or recurrent episode of acute sight-threatening uveitis should be treated with high-dose glucocorticoids, infliximab or interferon alpha. Intravitreal glucocorticoid injection is an option in patients with unilateral exacerbation as an adjunct to systemic treatment (Level of evidence: IIA; strength of recommendation: B).

Birdshot Retinochoroidopathy

In a retrospective case series, Artornsombudh et al (2013) reported the outcomes of infliximab treatment of birdshot retinochoroidopathy (BSRC) refractory to conventional immunomodulatory therapy. A total of 22 BSRC patients (44 eyes) who received infliximab between July 2005 and June 2012 were identified by retrospective chart review. All patients received 4 to 5 mg/kg infliximab at 4- to 8-week intervals. Data regarding patient demographics, use of immunosuppressive drugs, biologic agents, and reason for conventional therapy discontinuation were gathered. Disease activity markers, including signs of ocular inflammation, fluorescein angiography evidence of retinal vasculitis or papillitis, indocyanine green angiography evidence of active choroiditis, electroretinography parameters indicative of stable or worsening of retinal functions, and optical coherence tomography findings indicative of static or worsening macular edema were recorded. Main outcome measures included abolition of all evidence of active inflammation, visual acuity (VA), presence of cystoid macular edema at 6 months and 1 year, and adverse responses to infliximab. Mean duration of disease before starting infliximab was 58.6 months. Before infliximab therapy, all patients received and failed conventional immunosuppressive therapy; 10 patients had received another biologic agent. After initiating infliximab, control of inflammation was achieved in 81.8 % at 6 months and in 88.9 % at the 1-year follow-up. Three patients had active inflammation during therapy. The rate of cystoid macular edema decreased from 22.7 % at baseline to 13.9 % at 6 months and 6.7 % at 1 year after receiving the drug. Initial VA of 20/40 or better was found in 34 eyes (84.1 %). At 6 months and 1 year, 91.7 % and 94.4 % of eyes, respectively, had VA of 20/40 or better. Six patients had adverse events; infliximab therapy was discontinued in these patients because of neuropathy, drug-induced lupus, allergic reaction, or fungal infection. The authors concluded that these data suggested that infliximab is effective for controlling inflammation in otherwise treatment-refractory cases of BSRC. The main drawbacks of this study were its retrospective design, small sample size, and the lack of a control group. The authors stated that further prospective randomized controlled studies in larger populations are needed to ascertain the potential role of infliximab in the treatment of BSRC.

Cerebral Amyloid Angiopathy

An UpToDate review on "Cerebral amyloid angiopathy" (Greenberg, 2022) does not mention infliximab as a therapeutic option.

Chronic Cutaneous Sarcoidosis

Doherty and colleagues (2008) stated that although healthcare providers have arrived at a relatively comfortable zone of accepted clinical practice in the management of cutaneous sarcoidosis, virtually every treatment is based on minimal evidence-based data and relies almost exclusively on anecdotal information. Although it would be convenient to blame this state of affairs on the lack of certainty about disease etiology, the unavoidable fact is that little has been executed, even in the realm of well-designed comparative trials. Nonetheless, worldwide accepted standard therapies for sarcoidosis include the administration of corticosteroids, anti-malarials and methotrexate. A step-wise approach to patient care is appropriate, and potent topical corticosteroids (e.g., clobetasol) or repeated intralesional injections of triamcinolone (3 to 10 mg/ml) may be all that is needed in mild skin-limited disease. In patients requiring systemic therapy for recalcitrant or deforming skin lesions (or for widespread disease), corticosteroids (e.g., prednisone 40 to 80 mg/day, tapered accordingly) used alone or in combination with anti-malarials or methotrexate may be indicated. Anti-malarials and methotrexate are considered 2nd-line interventions and may be used as monotherapy for steroid-resistant sarcoidosis or in patients unable to tolerate steroids. Given the concern regarding ocular toxicity, the maximum dosages of chloroquine and hydroxychloroquine should not exceed 3.5 and 6.5 mg/kg/day, respectively. Methotrexate is given in weekly doses of 10 to 30 mg, with the caveat that hematological, gastro-intestinal (GI), pulmonary and hepatic toxicities are possible. Despite universal acceptance as standard care, the afore-mentioned treatments often result in an incomplete clinical response or unacceptable AEs. In such situations, more innovative treatment options may be used. Treatments that may well gain widespread future use include the TNF-alpha inhibitors infliximab and adalimumab. Experience was limited, but early reports are promising. Infliximab was administered via intravenous infusion at doses of 3 to 10 mg/kg at 0, 2 and 6 weeks and as indicated thereafter, whereas adalimumab was injected subcutaneously at doses of 40 mg either

weekly or every 2 weeks. Because adalimumab is not approved for the management of sarcoidosis, the optimum dose administration interval is uncertain. However, it has been given in both weekly and every other week regimens. Isotretinoin, 0.5 to 2 mg/kg/day, has been used successfully in a handful of reported cases. However, the teratogenic potential of isotretinoin is often prohibitive considering that the primary demographic group likely to develop sarcoidosis is women of child-bearing potential. Thalidomide at dosages of 50 to greater than 400 mg/day has limited, albeit promising, supporting data. However, access is restricted in many countries because of a deserved pregnancy category X rating. Melatonin (20 mg/day) and allopurinol (100 to 300 mg/day) are not well studied in cutaneous sarcoidosis, and the clinical experience with tetracycline derivatives has been mixed. That said, there are compelling reports of therapeutic benefit with both doxycycline and minocycline. Because neither of these agents is associated with the severe toxicity of cytotoxic drugs, they may serve as effective therapy in some patients. Pentoxifylline (400 mg, 3 times daily) has been of use in a small number of reported cases of pulmonary sarcoidosis, but there are no reports on its use in patients with primarily cutaneous disease. Both cyclosporine and chlorambucil have been largely abandoned given their associated toxicity and disappointingly unreliable efficacy. Finally, laser therapy is a newer modality that may provide patients with a quick and non-invasive treatment option for cutaneous sarcoidosis.

Toussirot and Pertuiset (2010) noted that increased production of TNF-α by alveolar macrophages and involvement of TNF-α in granuloma formation suggested that this cytokine is involved in the pathophysiology of sarcoidosis. The 3 available TNF-α blocking agents have been tested in sarcoidosis refractory to corticosteroids or immunosuppressive drugs. Data were available from isolated case reports or limited series of patients treated in open-label trials with favorable issue with anti-TNF-α monoclonal antibodies; 2 randomized, placebo-controlled studies evaluated the efficacy of infliximab in pulmonary and extra-pulmonary sarcoidosis, showing that infliximab improved significantly extra-pulmonary disease. There was no significant difference between infliximab and placebo in the treatment of pulmonary manifestations. Etanercept showed no efficacy for treating ocular sarcoidosis in a controlled trial and for pulmonary disease in an open-label trial. Paradoxical cases of proven sarcoidosis have been reported in patients

receiving anti-TNF-α agents for chronic inflammatory rheumatic diseases. A literature review identified 28 cases, including 16 with etanercept, 8 with infliximab and 4 with adalimumab. Although these cases were mainly reported with etanercept, paradoxical sarcoidosis has been reported with the 3 available anti-TNF-α agents, suggesting a class effect. Changes in the cytokine balance may be involved in these cases of induced sarcoidosis, which must be known by the clinician.

Drent et al (2014) stated that in severe refractory sarcoidosis cases not responding to conventional immunosuppressive treatment, the 3rd-line TNF-α inhibitors infliximab and adalimumab might be an alternative. However, appropriate studies to guide the clinician are lacking. These researchers established practical recommendations for the use of TNF-α inhibitors in the management of refractory sarcoidosis patients. Based on a literature search and the opinion of sarcoidosis experts worldwide, the recommendations were established. Studies conducted in sarcoidosis were supplemented with data obtained from relevant studies in other inflammatory diseases. A Delphi method of polling, using an online survey addressing 12 clinical questions, was performed amongst 20 of the world's leading sarcoidologists to investigate consensus in case of inadequate data to determine an objective answer. Of the 256 papers found, 101 were included. Randomized controlled trials (RCTs) about the use of TNF-α inhibitors in sarcoidosis were limited; 95 % (19 of 20) of the sarcoidologists contacted, completed the questionnaire (Europe 68 %, North America 32 %); 9 recommendations were formulated concerning general aspects of TNF-α inhibitor use. The authors concluded that based on earlier studies and consensus among world's leading sarcoidologists, practical recommendations for the use of TNF-α inhibitors in sarcoidosis were established. These recommendations, with emphasis on indications, dosage and discontinuation regimens, have been developed to support the clinician in the management of refractory sarcoidosis patients.

In a large, observational study, Heidelberger et al (2017) determined the safety and efficacy of anti-TNF in treating cutaneous sarcoidosis. STAT (Sarcoidosis Treated with Anti-TNF) is a French retrospective and prospective multi-center observational database that received data from teaching hospitals and referral centers, as well as several pneumology, dermatology, and internal medicine departments . Included patients had histologically proven sarcoidosis and received anti-TNF between January

2004 and January 2016. These researchers extracted data for patients with skin involvement at anti-TNF initiation. Response to treatment was evaluated for skin and visceral involvement using the ePOST (extra-pulmonary Physician Organ Severity Tool) severity score (from 0 [not affected] to 6 [very severe involvement]). Epidemiological and cutaneous features at baseline, efficacy, steroid-sparing, safety, and relapses were recorded. The overall cutaneous response rate (OCRR) was defined as complete (final cutaneous ePOST score of 0 or 1) or partial response (ePOST drop greater than or equal to 2 points from baseline but greater than 1 at last follow-up). Among 140 patients in the STAT database, 46 had skin involvement. The most frequent lesions were lupus pernio (n = 21 [46 %]) and nodules (n = 20 [43 %]). The median cutaneous severity score was 5 and/or 6 at baseline; 21 patients were treated for skin involvement and 25 patients for visceral involvement. Reasons for initiating anti-TNF were failure or adverse effects of previous therapy in 42 patients (93 %). Most patients received infliximab (n = 40 [87 %]), with systemic steroids in 28 cases (61 %) and immunosuppressants in 32 cases (69.5 %). The median (range) follow-up was 45 (3 to 103) months. Of the 46 patients with sarcoidosis and skin involvement who were treated with anti-TNF were included, median (range) age was 50 (14 to 78) years, and 33 patients (72 %) were women. The OCRR was 24 % after 3 months, 46 % after 6 months, and 79 % after 12 months. Steroid sparing was significant. Treatment was discontinued because of AEs in 11 patients (24 %), and 21 infectious events occurred in 14 patients (30 %). Infections were more frequent in patients treated for visceral involvement than in those treated for skin involvement (n = 12 of 25 [48 %] versus n = 2 of 21 [9.5 %], respectively; p = 0.02). The relapse rate was 44 % 18 months after discontinuation of treatment. Relapses during treatment occurred in 35 % of cases, mostly during anti-TNF or concomitant treatment tapering. The authors concluded that anti-TNF agents were effective but suspensive in cutaneous sarcoidosis. The risk of infectious events must be considered.

Furthermore, an UpToDate review on “Management of cutaneous sarcoidosis” (Prystowsky and Sanchez, 2019) states that “For patients who fail to improve with 1st- and 2nd-line therapies for cutaneous sarcoidosis, we suggest a trial of anti-TNF therapy (Grade 2C). Therapeutic options include infliximab, adalimumab, and thalidomide. Additional study is necessary to confirm the efficacy of these therapies”.

Chronic Pulmonary Sarcoidosis

There is limited evidence of infliximab's effectiveness in persons with chronic pulmonary sarcoidosis who remain symptomatic despite treatment with steroids or immunosuppressants. Baughman et al (2005) reported on a randomized controlled clinical trial comparing 2 doses of infliximab (3 or 5 mg/kg) to placebo in 138 patients with chronic (greater than 1-year duration) sarcoidosis who remain symptomatic (American Thoracic Society dyspnea score greater than 1) despite treatment with 3 or more months of prednisone (10 mg or more) or immunomodulator therapy or both, with evidence of parenchymal disease (Stage II or II) on chest x-ray, and a forced vital capacity (FVC) of greater than 50 % to less than 75 % predicted. The investigators reported significant (delta 2.5 %, p = 0.038) improvement in the percent of predicted FVC at week 24, the primary study endpoint, in the combined infliximab groups. The results did not differ significantly between infliximab doses. The investigators reported that subgroup analysis demonstrated greater benefit in patients with more extensive sarcoidosis disease burden, duration, activity, and severity.

Chronic Recurrent Multifocal Osteomyelitis (CRMO)

Deutschmann et al (2005) described the case of an 18-year old girl with chronic recurrent multifocal osteomyelitis (CRMO) over a period of 10 years. She had suffered predominantly from very painful recurrent swelling of her cheeks. Various therapeutic regimens including non-steroidal anti-inflammatory drugs (NSAIDs) and steroids had shown only a partial or temporary response. Because tumor necrosis factor-alpha-blocking agents have been successfully applied in Crohn's-associated CRMO and the related SAPHO (synovitis, acne, pustulosis, hyperostosis, and osteitis) syndrome, tumor necrosis factor-alpha-blocking therapy with infliximab was initiated. Thereafter, apart from 1 mild episode, no additional recurrences were observed during 21 months of follow-up. Infliximab was well-tolerated, and steroids were tapered off. The authors concluded that their observation indicated that infliximab may be an effective therapeutic option in CRMO.

Coronavirus Disease 2019 (COVID-19) Pneumonia

O'Halloran et al (2023) stated that immune dysregulation contributes to poorer outcomes in COVID-19. In a randomized, double-masked, placebo-controlled clinical trial, these investigators examined if abatacept, cenicriviroc, or infliximab would provide benefit when added to standard care for COVID-19 pneumonia. This study employed a master protocol to examine immunomodulators added to standard care for treatment of patients hospitalized with COVID-19 pneumonia. The results of 3 sub-studies were reported from 95 hospitals at 85 clinical research sites in the U.S. and Latin America. Hospitalized patients 18 years or older with confirmed SARS-CoV-2 infection within 14 days and evidence of pulmonary involvement underwent randomization between October 2020 and December 2021. Interventions entailed single infusion of abatacept (10 mg/kg; maximum dose, 1,000 mg) or infliximab (5 mg/kg) or a 28-day oral course of cenicriviroc (300-mg loading dose followed by 150 mg twice-daily). The primary outcome was time-to-recovery by day 28 evaluated using an 8-point ordinal scale (higher scores indicate better health). Recovery was defined as the 1st day the subject scored at least 6 on the ordinal scale. Of the 1971 subjects randomized across the 3 sub-studies, the mean (SD) age was 54.8 (14.6) years and 1,218 (61.8 %) were men. The primary endpoint of time-to-recovery from COVID-19 pneumonia was not significantly different for abatacept (recovery rate ratio [RRR], 1.12 [95 % CI: 0.98 to 1.28]; p = 0.09), cenicriviroc (RRR, 1.01; 95 % CI: 0.86 to 1.18; p = 0.94), or infliximab (RRR, 1.12; 95 % CI: 0.99 to 1.28; p = 0.08) compared with placebo. All-cause 28-day mortality was 11.0 % for abatacept versus 15.1 % for placebo (OR, 0.62; 95 % CI: 0.41 to 0.94), 13.8 % for cenicriviroc versus 11.9 % for placebo (OR, 1.18; 95 % CI: 0.72 to 1.94]), and 10.1 % for infliximab versus 14.5 % for placebo (OR, 0.59; 95 % CI: 0.39 to 0.90). Safety outcomes were comparable between active treatment and placebo, including secondary infections, in all 3 sub-studies. The authors concluded that time-to-recovery from COVID-19 pneumonia among hospitalized participants was not significantly different for abatacept, cenicriviroc, or infliximab versus placebo.

Crohn's Disease

Infliximab is indicated for the reduction in signs and symptoms of active Crohn's disease in patients who have had an inadequate response to conventional therapies (corticosteroids, sulfasalazine, mesalamine, olsalazine, or 6-mercaptopurine). The safety and efficacy of infliximab for patients with active Crohn's disease was demonstrated in a randomized controlled clinical trial. In clinical studies of infliximab for active Crohn's disease, all patients had experienced an inadequate response to prior conventional therapies, including corticosteroids, 5-aminosalicylates (5-ASA), and/or 6-mercaptupurine/ azathioprine (6-MP/AZA). Initial therapy consists of a single infusion of infliximab (5 mg/kg). In clinical studies, there was no evidence of a dose response; doses higher than 5 mg/kg did not result in a greater proportion of responders. Re-treatment with infliximab may be necessary and is covered. Although the optimal frequency of retreatment is uncertain, there is preliminary evidence to suggest that the optimal interval frequency of retreatments is 8-week intervals.

In a Cochrane review, Doherty et al (2009) examined the use of medical therapies for the prevention of post-operative recurrence of Crohn's disease. MEDLINE, EMBASE and the Cochrane Central Register of Controlled Trials (CENTRAL) were searched to identify relevant studies. References from selected papers and abstracts from Digestive Disease Week were also searched. Randomized controlled trials that compared medical therapy to placebo or other medical agents for the prevention of recurrence of intestinal Crohn's disease were selected for inclusion. Two authors reviewed all abstracts containing search terms, and those meeting inclusion criteria were selected for full data abstraction. Dichotomous data were summarized using relative risk and 95 % CI. A fixed-effects model was used, and sensitivity analysis performed. A total of 23 studies were identified for inclusion. Probiotics were not superior to placebo for any outcome measured. The use of nitroimidazole antibiotics appeared to reduce the risk of clinical (RR 0.23; 95 % CI: 0.09 to 0.57, number needed to treat [NNT] = 4) and endoscopic (RR 0.44; 95 % CI: 0.26 to 0.74, NNT = 4) recurrence relative to placebo. However, these agents were associated with higher risk of serious adverse events (RR 2.39, 95 % CI: 1.5 to 3.7). Mesalamine therapy was associated with a significantly reduced risk of clinical recurrence (RR 0.76; 95 % CI: 0.62 to

0.94, NNT = 12), and severe endoscopic recurrence (RR 0.50; 95 % CI: 0.29 to 0.84, NNT = 8) when compared to placebo. Azathioprine/6MP was also associated with a significantly reduced risk of clinical recurrence (RR 0.59; 95 % CI: 0.38 to 0.92, NNT = 7), and severe endoscopic recurrence (RR 0.64; 95 % CI: 0.44 to 0.92, NNT = 4), when compared to placebo. Neither agent had a higher risk than placebo of serious adverse events. When compared to azathioprine/6MP, mesalamine was associated with a higher risk of any endoscopic recurrence (RR 1.45, 95 % CI: 1.03 to 2.06), but a lower risk of serious adverse events (RR 0.51; 95 % CI: 0.30 to 0.89). There was no significant difference between mesalamine and azathioprine/6MP for any other outcome. The authors concluded that there are insufficient randomized controlled trials of infliximab, budesonide, tenovil and interleukin-10 to draw conclusions. Nitro-imidazole antibiotics, mesalamine and immunosuppressive therapy with azathioprine/6-MP or infliximab all appear to be superior to placebo for the prevention of post-operative recurrence of Crohn's disease. The cost, toxicity and tolerability of these approaches require careful consideration to determine the optimal approach for post-operative prophylaxis.

Infliximab is an intravenous medication that is indicated for the reduction in the number of enterocutaneous fistulas in adult patients with fistulating Crohn's disease. The safety and efficacy of infliximab in fistulizing Crohn's disease was demonstrated in a randomized, controlled study of patients with fistulizing Crohn's disease of at least 3 months duration.

Dermatomyositis and Polymyositis

Hengstman et al (2008) examined the efficacy of infliximab combined with weekly methotrexate in drug-naive recent-onset dermatomyositis (DM) and polymyositis (PM) . A multi-center, open-label, controlled trial was conducted. Disease activity was assessed using patient's and physician's disease activity assessment, manual muscle testing (MMT), handheld dynamometry, and serum CK. The primary objective was to assess the efficacy using MMT after a period of 26 weeks. The study was terminated prematurely because of a low inclusion rate and a high drop-out rate due to disease progression and the occurrence of an infusion reaction. The few patients who did reach the primary endpoint showed improvement in all aspects studied. The authors concluded that

infliximab combined with weekly methotrexate might be safe and well-tolerated in a small subgroup of patients with drug-naive recent-onset myositis. At present, the authors do not advocate the use of this treatment because treatment response can not be predicted beforehand.

Dastmalchi et al (2008) investigated the effect of the TNF blocking agent infliximab in patients with treatment-resistant inflammatory myopathies. A total of 13 patients with refractory PM, DM, or inclusion body myositis (IBM) were treated with 4 infliximab infusions (5 mg/kg body weight) over 14 weeks. Outcome measures included myositis disease activity score with improvement defined according to The International Myositis Assessment and Clinical Studies Group (IMACS), and MRI. Repeated muscles biopsies were investigated for cellular infiltrates, major histocompatibility complex (MHC) class I and II, TNF, interleukin (IL)1alpha, IL6, high mobility group box chromosomal protein 1 (HMGB-1), interferon gamma (IFNgamma), myxovirus resistance protein A (MxA) and membrane attack complex (MAC) expression. Type I IFN activity was analyzed in sera. Nine patients completed the study. Three patients discontinued due to adverse events and one due to a discovered malignancy. Three of the completers improved by greater than or equal to 20 % in 3 or more variables of the disease activity core set, 4 were unchanged and 2 worsened greater than or equal to 30 %. No patient improved in muscle strength by manual muscle test. At baseline, 2 completers had signs of muscle inflammation by MRI, and 5 at follow-up. T lymphocytes, macrophages, cytokine expression and MAC deposition in muscle biopsies were still evident after treatment. Type I IFN activity was increased after treatment. The authors concluded that infliximab treatment was not effective in refractory inflammatory myopathies. In view of radiological and clinical worsening, and activation of the type I IFN system in several cases, infliximab is not an alternative treatment in patients with treatment-resistant myositis.

Also, an UpToDate review on "Treatment of recurrent and resistant dermatomyositis and polymyositis in adults" (Miller and Rudnicki, 2012) states that "[d]ata related to the efficacy of tumor necrosis factor inhibition in DM and PM are mixed. Only small studies of etanercept and infliximab have been reported, and further studies will be required to further define their role".

Graft-Versus-Host Disease

In a phase III clinical trial, Couriel et al (2009) examined if the addition of infliximab to steroids could improve results for patients with newly diagnosed grade II to IV acute graft-versus-host disease (aGVHD). A total of 63 patients were randomized either to 2 mg/kg/day methylprednisolone (MP) or infliximab + MP. Average age was 47 years (range of 20 to 70 years); 64 % were male. Fifty-three percent and 51 % of patients received a matched-sibling and/or bone marrow (BM) graft. Sixty-seven percent had grade II, 33 % grade III to IV aGVHD; 62 % had skin, 53 % gastrointestinal (GI), and 7 % had liver involvement. At days 7 and 28, the response rate for infliximab + MP versus MP was 52 % versus 78 %, p = 0.03 and 62 % versus 58 %, p = 0.7, respectively. Cumulative incidences of GVHD-related mortality, non-relapse mortality (NRM), and overall survival (OS) were not significantly different between the 2 groups (GVHD-related mortality: 38 % versus 32 %, p = 0.6; NRM: 52 % versus 36 %, p = 0.3; OS: 17 % and 28 %, p = 0.4 for infliximab + MP versus MP, respectively). Patients with newly diagnosed aGVHD derive no benefit from the addition of anti-TNF-alpha therapy with infliximab when compared to corticosteroids alone.

Pidala et al (2009) examined the effectiveness of infliximab as a salvage therapy for steroid-refractory aGVHD. In a series of 52 patients, 71 % of whom had grade III to IV aGVHD, only 15 % achieved complete remission (CR) with the use of infliximab alone as salvage therapy. Complete remission of aGVHD differed according to overall aGVHD grade at salvage (grade II, 5/15; grade III, 2/17; grade IV, 1/20; p = 0.03). Median OS was only 1.7 months (95 % CI: 0.99 to 2.3 months). Complete remission of aGVHD was significantly associated with OS, with a hazard ratio of 8.4 for death in those without CR (95 % CI: 3.6 to 19.6; p < 0.0001). The authors concluded that this series demonstrated the limited activity of infliximab in patients with high-grade aGVHD. They stated that further work is needed to identify effective therapy for aGVHD.

In a retrospective analysis, Motllo et al (2011) evaluated the activity and toxicity of infliximab in 9 patients with acute or chronic steroid-resistant GVHD. Six patients had acute GVHD and 3 chronic GVHD. The source of progenitors was peripheral blood in all patients except 1. Six received matched-related stem cell transplant and 3 unrelated stem cell

transplant. Patients received between 2 and 7 doses of infliximab, and 5 achieved a partial response. All patients presented infections: 4 developed pseudomonas aeruginosa septicemia and 5 probable or confirmed infection by aspergillus fumigatus. The authors concluded that infliximab provides transient response in steroid-resistant GVHD. However, it is associated with a high rate of infections. The authors stated that earlier administration of infliximab should be explored to reduce the frequency of infections.

Rager et al (2011) noted that treatment options for steroid-refractory GVHD are unsatisfactory and prognosis is poor. Inflammatory cytokines IL-2 and TNF-α are important mediators of GVHD and may be critical targets for therapy. These researchers retrospectively reviewed their experience using combination anti-cytokine therapy of daclizumab and infliximab. A total of 17 evaluable patients had a median age of 47 years (range of 35 to 63). The conditioning regimen was myeloablative in 13 and non-myeloablative in 4 cases. Graft-versus-host-disease occurred at a median of 49 days after transplant in 12 patients (range of 21 to 231 days) and at a median of 46 days (range of 25 to 119 days) after donor lymphocyte infusion in 5 patients. All patients had persistent or progressive GVHD despite 1 to 2 mg/kg/day of corticosteroids for a median of 7 days (range of 2 to 26 days). They received a combination of daclizumab and infliximab for acute GVHD Center for International Blood and Marrow Transplant Research (CIBMTR)severity index B (3), C (10) or D (4). Of the 17 patients analyzed, 47 % responded to treatment, 24 % had complete resolution of symptoms and 24 % had partial responses. Survival was limited and all the patients died a median of 6.7 months (range of 1.6 to 26) from transplant and 35 days from initiation of daclizumab/infliximab. The findings of this retrospective analysis suggested that combination anti-cytokine therapy with daclizumab/infliximab has significant activity in steroid-refractory GVHD, but outcomes remain poor. New methods to prevent and treat GVHD are urgently needed. The authors do no recommend combination of daclizumab and infliximab for advanced steroid-refractory GVHD.

Uhlving et al (2012) stated that bronchiolitis obliterans following allogeneic hematopoietic SCT is a serious complication affecting 1.7 to 26.0 % of the patients, with a reported mortality rate of 21 to 100 %. It is considered a manifestation of chronic graft-versus-host disease, but the

knowledge of etiology and pathogenesis is still limited. The authors noted that there is only limited evidence of infliximab on bronchiolitis obliterans (case study and case-series studies).

The National Comprehensive Cancer Network Drugs and Biologics Compendium (NCCN, 2020) provide a category 2A recommendation for use of infliximab in acute graft-versus-host disease (GVHD) as additional therapy in conjunction with systemic corticosteroids following no response (steroid-refractory disease) to first-line therapy options. Therapy for steroid-refractory acute GVHD is often used in conjunction with the original immunosuppressive agent.

The NCCN Clinical Practice Guidelines in Oncology for "Hematopoietic cell transplantation (HCT): Pre-transplant recipient evaluation and management of graft-versus-host disease" (version 2.2020) states that infliximab is often used in conjunction with the original immunosuppressive agent for acute GVHD. "In a retrospective evaluation of 21 patients with steroid-refractory aGVHD who had received treatment with single-agent infliximab (10 mg/kg once weekly for at least 4 doses), the ORR was 67% with 62% of patients achieving CR". A "retrospective analysis involving 35 patients with steroid-refractory aGVHD reported an ORR of 40% for infliximab administered intravenously at 10 mg/kg weekly for a medial of 4 doses, with 83% of patients developing infectious complications. These data suggest that infliximab is active in the treatment of steroid-refractory aGVHD; however, the potential for excessive infections should be evaluated".

Herniated Disc and Sciatic Pain

A systematic evidence review in BMJ Clinical Evidence found that the effectiveness of infliximab in treatment of herniated disc is unknown (Jordan et al, 2008). The assessment identified 1 randomized controlled clinical trial of 41 people with acute or subacute sciatic pain, caused by herniated discs confirmed by magnetic resonance imaging, comparing infliximab, given as a single intravenous infusion) versus control (saline infusion over 2 hours) (Korhonen et al, 2005). The randomized controlled trial found no significant difference between infliximab and control in leg or back pain score improvements at 12 weeks or median reduction in back pain score at 12 weeks. The trial also found no significant

difference between groups in reduction of Oswestry Disability Index scores, median cumulative sick leave, or the proportion of people undergoing discectomy.

Hidradenitis Suppurativa

Haslund and associates (2009) noted that hidradenitis suppurativa (HS) is a common inflammatory skin disease. Medical treatment is often disappointing and in severe disease surgery remains the therapy of choice. Extensive surgery may be effective but also mutilating. Patients experience a significant reduction in quality of life and the need for new treatment modalities are urgent. In recent years, patients with HS have been treated off-label with TNF-alpha inhibitors with a varying degrees of effect. These researchers performed a systematic review of papers retrieved from 2 databases (PubMed and Web of Science) using the following keywords: hidradenitis suppurativa, acne inversa, infliximab, etanercept, and adalimumab. A total of 34 publications were retrieved, describing treatment of 105 patients. Most cases reported treatment with infliximab (52/105). A positive treatment outcome was reported in 90/105 cases, with only 7/105 non-responders and 8/105 patients experiencing side-effects. The side-effects were comparable to those seen in other TNF-alpha inhibitor studies. In the majority of cases, the treatment was effective when given as a suppressive therapy, but 15/105 cases were described with long-term remission (greater than or equal to 3 months) after the end of therapy. In most publications follow-up was, however, insufficient to allow a systematic exploration of this. The authors concluded that TNF-alpha inhibitors seem to be effective in the treatment of HS. However, several questions remain to be answered through specific studies. This review has also identified a need for more standardized reporting of the outcomes as well as randomized controlled trials in this disease.

Rambhatla et al (2012) conducted a systematic review of the effectiveness of various modalities to treat hidradenitis suppurativa (HS). The authors reported that treatments of HS shown to be effective were a clindamycin-rifampin combination regimen, a course of infliximab, monthly Nd:YAG laser sessions, and surgical excision and primary closure with a gentamicin sulfate-collagen sponge. The authors stated

that most therapies used to treat HS were supported by limited or weak scientific evidence. The authors emphasized the need for large randomized controlled trials to evaluate treatment options for HS.

Randomized controlled clinical trials of tumor necrosis factor inhibitors for hydradenitis suppurativa have shown inconsistent results (Jemec, 2012). Grant, et al. (2011) reported positive results with infliximab therapy for patients with moderate to severe hidradenitis suppurativa in a randomized, double-blind, placebo-controlled crossover trial. Study subjects received infliximab or placebo during an eight-week double-blind treatment phase, followed by an open-label phase where subjects taking placebo were given the opportunity to cross over to infliximab, and an observational phase. Primary treatment efficacy was based on HS Severity Index. Secondary endpoints included Dermatology Life Quality Index, visual analog scale, and Physician Global Assessment scores. Inflammatory markers erythrocyte sedimentation rate and C-reactive protein were also assessed. The investigators found that more patients in the infliximab than in the placebo group showed a 50 percent or greater decrease from baseline HS Severity Index score. In addition, statistically and clinically significant improvement from baseline was observed at week 8 in Dermatology Life Quality Index score, visual analog scale score, erythrocyte sedimentation rate, and C-reactive protein compared with placebo. The investigators reported that patients in the placebo group treated with infliximab after week 8 (crossover) responded similarly to the original infliximab group. The investigators noted that many patients withdrew during the observational phase to continue anti-tumor necrosis factor-alfa therapy. No unexpected serious adverse events were observed. The investigators noted limitations of this study, including that patients were treated by a single physician at a single center, and some patients did not return after their last infusion, and the HS Severity Index requires validation. The investigators reported that infliximab was well tolerated, no unexpected safety issues were identified, and improvements in pain intensity, disease severity, and quality of life were demonstrated with concomitant reduction in clinical markers of inflammation.

However, another randomized, double-blind, placebo-controlled trial failed to show a significant benefit of the tumor necrosis factor inhibitor etanercept with the use of a physician's global assessment scale (Adams,

et al., 2010). In a third randomized, controlled trial, the tumor necrosis factor inhibitor adalimumab resulted in significant improvement on the basis of a score that reflected the extent and severity of disease at 6 weeks, but this benefit was not maintained at 12 weeks (the primary outcome of the trial).

In a comparative study, Rappard et al (2012) compared the outcomes of the tumor necrosis factors infliximab and adalimumab HS, and found that infliximab was more effective than adalimumab. The authors performed a retrospective study to compare two cohorts of 10 adult patients suffering from severe, recalcitrant HS. Ten patients were treated with intravenous infliximab, and subsequently 10 other patients were treated in the same hospital with subcutaneous adalimumab. Both cohorts were followed up for one year using identical evaluation methods [Sartorius score, quality of life index, reduction of erythrocyte sedimentation rate and C-reactive protein, patient and doctor global assessment, and duration of efficacy]. The authors reported that 19 patients completed the study. In both groups, the severity of the HS diminished. The investigators found that infliximab performed better than adalimumab in all aspects. The average Sartorius score was reduced to 54 percent of baseline for the infliximab group and 66 percent of baseline for the adalimumab group. The authors concluded that subcutaneous adalimumab is less effective than intravenous infliximab.

Shuja et al (2010) noted that there are various treatment options for HS, but none is wholly satisfactory or effective. There have been many reports on the efficacy of biologic drugs for the treatment of HS; this article was an in-depth review of the published evidence on this topic. Because the current evidence is limited, randomized, double-blinded, placebo-controlled trials are needed to better elucidate the future of these drugs for the treatment of HS.

Idiopathic Membranous Nephropathy and Idiopathic Nephrotic Syndrome

UpToDate reviews on "Membranous nephropathy: Treatment and prognosis" (De Vriese and Cattran, 2021) and "Steroid-resistant idiopathic nephrotic syndrome in children: Management" (Niaudet et al, 2021) do not mention the use of infliximab as a therapeutic option.

IgG4-Related Disease

Karim et al (2017) stated that IgG4-related disease (IgG4-RD) is a systemic fibro-inflammatory condition with unclear pathophysiology. It may occur as a single organ disorder, but multi-organ presentation is common and can mimic several conditions. The preferred therapy consists of steroids, but definite maintenance strategy remains unclear. These researchers described a case of a 61-year old woman, initially diagnosed with idiopathic orbital inflammation refractory to multiple immunosuppressive agents. The disease was complicated with epilepsy, vision loss, and trismus. Treatment with various immunosuppressive agents was unsuccessful. Eventually, the patient was effectively treated with infliximab. This is the 2nd case of IgG4-RD treated with a TNF-blocker documented in literature and the 1st description to demonstrate its superiority over steroid sparing agents. Although speculative, TNF-blockers might exert their effect in IgG4-RD by interfering with the possible overexpressed TNF-alpha due to fibrosis in this disease. The authors concluded that treatment with infliximab appeared a good alternative for refractory IgG4-RD; however, further studies are needed to define the value of infliximab in IgG4-RD.

Immune Checkpoint Inhibitor-Related Toxicities

Immunologic checkpoint inhibition agents such as cytotoxic T-lymphocyte-associated antigen 4 (CTLA-4) (e.g., ipilimumab), programmed cell death receptor 1 (PD-1) (e.g. nivolumab , pembrolizumab), and anti-programmed cell death ligand 1 (PD-L1) agents (e.g. atezolizumab), are indicated for treatment of various malignant diseases (e.g., melanoma, renal cell carcinoma, non-small cell lung cancer, urothelial carcinoma, Hodgkin lymphoma, Merkel cell carcinoma, and microsatellite instability-high or mismatch repair deficient [dMMR] solid tumors). Treatment with these medications can be associated with immune-related adverse events (irAEs) that can become severe or fatal (Postow and Wolchok, 2018).

The American Society of Clinical Oncology (ASCO) and the National Comprehensive Cancer Network (NCCN) have published guidelines and recommendations for the management of immunotherapy-related toxicities. The management approach to irAEs is based upon clinical

experience, since no prospective trials have been conducted to guide the treatment of irAEs. The majority of data are derived from patients with advanced melanoma who were treated with ipilimumab, nivolumab, and pembrolizumab. In general, treatment of moderate or severe irAEs requires interruption of the checkpoint inhibitor and the use of corticosteroid immunosuppression. Treatment is based upon the severity of the observed toxicity. (Postow and Wolchok, 2018).

- Moderate or severe colitis if no improvement after 48 hours of methylprednisolone.
- Severe pneumonitis if no improvement after 48 hours of methylprednisolone
- Severe or life-threatening renal failure/elevated serum creatinine if toxicity remains greater than G2 (creatinine remaining greater than 2 to 3 times above baseline) after 1 week of corticosteroids
- Life-threatening myocarditis, pericarditis, arrhythmias, or impaired ventricular function
- Refractory/severe inflammatory arthritis not responding to corticosteroids or anti-inflammatory agents.

Hillock and associates (2017) reviewed the outcomes of metastatic melanoma patients treated with infliximab for severe steroid-refractory colitis secondary to ipilimumab therapy. These researchers retrospective reviewed evidence of patients administered infliximab for ipilimumab-induced colitis at South Australian public hospitals between October 2011 and April 2015. Main outcome measures included resolution of colitis/diarrhea, duration of hospital stay, dosage regimen of infliximab used (single dose versus multiple dose) and surgical intervention if needed. Between October 2011 and April 2015, a total of 106 patients were dispensed ipilimumab from South Australian public hospitals for the treatment of metastatic melanoma; 13 were administered infliximab for severe, steroid-refractory colitis secondary to ipilimumab; 62 % received a single-dose of infliximab only; 4 patients achieved resolution of colitis symptoms at 1 month post-infliximab; 33 % required surgical intervention despite treatment with infliximab; 1 patient declined surgery and subsequently died due to bowel perforation. The average number of over-night bed days due to colitis was 27. The authors concluded that the findings of this series of patients with severe ipilimumab-induced colitis suggested that despite treatment with infliximab a high proportion

of patients did not achieve resolution of symptoms. Review of these cases has highlighted the absence of evidence-based guidelines to treat severe, steroid refractory colitis secondary to ipilimumab. They stated that further prospective studies may clarify the role of infliximab for the treatment of ipilimumab-induced colitis.

Wang et al. (2018) conducted a retrospective analysis to evaluate immune-checkpoint inhibitor-induced diarrhea and colitis in patients with advanced malignancies. The authors state that immune checkpoint inhibitors (ICPIs) treatment can be complicated by diarrhea and colitis, in which systemic steroids are the first line treatment. Infliximab is reserved for severe refractory cases. The study included 327 cancer patients who received ICPIs between 2011 and 2017. Patients with ICPI-induced toxicities in other organs were excluded. The authors used descriptive analysis to compare different groups based on the occurrence and the treatment of diarrhea and colitis. Kaplan-Meier and log-rank test were used to estimate and compare overall survival durations between groups. The authors found that diarrhea was recorded in 117 (36%) patients; 79 (24%) of them required immunosuppressive treatment of either systemic corticosteroid without infliximab (n = 44) or with infliximab (n = 35). Caucasian ethnicity, melanoma, stage 3 cancer, and ipilimumab were predictors of colitis that requires immunosuppression. Patients who required immunosuppressants had better overall survival than those who did not require treatment for colitis or diarrhea ($p < 0.001$). Immunosuppression for diarrhea or colitis did not affect the overall survival significantly ($p = 0.232$), nor did the choice of treatment (corticosteroids with vs. without infliximab; $p = 0.768$). Diarrhea was an independent predictor of a favorable overall survival ($p < 0.001$), irrespective of treatment need ($p = 0.003$). Patients who received long duration of steroid treatment (> 30 days) had numerically higher infection rate than those who received steroid for shorter duration ($p = 0.160$). Likewise, long duration of steroid without infliximab was associated with increased risk of infection compared to short duration of steroid with infliximab ($p = 0.089$). The authors concluded that immunosuppressive treatment for diarrhea did not significantly affect overall survival; however, infection rates were numerically higher among patients who received steroids for a long duration. Therefore, early non-steroid immunosuppressive therapy may ensure a more favorable overall

outcome. Future prospective studies are required to further elucidate the relationship between ICPI-induced GI tract adverse events and tumor response and patient survival.

The National Comprehensive Cancer Network Drugs & Biologics Compendium (NCCN, 2023) provides a category 2A recommendation for use of infliximab in immune checkpoint inhibitor-related toxicities. NCCN compendia states to consider adding infliximab for the management of immunotherapy-related for the following: myocarditis as a further intervention if no improvement within 24-48 hours of starting high-dose methylprednisolone; mild (G1) diarrhea or colitis if persistent or progressive symptoms and positive lactoferrin/calprotectin; moderate (G2) and strongly consider for severe (G3-4) diarrhea or colitis; moderate or severe inflammatory arthritis as additional disease modifying antirheumatic drug (DMARD) therapy if no improvement after holding immunotherapy and treating with oral corticosteroids or if unable to taper corticosteroids, or no response to conventional synthetic (cs)DMARDs; G1 -4 uveitis that is refractory to high-dose systemic corticosteroids (treatment guided by ophthalmology); moderate (G2) pneumonitis if no improvement after 48-72 hours of corticosteroids or severe (G3-4) pneumonitis if no improvement after 48 hours of methylprednisolone; or stage 3 acute kidney injury/elevated serum creatinine if toxicity remains greater than stage 2 after 4-6 weeks of corticosteroids or if creatinine increases during steroid taper (or once off steroids).

Immune-mediated Enterocolitis

Ipilimumab (Yervoy) is a fully human IgG1 anti-CTLA-4 monoclonal antibody indicated for metastatic melanoma.These immune-mediated reactions may involve any organ system; however, the most common severe immune-mediated adverse reactions are gastrointestinal. Case series support the use of infliximab for immune-mediated enterocolitis from ipilimumab (Cheng, et al., 2015; Merrill, et al., 2014; Slangen, et al., 2013; Pagès, et al., 2013; Lord, et al., 2010; Minor, et al., 2009; Johnston, et al., 2009; Phan, et al., 2008; Beck, et al., 2006). The prescribing information for Yervoy recommends antidiarrheals, steroids, interrupting treatment, and permanent discontinuation in severe cases (BMS, 2013). The Yervoy REMS program recommends alternative

immunosuppressive therapy in cases of severe or life-threatening enterocolitis (defined as ≥ 7 stools/day over baseline, peritoneal signs consistent with bowel perforation, ileus or fever) that persists despite permanent discontinuation of Yervoy and administration of systemic corticosteroids (BMS, 2011). Sznol, et al. (2013) recommends administration of a single dose of infliximab 5 mg/kg in this situation unless infliximab is contraindicated or perforation or sepsis is present.

Inflammatory Bowel Disease

Yanai and Hanauer (2011) appraised the latest evolution in treatment strategies in IBD and suggested an evidence-based approach and risk stratification while coping with cases of non-responders or loss of response to biological therapies. These researchers conducted a literature search of English publications listed in the electronic databases of Medline (source PubMed) and constructed an analytical review based on definitions of response and loss of response, considering potential responsible mechanisms, clinical assessment tools, and finally recommending a practical approach for its prevention and management. Favorable clinical outcome appeared to be the consequence of sustained therapeutic drug levels, and the current literature supports a practice of dose adjustments. When immunogenicity develops to a single biological agent, response can be regained by introduction of an alternative biological agent of the same or different class. Efficacy is reduced with 2nd-line agents either within or across classes compared with naive patients. In the absence of direct measurement of drug levels and anti-drug antibodies, clinical judgment is necessary to assess the mechanisms of loss of response, and more empiric decision-making may be necessary to determine the choice of 2nd-line biological agents. Optimal treatment strategies are still controversial. The authors concluded that it is essential to recognize the spectrum of mechanisms affecting response and loss of response to form a logical and efficient management algorithm, and, perhaps, it is time to incorporate the measurement of trough levels and anti-drug antibodies in the strategy of such an assessment. They stated that prospective controlled trials are direly needed to investigate the optimal tailored management in individual patients who lose response. They noted that “Thus, at present, we must

rely on clinical symptoms, signs, endoscopic responses, and generic biomarkers (CRP and fecal calprotectin) to assess individual patient responses and to correlate with dosing strategies".

Wang et al (2012) stated that antibody-based drugs such as infliximab (IFX) are effective for the treatment of inflammatory bowel disease (IBD) and other immune-mediated disorders. The development of antibodies against these drugs may result in unfavorable consequences, including the loss of drug efficacy, hypersensitivity reactions, and other adverse events. Therefore, accurate monitoring of serum drug and anti-drug antibody levels should be an important part of therapy for patients being treated with an antibody-based drug. Current methods for the assessment of anti-drug antibodies and drug levels, involving various bridging ELISA and radioimmunoassay techniques, are limited by their sensitivity, interference, and/or complexity. To overcome these limitations, these researchers have developed a non-radiolabeled homogeneous mobility shift assay (HMSA) to measure the antibodies-to-infliximab (ATI) and IFX levels in serum samples. Full method validation was performed on both the ATI- and IFX-HMSA, and the clinical sample test results were also compared with those obtained from a bridging ELISA method to evaluate the difference in performance between the 2 assays. Validation of the ATI-HMSA revealed a lower limit of quantitation of 0.012 μg/ml in serum. The linear range of quantitation was 0.029 to 0.54 μg/ml. The intra- and inter-assay precision was less than 20 % of coefficient of variation (CV), and the accuracy (% error) of the assay was less than 20 %. In serum samples, ATI as low as 0.036 μg/ml can be measured, even in the presence of 60 μg/ml of IFX in the serum. Sera from 100 healthy subjects were tested to determine the cut-point of the assay. ATI-positive samples that had been previously analyzed by using a bridging ELISA from 100 patients were also measured by the new method. There was a high correlation between the 2 methods for ATI levels ($p < 0.001$). Significantly, the new method identified 5 false-positive samples from the bridging ELISA method. Validation of the mobility shift IFX assay also showed high assay sensitivity, precision and accuracy. The authors concluded that the HMSA method may also be applied to other protein-based drugs to accurately detect serum drug and anti-drug antibody levels. This article did not appear to address Prometheus Anser IFX test; and the authors stated that additional patient samples are needed to verify the clinical utility of ATI- and IFX-HMSA.

Ordas et al (2012) stated that “preliminary evidence suggests that multiple factors influence the PK of mAbs. Understanding the determinants of the PK of mAbs has great potential s to improve and optimize the management of patients with IBD“. This article was a review and the authors concluded that “Incorporation of therapeutic drug monitoring into clinical practice may allow clinicians to optimize treatment by maintaining effective drug concentrations over time”.

Joosse et al (2015) stated that anti-TNFα therapy is effective in pediatric patients with IBD but associated with a risk of developing anti-drug antibodies, which lower the efficacy. Incorporating measurement of trough levels and anti-drug antibodies (therapeutic drug monitoring) may prevent the development of neutralizing anti-drug antibodies or could contribute to more optimal treatment strategies if anti-drug antibodies are already formed. These researchers examined the role of therapeutic drug monitoring in children and adolescents with IBD exposed to anti-TNFα agents. They performed a literature search and identified publications that measured anti-TNFα drug trough levels and/or anti-drug antibodies in children or adolescents with IBD. Studies were eligible when (i) article was written in English, (ii) original data were available, (iii) full text article or abstract was available, (iv) measurement of antibodies against anti-TNFα drugs or trough level of anti-TNFα drugs were reported, and (v) levels were measured in pediatric patients with IBD. The search yielded 811 articles, of which 795 articles were excluded based on title or abstract. A total of 14 studies were included in the review. The authors concluded that therapeutic drug monitoring within the pediatric IBD population certainly had a potential benefit. As occurrence of immune reactions to anti-TNFα agents varied widely, incorporating measurement of IFX trough levels at week 8 or week 14 predicted therapy response and allowed for dose adjustments to reach therapeutic drug concentrations. However, they stated that a clinically relevant cut-off level for adalimumab (ADA) has not been defined yet, and the optimal intervention strategy still has to be determined.

Felice et al (2015) noted that the elaboration of effective and cost-saving algorithms for anti-TNF optimization in patients with IBD represents one of the biggest challenges over the last few years. Recently, many studies have been focused on the identification of an optimal trough level (TL) for

most used anti-TNF agents and on the role of anti-drug antibodies, especially in the management of patients who lose response to biological treatments. Therapeutic drug monitoring may potentially help to also prevent loss of clinical benefit overtime and to reduce health-related costs. These investigators reviewed the current evidence about the correlation between clinical outcomes and anti-TNF TLs, the role of anti-drug antibodies in the context of safety and loss of response to anti-TNF, as well as the utility of therapeutic drug monitoring in clinical practice. The authors concluded that the data available so far support the utility of TL and anti-drug antibodies measurement for the management of IBD patients with loss of response to anti-TNF; but did not currently authorizes a routine application in clinical practice of proactive therapeutic monitoring in patients in clinical remission. They stated that this remained a promising approach to optimize anti-TNF therapies and possibly to reduce health-related costs, and further prospective studies are strongly expected.

Infliximab Serum Levels and Antibodies

For measurement of serum levels of infliximab and/or antibodies to infliximab, see CPB 1091 - Therapeutic Drug Monitoring of Anti-Tumor Necrosis Factor Blocking Agents (../1000_1099/1091.html).

Juvenile Rheumatoid Arthritis and Juvenile Idiopathic Arthritis

Evidence suggests that infliximab may be effective in juvenile rheumatoid arthritis. A randomized, controlled clinical study found a non-significant trend in favor of infliximab in polyarticular juvenile idiopathic arthritis (Ruperto et al, 2007). In this multi-center, randomized, placebo-controlled study, 122 children with persistent polyarticular juvenile idiopathic arthritis despite prior methotrexate therapy were randomized to receive infliximab or placebo for 14 weeks, after which all children received infliximab through week 44. Patients received methotrexate plus infliximab 3 mg/kg through week 44, or methotrexate plus placebo for 14 weeks followed by methotrexate plus infliximab 6 mg/kg through week 44. The investigators reported that a higher proportion of patients in the 3 mg/kg infliximab group than in the placebo group had achieved responses according to the American College of Rheumatology (ACR) Pediatric 30 (Pedi 30) criteria for improvement at week 14 (63.8 % and

49.2 %, respectively), but the between-group difference in this primary efficacy end point was not statistically significant (p = 0.12). The investigators reported that, by week 16, after the crossover from placebo to infliximab 6 mg/kg when all patients were receiving infliximab, an ACR Pedi 30 response was achieved in 73.2 % of all patients. By week 52, ACR Pedi 50 and ACR Pedi 70 responses had been reached in 69.6 % and 51.8 %, respectively, of patients.

An open-label extension of this study of juvenile idiopathic arthritis concluded that infliximab was safe and effective in the long-term, but had a high discontinuation rate (Ruperto et al, 2010). Seventy-eight of the 122 subjects (64 %) entered this open-label extension study. Of these, 42 discontinued infliximab, most commonly due to consent withdrawal (11 patients), lack of efficacy (8 patients) or patient/physician/sponsor requirement (8 patients). The authors reported that infliximab at a mean dose of 4.4 mg/kg per infusion was generally well-tolerated. Infusion reactions occurred in 32 % (25/78) of patients, with a higher incidence in patients positive for antibodies to infliximab (58 %, 15/26). At 4 years, the proportions of patients achieving ACR-Pedi-30/50/70/90 response criteria and inactive disease status were 44 %, 40 %, 33 %, 24 %, and 13 %, respectively.

An UpToDate review on "Systemic juvenile idiopathic arthritis: Treatment" (Kimura, 2020) state that TNF inhibitors, such as infliximab, offer a wider range of dosing options, and high doses may be more effective for treating sJIA, but this class of biologics is not as dramatically effective as the other biologics. Infliximab has been effective in patients who failed to respond to etanercept, although there are limited published data.

Kawasaki Disease

Han and Zhao (2018) stated that Kawasaki disease (KD) is a serious disease characterized by systemic lesions of the skin and mucous membranes, as well as lymphomas and vascular inflammation; KD threatens the health and lives of children, especially young ones. These investigators compared the therapeutic effects of single intravenous immunoglobulin gamma (IVIG) versus a combination of IVIG and infliximab in young children with KD. A total of 154 children with KD, younger than 5 years old, were enrolled in the study from January 2013

to January 2017. Participants were randomly divided into an IVIG group and a combination of IVIG and infliximab treatment group. After systematic treatments, the therapeutic indicators of the 2 groups were compared. During the treatment process, body temperature and other important inflammatory indicators, including CRP, white blood cell count (WBC), and TNF-α, were monitored in the first 4 days. There were fewer refractory KD patients in the combined treatment group than in the IVIG group (4 versus 14, $p < 0.001$); KD patients in the combined treatment group had better outcomes with shorter fever durations and hospital stays, as well as less coronary artery (CA) dilation. However, there was no obvious differences in the incidence rate of CA aneurysms (CAA) between the 2 groups ($p > 0.05$). Costs of administration were similar between groups ($p > 0.05$). Body temperature, CRP, WBC, and TNF-α in the combined therapy group all showed an earlier drop than in the IVIG group, indicating a more effective anti-inflammation effect. The authors concluded that the introduction of IVIG combined with infliximab in the treatment of young children with KD had more advantages than single IVIG therapy and can be considered as a preferred treatment for KD. However, these investigators stated that this study had certain limitations that need to be considered. The therapeutic effect of IVIG and IVIG plus Infliximab treatments on atypical or incomplete KD patients has not been examined. Other therapeutic regimens should also be considered and compared. They stated that it is necessary to further examine if there is a significant difference in CAA frequency and long-term outcomes between these 2 strategies among a larger number of patients.

Eun (2019) stated that as TNF-alpha is a key inflammatory cytokine involved in KD. Recently, infliximab has been introduced and used for IVIG-resistant KD. This revealed early addition of infliximab treatment to be safe and well-tolerated and reduced fever duration, some markers of inflammation, left anterior descending coronary artery Z score. Meanwhile, there have been opposite attention about treatment with infliximab. The complications of infliximab administration include the reactivation of latent TB and an increased risk of bacterial sepsis, with concern about possible malignancy risk. Luckily, it has been reported that infliximab exposure is not associated with an increased risk of malignancy in children patients with IBD. In patients with refractory KD, infliximab is supposed to be administered only once, not repeatedly, which may not cause the worrisome complication, different with the CD or

juvenile idiopathic arthritis (JIA)'s repetitive administration. According to the investigation of infliximab in refractory KD patients, it overall appeared to cause rapid defervescence, resulting in a shorter length of hospital stay, and is relatively well-tolerated. Retrospective studies have reported response rates (defined by a reduction in fever and C-reactive protein level) of 81.3 to 91.7 % when infliximab was used as a 2nd-line agent. Thus, infliximab might be a noticeable promising treatment for IVIG-resistant KD to potential decrease coronary artery abnormalities, although, the most optimal choice of adjunctive therapy will be different for each KD patient. The author concluded that further studies are needed to classify the subsets of KD patients according to immune status.

In a multi-center study, Kur and associates (2019) examined the status of infliximab use in IVIG-resistant KD patients and the incidence of CAAs according to treatment regimens. Between March 2010 and February 2017, a total of 16 hospitals participated in this study; 102 (32.3 ± 19.9 months, 72 males) who received infliximab at any time after 1st IVIG treatment failure were enrolled. Data were retrospectively collected using a questionnaire. Subjects were divided into 2 groups according to the timing of infliximab administration. Early treatment (group 1) had shorter fever duration (10.5 ± 4.4 days) until infliximab infusion than that in late treatment (group 2) (16.4 ± 4.5 days; $p < 0.001$). These investigators examined the response rate to infliximab and the incidence of significant CAA (z-score of greater than 5). Overall response rate to infliximab was 89/102 (87.3 %) and the incidence of significant CAA was lower in group 1 than in group 2 (1/42 [2.4 %] versus 17/60 [28.3 %], $p < 0.001$). The authors concluded that the findings of this study suggested that the early administration of infliximab may reduce the incidence of significant CAA in patients with IVIG-resistant KD. However, these researchers stated that further prospective randomized studies with larger sample sizes are needed.

Furthermore, an UpToDate review on “Refractory Kawasaki disease” (Sundel, 2019) states that “Anti-tumor necrosis factor (TNF) alpha agents such as infliximab and etanercept have been studied for the treatment of KD because elevated levels of TNF are a characteristic of KD. TNF inhibition or blockade is a promising treatment for TNF-driven signs and symptoms in KD, based upon its relative safety, rapidity of action, and

reported efficacy compared with plasmapheresis, cyclosporine, and cyclophosphamide … Therapeutic alternatives, which have demonstrated effectiveness for other forms of vasculitis, include inhibitors of tumor necrosis factor (TNF) alpha, other immunosuppressive agents, and plasmapheresis. However, there are limited data concerning the risks and benefits of these agents for refractory KD and only a few, small trials with which to evaluate them. The most commonly used of these agents is infliximab, a monoclonal antibody against TNF-alpha. In multiple trials, it effectively lowers fever, laboratory markers of inflammation, and signs of mucocutaneous inflammation, but there is minimal evidence that it improves coronary artery outcomes. Infliximab (5 mg/kg) is typically given to children who have evidence of active vasculitis despite receiving 2 courses of IVIG (total 4 mg/kg) and 1 to 3 doses of IV methylprednisolone".

In a systematic review and meta-analysis, Kabbaha et al (2023) examined the safety and impact of IFX versus IVIG on the incidence of CAA and treatment resistance in children with refractory KD. The Medline/PubMed, Embase, CINAHL, Cochrane Central Register of Controlled Trials and clinical trials registries were searched to December 2021; RCTs comparing infliximab as 2nd-line therapy to a 2nd dose of IVIG in children with refractory KD, reported in abstract or full text, were included. Studies were selected and assessed for risk of bias by 2 reviewers. Data were extracted and pooled using conventional random-effects meta-analysis; and the certainty of evidence was assessed using the GRADE system. A total of 199 subjects from 4 RCTs were included. The pooled RR for the incidence of treatment resistance in patients treated with IFX was 0.40 (95 % CI: 0.25 to 0.64). For incidence of CAA, RR was 1.20 (95 % CI: 0.54 to 2.63), the incidence of adverse effect "infusion reactions" RR was 0.48, (95 % CI: 0.12 to 1.92), and for "infections" RR was 0.55 (95 % CI: 0.27 to 1.12). Overall, the GRADE strength of evidence for the primary outcomes was low. Evidence on the duration of fever and inflammatory biomarkers was sparse, heterogeneous, and inconclusive. The authors concluded that moderate-certainty evidence showed that IFX may lower the incidence of treatment resistance in children with refractory KD; however, the limited strength of evidence warrants further investigations.

Localized Scleroderma/Morphea

Diab and colleagues (2010) reported the case of a 66-year old white woman with a history of hypertension and degenerative arthritis was seen in July 2003 by her primary care physician (PCP) for erythematous and sclerotic patches on her trunk. A skin biopsy specimen taken by the PCP at that time was interpreted (by an outside pathology laboratory) to indicate granuloma annulare. The patient began treatment with systemic corticosteroids and ceased taking her anti-hypertensive medication. However, her cutaneous symptoms continued to progress, and she was referred to the authors' clinic in January 2004. TNF therapy was considered in light of the demonstration of TNF within the vascular endothelium of her biopsy specimen. After her tuberculosis treatment was completed, these investigators initiated infliximab therapy at 5 mg/kg monthly. The patient reported a halt in the progression of her skin lesions along with a decrease in skin tightness after the second infusion of infliximab. The authors noted that their patient with GM demonstrated remarkable improvement after treatment with infliximab. This agent may serve as an alternative therapeutic approach for patients with similar presentations. However, it is important to note that the natural course of all sclerosing disorders is to "burn out" over time, and so randomized controlled studies are needed. These studies are currently lacking, and all treatments that have appeared to show improvement in sclerosing disorders have failed to do so in the context of controlled studies.

Ferguson et al (2015) reported successful treatment of an adolescent Caucasian female with severe progressive localized scleroderma (mixed subtype, including generalized morphea and linear scleroderma of the trunk/limb) using infliximab and leflunomide. The patient demonstrated improvement after the first 9 months of therapy based on her clinical examination, objective measures, and patient and parent global assessments. The authors concluded that infliximab is a potential treatment option for pediatric localized scleroderma patients who have progression of disease or who are unable to tolerate the side effect profile of more standard systemic therapy. They stated that larger longitudinal studies or case series are needed to confirm and further investigate infliximab's role in localized scleroderma.

Lupus Erythematosus

In a Cochrane review, Jessop et al (2009) assessed the effects of drugs for discoid lupus erythematosus. These investigators updated their searches of the Cochrane Skin Group Specialised Register, the Cochrane Central Register of Controlled Trials (CENTRAL) in The Cochrane Library (Issue 2, 2009), MEDLINE, EMBASE, LILACS, and online ongoing trials registers. The reference lists of relevant reviews were searched. Index Medicus (1956 to 1966) was hand searched and we approached authors for information about unpublished trials. They included all randomized trials of drugs to treat people with discoid lupus erythematosus. Drugs included in the search were azathioprine, chloroquine, clofazimine, corticosteroids, (oral and topical), dapsone, gold, interferon alpha-2a, methotrexate, phenytoin, retinoids, sulphasalazine, thalidomide, topical calcineurin blockers (pimecrolimus and tacrolimus), and biological agents (etanercept, efalizumab, infliximab, and rituximab). Two reviewers independently examined each retrieved study for eligibility. Two trials involving 136 participants were included. No new trials were included in this update. In a cross-over study of 12 weeks duration, fluocinonide 0.0 5 % cream (a potent topical corticosteroid), appeared to be better than hydrocortisone 1 % cream (a mild corticosteroid) when the first arm of the trial involving 78 subjects was analyzed at 6 weeks. Clearing or excellent improvement was seen in 27 % of individuals using fluocinonide and in 10 % of those using hydrocortisone, giving a 17 % absolute benefit in favor of fluocinonide (95 % CI: 0.0 to 0.34, NNT = 6). In the second trial, acitretin (50 mg/day) was compared with hydroxychloroquine (400 mg/day) in 58 people in a parallel trial of 8 weeks duration. There was marked improvement or clearing in 46 % of people using acitretin and in 50 % of those on hydroxychloroquine, but there was no significant difference between the 2 interventions. The adverse effects were more frequent and more severe in the acitretin group. In this trial clearing of erythema was measured and found to be better in the hydroxychloroquine group (RR 0.61, 95 % CI: 0.36 to 1.06). The authors concluded that fluocinonide cream may be more effective than hydrocortisone in treating people with discoid lupus erythematosus. Hydroxychloroquine and acitretin appear to be of equal efficacy, although adverse effects are more frequent and more severe with acitretin. There is not enough reliable evidence about other drugs used to treat discoid lupus erythematosus.

Lateef and Petri (2010) stated that the pathogenesis of systemic lupus erythematosus (SLE) involves aberrations in multiple components of the immune system including B cells, T cells, cytokines and growth factors. Therapeutic agents targeting these mediators selectively have been tested for the treatment of SLE. This review summarized the recent advances in these biological therapies. The 2 large phase II/III randomized placebo-controlled trials of B-cell depletion, using rituximab, in SLE, reported unexpected negative results. On the contrary, 2 large phase III trials of belimumab, the monoclonal antibody against B-lymphocyte stimulator (BLyS), showed significant clinical benefit. Response rates were 57.6 and 43.2 % for 10 mg/kg belimumab, compared with 43.6 and 33.8 % for placebo in BLISS-52 and BLISS-76, respectively. Studies of a co-stimulation blocker (abatacept), TNF inhibitor (infliximab), and interleukin-6 inhibitor (tocilizumab) were either negative (abatacept) or were associated with high rates of adverse events. Studies of T cell and interferon inhibition remain in the early development phase. The authors concluded that despite the enthusiasm in the field of biologic therapies, the majority of these new modalities have fallen short of expectations for various reasons. Only belimumab has recently met its primary outcome in 2 phase III trials.

Microscopic Colitis

Microscopic colitis may be defined as a clinical syndrome, of unknown etiology, consisting of chronic watery diarrhea, with no alterations in the large bowel at the endoscopic and radiologic evaluation (Ianiro, et al., 2012). Therefore, a definitive diagnosis is only possible by histological analysis. Microscopic colitis includes two histological subtypes [collagenous colitis (CC) and lymphocytic colitis (LC)] with no differences in clinical presentation and management. Collagenous colitis is characterized by a thickening of the subepithelial collagen layer that is absent in LC. The main feature of LC is an increase of the density of intra-epithelial lymphocytes in the surface epithelium. Microscopic colitis typically presents as chronic or intermittent watery diarrhea, that may be accompanied by symptoms such as abdominal pain, weight loss and incontinence.

Evidence supporting the use of infliximab for microscopic colitis is limited to case reports and small case series (Pola, et al., 2013; Esteve, et al., 2011). In a review on the management of microscopic colitis, Ianiro, et al. (2012) concluded that the use of anti-tumor necrosis factor-α agents, infliximab and adalimumab, constitutes a new, interesting tool for the treatment of microscopic colitis, but larger, adequately designed studies are needed to confirm existing data.

Necrobiosis Lipoidica Diabeticorum

Basoulis and colleagues (2016) stated that necrobiosis lipoidica diabeticorum (NLD) is a rare degenerative connective tissue disorder associated with diabetes mellitus, which usually presents with red papules or plaques with raised edges and occasional ulceration. Ulcerating NLD is notoriously difficult to treat. These investigators described the case of a young patient with ulcerative NLD who was successfully treated with infliximab. Case presentation is followed by a review of therapeutic TNFα blockade in NLD. The subject was a 17-year old woman with type 1 diabetes since the age of 8, presented with a long-standing and extensively ulcerated and infected NLD lesion on her left shin. After achieving better glycemic control and treating her for infection of the wound, several NLD treatments failed to help, including corticosteroids and hyperbaric oxygen. She was treated successfully with 4 monthly sessions of 5 mg/kg body weight intravenous infliximab, achieving complete resolution of ulceration. A multitude of available treatments have been suggested for NLD over the past decades, based on 2 axes, one through wound healing and the other through immunosuppression. Anti-TNFα agents are relatively new drugs that brought a revolution in chronic inflammatory diseases and have been on the rise as novel potential treatments for NLD; 3 out of 5 available anti-TNFα agents have been safely tested so far, both topically and systematically, with mostly favorable results. The authors concluded that intravenous infliximab was successful in the treatment of recalcitrant ulcerating NLD in this patient. These researchers stated that taken together with an increasing number of similar reports revealing a pathogenetic role of TNFα in NLD, they suggested that anti-TNFα agents are promising drugs in the management of this condition.

Neurosarcoidosis (CNS Sarcoidosis)

Moravan and Segal (2009) described the effects of infliximab, and the anti-proliferative immunosuppressant, mycophenolate mofetil, in refractory neurosarcoidosis. These researchers treated patients with biopsy-proven sarcoidosis and central nervous system (CNS) involvement, who had failed treatment with steroids, with infliximab (5 mg/kg on weeks 0, 2, and 6, and then every 6 to 8 weeks thereafter). Six out of 7 patients were co-treated with mycophenolate mofetil (1,000 mg PO BID). Patients underwent a review of symptoms and complete neurological examination every 3 months and MRI scanning before and after 3 to 4 infusions of infliximab. All patients reported significant symptomatic improvement by the 4th infusion of infliximab, including relief of headache and neuropathic pain, reversal of motor, sensory, or coordination deficits, and control of seizure activity. Furthermore, infliximab therapy was universally associated with a decrease in lesion size or suppression of gadolinium enhancement as documented by MRI. A positive treatment response was attained irrespective of location or distribution of CNS involvement by sarcoidosis (dural/leptomeningeal based versus intra-parenchymal; cord versus brain; single lesion versus multi-focal). There were no serious adverse effects in a follow-up period spanning 6 to18 months. The authors concluded that combination treatment with mycophenolate mofetil and infliximab is a promising therapeutic approach for neurosarcoidosis.

Gelfand and colleagues (2017) described clinical and imaging responses in neurosarcoidosis to infliximab. Investigators at 6 US centers retrospectively identified patients with CNS sarcoidosis treated with infliximab, including only patients with definite or probable neurosarcoidosis following rigorous exclusion of other causes. Of 66 patients with CNS sarcoidosis (27 definite, 39 probable) treated with infliximab for a median of 1.5 years, the mean age was 47.5 years at infliximab initiation (SD 11.7, range of 24 to 71 years); 56.1 % were women; 62.1 % were white, 37.0 % African American, and 3 % Hispanic. Sarcoidosis was isolated to the CNS in 19.7 %. Using infliximab doses ranging from 3 to 7 mg/kg every 4 to 8 weeks, MRI evidence of a favorable treatment response was observed in 82.1 % of patients with imaging follow-up (n = 56), with complete remission of active disease in 51.8 % and partial MRI improvement in 30.1 %; MRI worsened in 1

patient (1.8 %). There was clinical improvement in 77.3 % of patients, with complete neurologic recovery in 28.8 %, partial improvement in 48.5 %, clinical stability in 18.2 %, worsening in 3 %, and 1 lost to follow-up. In 16 patients in remission when infliximab was discontinued, the disease recurred in 9 (56 %), typically in the same neuroanatomic location. The authors concluded that most patients with CNS sarcoidosis treated with infliximab exhibited favorable imaging and clinical treatment responses, including some previously refractory to other immunosuppressive treatments. This study provided Class IV evidence that for patients with CNS sarcoidosis infliximab was associated with favorable imaging and clinical responses.

The authors stated that drawbacks of this study include potential biases inherent in any retrospective case-series, including the absence of a comparison of outcomes to a non-infliximab-treated group. However, the return of disease activity in 56 % of patients who achieved remission on infliximab at a mean of 5.7 months after infliximab discontinuation provided additional support for the presumed benefit of infliximab in neurosarcoidosis. In addition, the favorable treatment response to infliximab in some patients who failed to respond to other immunosuppressive agents was supportive of an infliximab benefit. Another drawback of this study was that infliximab treatment protocols were not standardized within or across the centers, and variability in infusion protocols and use of concurrent immunotherapies could have had some influence on safety and efficacy. In this series, infliximab was most commonly used as combination therapy, rather than monotherapy, and some of the treatment benefit could relate to synergistic effects of combination immunosuppression. Variability in practice patterns related to timing of clinical and imaging follow-up in this retrospective case series precluded more specific time to event analyses, and future studies with standardized clinical and imaging follow-up would be helpful to clarify the specifics of timing of treatment of response with various dosing regimens. As the authors' centers have a special interest in caring for patients with neurosarcoidosis, the patient sample could reflect referral bias. Moreover, these researchers stated that further investigation is needed to determine how combination therapy compares with monotherapy in neurosarcoidosis and how various combinations compare in terms of efficacy, safety, and tolerability. The optimal duration of therapy also remains to be determined, but instances of relapse after

long-term infliximab treatment together with the observation that over 50 % of patients in remission on infliximab relapse after infliximab discontinuation supports the use of prolonged infliximab therapy in some patients and the need for close clinical and imaging follow-up if electing to discontinue infliximab. Many questions remain regarding longer-term management strategies and maintenance therapy. For example, after achieving remission, should the total dose or frequency of infliximab infusions be decreased? And when using combination regimens, what is the best strategy for tapering? Moreover, they stated that prospective studies are needed to determine the comparative risk of infliximab to other disease-modifying agents in neurosarcoidosis and what dose, frequency, and treatment combinations are optimal for neurosarcoidosis.

Van Sanford et al (2019) noted that CNS involvement in sarcoidosis is observed in 5 to 10 % of cases. Long-term treatment entails steroids and other immunomodulatory agents, including infliximab. Chronic immunosuppression can result in increased patient susceptibility to opportunistic infections. These investigators presented a case of fatal aspergillosis in a patient with neurosarcoidosis treated with infliximab. A 55-year old woman with neurosarcoidosis on infliximab (started 4 months prior) and dexamethasone, presented with progressive cognitive decline. Examination revealed impaired attention and disorientation with preserved language. Brain MRI showed multiple, bilateral cortical and sub-cortical ring-enhancing lesions. These researchers held immunosuppression due to suspicion of infection; empiric Amphotericin B was given early in the hospital course. The patient rapidly deteriorated from a neurological and respiratory stand-point, requiring intubation; cerebrospinal fluid (CSF) analysis showed elevated protein of 511 and normal glucose of 104 (67 % serum), with lymphocytic pleocytosis (25 cells, 96 % lymphocytes). Systemic and CNS microbiological studies were negative. On hospital day 13, bronchial fluid grew Aspergillus fumigatus, prompting a switch to voriconazole. Despite early empiric anti-fungal treatment, she died from respiratory failure; autopsy revealed systemic and CNS aspergillosis with multiple brain abscesses. The authors concluded that this case represented an example of a fatal complication of infliximab therapy, which was recently shown to be effective in neurosarcoidosis in 1 study. It also served to high-light the challenges faced in diagnosing ring-enhancing lesions, especially in patients with pre-existing brain disorders. Finally, it high-lighted the

difficulty in treating invasive aspergillosis. These investigators stated that further studies are needed to identify risks associated with infliximab therapy and potential early interventions to improve outcomes.

Prevention of Post-Operative Recurrence of Crohn's Disease

In a meta-analysis, Huang and colleagues (2018) examined the safety and efficacy of infliximab for prevention of post-operative recurrence in patients with CD. The Medline, Embase, PubMed, and Web of Science databases were systematically searched for suitable studies. A meta-analysis of enrolled studies was performed to analyze the efficacy of infliximab on outcomes regarding the prevention of post-operative recurrence of CD. A Galbraith radial plot was used to quantify the heterogeneity. Funnel plot and Egger test were performed to describe the bias of publication. A Forest plot was prepared to indicate the efficacy outcomes. A total of 7 prospective trials were included in this meta-analysis (n = 455). The Funnel plot and Egger test showed there was no significant bias in the included publications. The Cochrane collaboration tool indicated that all 7 prospective trials were of high quality. The results of Galbraith radial plot showed that no study was the source of heterogeneity. Compared with the placebo group, infliximab decreased the rates of endoscopic recurrence (RR = 0.421; 95 % CI: 0.328 to 0.539; $p < 0.001$), and there was a significant reduction in rates of clinical recurrence in the infliximab-treated group (RR = 0.519; 95 % CI: 0.349 to 0.774; $p = 0.001$). Furthermore, infliximab treatment did not show adverse effects as other systematic therapeutic drugs, indicating that infliximab treatment was effective and well-tolerated. The authors concluded that compared with the controls, infliximab was a promising therapeutic agent for the management of CD patients. These researchers stated that further investigations should aim to prove the long-term stability, safety, and efficacy of Infliximab in the treatment of CD.

The authors stated that a limitation of this meta-analysis was the limited number of trials included (n = 7). However, all the trials were prospective studies and are therefore of high quality. Furthermore, although almost all the publications included in this study were from top journals with high impact factors, risks of bias, e.g., due to funding from the pharmaceutical industry, may exist. Due to ethical limitations, it is difficult to perform

clinical trials on healthy people, which play important role in control group, and it would be fine to compare low-risk population (the first resection/non-penetrating disease) and high-risk CD populations on the efficacy of infliximab (very relevant for clinical practice); future studies should address the issue in these patient's groups.

Primary Systemic Vasculitis

In a retrospective descriptive case series, Eleftheriou and colleagues (2009) reported the safety and effectiveness of biologic therapies in a multi-center series of children with primary systemic vasculitis (PSV). Primary retrospective outcome assessment measures were: daily corticosteroid dose; Birmingham Vasculitis Activity Score (BVAS); and adverse events (including infection rate). A total of 25 patients median age 8.8 (range of 2.4 to 16) years; 11 males with active PSV (n = 6 with anti-neutrophil cytoplasmic antibody associated vasculitides, n = 11 with polyarteritis nodosa, n = 7 with unclassified vasculitis and n = 1 with Behçet's disease) were treated with biologic agents including infliximab (n = 7), rituximab (n = 6), etanercept (n = 4), adalimumab (n = 1) or multiple biologics sequentially (n = 7). Overall, there was a significant reduction in BVAS from a median of 8.5 (range of 5 to 32) at start of therapy to 4 (range of 0 to 19) at median 32 months follow-up (p = 0.003) accompanied by significant reduction in median daily prednisolone requirement from 1 (range of 0.2 to 2) to 0.25 (range of 0 to 1) mg/kg/day, p = 0.000. For those receiving multiple biologic agents sequentially, a similar clinical improvement was observed with corticosteroid sparing. Infections occurred in 24 %, the most severe in those receiving infliximab. The authors concluded that these findings provided retrospective evidence of efficacy of these agents, and high-lighted the associated infectious complications. They stated that further multi-center standardization of treatment protocols and data collection of biologic therapy in systemic vasculitis of the young is needed.

Psoriatic Arthritis and Plaque Psoriasis

The U.S. Pharmacopeial Convention (2003) has concluded that psoriatic arthritis and psoriasis are accepted indications for infliximab. A controlled clinical study (Chaudhari et al, 2001) has demonstrated the short-term effectiveness of infliximab in plaque psoriasis. In a controlled clinical trial

in which patients and investigators were blinded for the first 10 weeks, participants were assigned to either of 2 doses of infliximab (5 mg/kg or 10 mg/kg at baseline, 2 weeks, and 6 weeks) or to placebo. Nineteen of 22 patients assigned to infliximab achieved good or better physician's overall assessments, compared with 2 of 11 patients assigned to placebo. In initial studies, remissions seemed to be durable, with many patients improving for 6 months or longer. Subsequent randomized controlled clinical studies have confirmed the efficacy of infliximab for treatment of plaque psoriasis (Reich et al, 2005; Feldman et al, 2005; Gottlieb et al, 2004).

Guidelines from the British Association of Dermatologists (2005) stated that infliximab is useful in clinical circumstances requiring rapid control of psoriasis, e.g., in unstable erythrodermic or pustular psoriasis, due to its very rapid onset of action and high response rate.

Infliximab (Remicade/Avsola/Inflectra/Renflexis) is FDA approved for treatment of reducing signs and symptoms of active arthritis, inhibiting the progression of structural damage, and improving physical function in persons with psoriatic arthritis. Furthermore, it has the FDA approval for treatment of adults with chronic severe (i.e., extensive and/or disabling) plaque psoriasis who are candidates for systemic therapy and when other systemic therapies are medically less appropriate.

Pyoderma Gangrenosum

A clinical trial has demonstrated that infliximab is superior to placebo in the treatment of pyoderma gangrenosum. Pyoderma gangrenosum is a chronic ulcerating skin condition that often occurs in association with inflammatory bowel disease. Pyoderma gangrenosum is treated by managing the underlying cause. Topical and oral steroids are used. If resistant, oral dapsone or minocycline, and occasionally cyclosporine is used. Brooklyn et al (2006) reported on the results of a randomized controlled trial of infliximab. Patients 18 years of age and older with pyoderma gangrenosum were randomized to receive infliximab at 5 mg per kilogram (n = 13) or placebo (n = 17). Patients were assessed 2 weeks later and, during the open label portion of the study, non-responders were offered open labeled infliximab, and were re-assessed 4 weeks later. At week 2, significantly more patients in the infliximab group

had improved (46 % (6/13) compared with the placebo group (6 % (1/17) (p = 0.025). Overall, 29 patients received infliximab with 69 % (20/29) demonstrating a clinical beneficial response. Remission rate at week 6 was 21 % (6/29). There was no response in 31 % (9/29) of patients. The authors concluded that infliximab is superior to placebo in the treatment of pyoderma gangrenosum.

Relapsing Polychondritis

Rovensky and Sedlackova (2012) noted that relapsing polychondritis (RP) is an unusually rare disease involving multiple organs. It has an episodic course, occasionally also progressing. Typically, inflammation of cartilaginous tissues and tissues rich in glycosaminoglycans is present. Clinical symptoms are concentrated in auricula, nose, larynx, upper respiratory tract, joints, heart, blood vessels, inner ear, cornea and sclera. Manifestations include: (i) chondritis of auricular, nasal, laryngo-tracheal, costal and joint cartilages, (ii) inflammation of the eyes and inner ear, (iii) collapse of laryngo-tracheal structures and structures in the subglottic area resulting in increased susceptibility to upper respiratory tract infections, and (iv) diversity of clinical manifestations, of the disease course and also of the treatment response.

Concurrent systemic vasculitis or glomerulonephritis may contribute to higher morbidity and premature mortality. In about 30 % of cases the RP is secondary, accompanied by other systemic connective tissue disorders as RA, SLE, Sjogren's syndrome, thyroiditis, ulcerative colitis, psoriasis and Behcet's syndrome. Diagnosis is based on 1986 diagnostic criteria from Minnesota and RP has to be suspected when the inflammatory bouts involve at least 2 of the typical sites -- auricular, nasal, laryngo-tracheal or 1 of the typical sites and 2 other -- ocular, stato-acoustic disturbances (hearing loss and/or vertigo) and arthritis. Conventional treatments entail corticoids and non-steroidal anti-inflammatory drugs, corticoids combined with immunosuppressive therapy (cyclophosphamide, azathioprine, chlorambucil, cyclosporine). More recently, also biologic therapy is used in RP (infliximab, adalimumab, etanercept, tocilizumab, rituximab). It is necessary to underscore that biologic therapy for RP is only a research modality used in very severe refractory forms of RP. Preliminary results suggested that biologic therapy will have its place in severe refractory relapsing forms of RP.

Also, an UpToDate review on "Treatment of relapsing polychondritis" (Michet, 2013) states that "[t]he major second-line drugs in RPC include cyclophosphamide, azathioprine, cyclosporine, pulse methylprednisolone, methotrexate, and other newer therapies. Response to anti-inflammatory/immunomodulatory compounds with potential efficacy, such as leflunomide, soluble tumor necrosis factor alpha receptor (etanercept) or anti-tumor necrosis factor antibody (infliximab, adalimumab), and anti-interleukin 1 receptor (anakinra), have been largely limited to individual case reports …. Successful treatments with infliximab (5 mg/kg by intravenous infusion at two, four, then eight-week intervals) and etanercept (50 mg weekly) have been reported. Potential efficacy has been shown in refractory RPC associated laryngo-tracheal disease, necrotizing vasculitis, episcleritis/scleritis, aortic valve disease, synovitis, and auricular and nasal cartilage inflammation. These promising uncontrolled results warrant further investigation to assess the role of anti-TNF alpha therapies".

Renal Cell Carcinoma

Harrison and colleagues (2007) stated that tumor necrosis factor-alpha (TNF-alpha) may play a role in renal cell carcinoma (RCC). These researchers performed 2 sequential phase-II clinical trials of infliximab in patients with immunotherapy-resistant or refractory RCC. Patients progressing after cytokine therapy were treated with intravenous infliximab as follows: study 1 (19 patients), 5 mg/kg at weeks 0, 2, and 6, and then every 8 weeks; study 2 (18 patients), 10 mg/kg at weeks 0, 2, and 6, and then every 4 weeks. Treatment continued until disease progression (PD). Response was assessed according to Response Evaluation Criteria in Solid Tumors. Plasma levels of TNF-alpha, CCL2, and interleukin-6 (IL-6) were measured before and during treatment. TNF-alpha and its receptors were detected in malignant cells in RCC biopsies. In study 1, 3 patients (16 %) achieved partial response (PR) and 3 patients (16 %) achieved stable disease (SD). Median duration of response (PR + SD) was 7.7 months (range of 5.0 to 40.5+ months). In study 2, 11 patients (61 %) achieved SD. Median duration of response was 6.2 months (range of 3.5 to 24+ months); 1 patient developed grade-3 hypersensitivity and another died as a result of pulmonary infection/sepsis. Enzyme-linked immunosorbent assay (ELISA) analysis of plasma revealed that higher levels of TNF-alpha at baseline and higher

levels of CCL2 during treatment were associated with PD. There were also correlations between higher levels of TNF-alpha, IL-6, and CCL2 and poor survival (less than 12 months). The authors concluded that this was the first direct clinical evidence suggesting that TNF-alpha may be a therapeutic target in RCC. Plasma levels of TNF-alpha, IL-6, and CCL2 may have predictive and prognostic significance.

Larkin et al (2010) noted that there is clinical evidence to suggest that TNF-α may be a therapeutic target in RCC. Multi-targeted kinase inhibitors, such as sorafenib and sunitinib, have become standard of care in advanced RCC. The anti-TNF-α monoclonal antibody infliximab and sorafenib have differing cellular mechanisms of action. These researchers conducted a phase I/II clinical trial to determine the safety and efficacy of infliximab in combination with sorafenib in patients with advanced RCC. Eligible patients were systemic treatment-naive or had received previous cytokine therapy only. Sorafenib and infliximab were administered according to standard schedules. The study had 2 phases: in phase-I, the safety and toxicity of the combination of full-dose sorafenib and 2 dose levels of infliximab were evaluated in 3 and 3 patients, respectively, and in phase-II, further safety, toxicity and efficacy data were collected in an expanded patient population. Acceptable safety was reported for the 1st 3 patients (infliximab 5 mg kg^{-1}) in phase-I. Sorafenib 400 mg twice-daily and infliximab 10 mg kg^{-1} were administered to a total of 13 patients (3 in phase-I and 10 in phase-II). Adverse events included grade-3 hand-foot syndrome (31 %), rash (25 %), fatigue (19 %) and infection (19 %). Although manageable, toxicity resulted in 75 % of the patients requiring at least 1 dose reduction and 81 % requiring at least 1 dose delay of sorafenib; 4 patients were progression-free at 6 months (PFS_6 31 %); median PFS and overall survival(OS) were 6 and 14 months, respectively. The authors concluded that sorafenib and infliximab can be administered in combination, but a significant increase in the numbers of adverse events requiring dose adjustments of sorafenib was observed. There was no evidence of increased efficacy compared with sorafenib alone in advanced RCC. These investigators stated that the combination of sorafenib and infliximab did not warrant further evaluation in patients with advanced RCC. Moreover, they noted that the putative anti-tumor activity of infliximab that has been previously

demonstrated in advanced RCC warrants further investigation and combination with alternative agents or in subgroups of patients should be considered.

An UpToDate review on “Overview of the treatment of renal cell carcinoma” (Atkins, 2019) does not mention infliximab as a therapeutic option.

Furthermore, National Comprehensive Cancer Network’s clinical practice guideline on “Kidney cancer” (Version 1.2024) does not mention infliximab as a therapeutic option.

Retinal Vasculitis

Cheema et al (2011) reported their experience in treating 2 patients of idiopathic retinal vasculitis, aneurysm, and neuroretinitis (IRVAN) syndrome with infliximab who showed a very favorable response to treatment. Two patients with clinical diagnosis of IRVAN syndrome were included in the study. The visual acuity (VA) was affected due to ocular inflammation and presence of macular edema due to exudation around the optic nerve. The patients did not respond to initial treatment with oral steroids, and VA continued to deteriorate due to macular exudation. Infliximab therapy resulted in prompt resolution of the inflammatory reaction and retinal exudation, with improvement in VA, that was subsequently maintained with maintenance therapy. The intravenous infliximab infusions were scheduled at 0, 4, 8, and 12 weeks initially, and every 2 months thereafter. Retinal neovascularization in each patient was managed by pan retinal photocoagulation. The authors concluded that infliximab therapy may be useful in reducing inflammation and leakage from the optic nerve in patients with IRVAN syndrome. This may help preserve or improve VA.

The American Academy of Ophthalmology (AAO) EyeWiki’s review on “Retinal vasculitis” (Agarwal, 2014) stated that “Unlike systemic vasculitis that can be managed with colchicines and non-steroidal anti-inflammatory agents, retinal vasculitis due to causes such as Adamantiades-Behcet’s disease (ABD) requires a more aggressive approach. Available immunosuppressive agents include cyclosporine, azathioprine, cyclophosphamide, mycophenolate mofetil or biologic agents such as

infliximab or etanercept. Cyclosporine has been used as a drug of choice in previous studies. In eyes with vasculitis associated with birdshot chorioretinopathy, sarcoidosis and Harada's disease, azathioprine has been used for treatment. Alkylating agents such as chlorambucil and cyclophosphamide have also been used in combination with corticosteroids. Biologic agents have been increasingly used to achieve remission in eyes with retinal vasculitis. These include infliximab, tacrolimus and adalimumab, apart from other agents targeting molecules such as tumor necrosis factor and interleukins".

Furthermore, an UpToDate review on "Retinal vasculitis associated with primary ocular disorders" (Tolentino and Dana, 2019) states that "Treatment of idiopathic retinal vasculitis is usually reserved for those who develop poor vision. Systemic glucocorticoids, azathioprine, methotrexate, cyclosporine, and plasma exchange have all been used with variable efficacy, but none has been evaluated in a randomized fashion. One retrospective study of 29 patients with vision-threatening disease found that the administration of prednisone (initial dose 1 mg/kg per day) was associated with visual improvement in 60 % of patients. Laser therapy is reserved for complications of retinal vasculitis, such as rubeosis iridis, recurrent vitreous hemorrhage, or retinal neovascularization. Other local procedures are warranted in selected patients. Vitrectomy has been indicated in dense vitreous inflammation, vitreous hemorrhage, retinal detachment, or epiretinal membrane formation. In some patients, filtering procedures (to increase drainage of aqueous) must be performed to treat the secondary glaucoma that may accompany retinal vasculitis and ocular inflammation. Secondary glaucoma can occur because of inflammation and/or new blood vessel growth on the angle (drainage portion of the eye). In patients in whom macular edema develops, bevacizumab has been shown to reduce the edema and to improve vision. For idiopathic retinal vasculitis aneurysm neuroretinitis (IRVAN), a single systemic dose of infliximab and ranibizumab appears to be helpful in improving vision and in suppressing complications related to this rare form of retinal vasculitis".

Rheumatoid Arthritis

Consistent with FDA-approved product labeling, infliximab, in combination with methotrexate, is considered medically necessary for the reduction of signs and symptoms of RA in patients who have had an inadequate response to methotrexate monotherapy. The safety and efficacy of infliximab for RA was demonstrated in a 30-week-long, multi-center, randomized, controlled clinical study involving RA patients who had failed to adequately respond to 6 or more months of methotrexate monotherapy. When used for treatment of RA, infliximab is administered intravenously in 3 mg/kg doses at 0, 2 and 6 weeks and then every 8 weeks thereafter. Infliximab may be administered as frequently as every 4 weeks in patients with an inadequate response to less frequent dosing. However, in clinical trials, higher doses and/or more frequent administrations did not result in higher response rates.

In a double-blind, placebo-controlled, phase III clinical trial, Maini and colleagues (1999) examined if infliximab would provide additional clinical benefit to patients who had active RA despite receiving methotrexate. A total of 428 patients who had active RA and had received continuous methotrexate for at least 3 months and at a stable dose for at least 4 weeks, were randomized to placebo (n = 88) or one of four regimens of infliximab at weeks 0, 2, and 6. Additional infusions of the same dose were given every 4 or 8 weeks thereafter on a background of a stable dose of methotrexate (median 15 mg/week for greater than or equal to 6 months, range 10 to 35 mg/wk). Patients were assessed every 4 weeks for 30 weeks. At 30 weeks, the American College of Rheumatology (ACR)-20 response criteria, representing a 20 % improvement from baseline, were achieved in 53, 50, 58, and 52 % of patients receiving 3 mg/kg every 4 or 8 weeks or 10 mg/kg every 4 or 8 weeks, respectively, compared with 20 % of patients receiving placebo plus methotrexate ($p < 0.001$ for each of the four infliximab regimens versus placebo). A 50 % improvement was achieved in 29, 27, 26, and 31 % of infliximab plus methotrexate in the same treatment groups, compared with 5 % of patients on placebo plus methotrexate ($p < 0.001$). Infliximab was well-tolerated; withdrawals for adverse events as well as the occurrence of serious adverse events or serious infections did not exceed those in the placebo group. The authors concluded that during 30 weeks, treatment

with infliximab plus methotrexate was more effective than methotrexate alone in patients with active RA not previously responding to methotrexate.

The American College of Rheumatology (ACR) conducted a systematic review to synthesize the evidence for the benefits and harms of various treatment options. Their goal was to develop evidence-based, pharmacologic treatment guideline for rheumatoid arthritis. The 2015 American College of Rheumatology Guidelines for the Treatment of Rheumatoid Arthritis provided “strong” recommendations for established RA and symptomatic early RA.

For established RA, the guidelines state “if the disease activity is low, in patients who have never taken a DMARD, the recommendation is to use DMARD monotherapy (methotrexate preferred) over TNFi”. “If disease activity remains moderate or high despite DMARD monotherapy, the recommendation is to use combination traditional [conventional] DMARDs or add a TNFi or a non-TNF biologic or tofacitinib (all choices with or without methotrexate, in no particular order of preference), rather than continuing DMARD monotherapy alone”. Recommendations for patients with symptomatic early RA state that “if disease activity is low, in patients who have never taken a DMARD, use DMARD monotherapy (methotrexate preferred) over double or triple therapy”. “If disease activity remains moderate or high despite DMARD monotherapy (with our without glucocorticoids), use combination DMARDs or a TNFi or a non-TNF biologic (all choices with our without methotrexate, in no particular order of preference), rather than continuing DMARD monotherapy alone”. A strong recommendation means that the panel was confident that the desirable effects of following the recommendation outweigh the undesirable effects (or vice versa), so the course of action would apply to most patients, and only a small proportion would not want to follow the recommendation (Singh et al., 2016).

The 2022 EULAR (European Alliance of Associations for Rheumatology) guidelines for RA management emphasize a treat-to-target approach, aiming for sustained remission or low disease activity through early intervention, regular monitoring, and shared decision-making. Recommendations include use of methotrexate (MTX) as the preferred first-line therapy, typically initiated at 15 mg/week and escalated up to 25-

30 mg/week as tolerated, with folic acid supplementation to reduce side effects. If MTX is contraindicated or not tolerated, alternatives such as leflunomide, sulfasalazine, or hydroxychloroquine can be considered. Hydroxychloroquine is particularly appropriate for patients with mild disease or when used in combination with other DMARDs, but is not recommended as monotherapy in individuals with poor prognostic indicators due to its limited efficacy in preventing joint damage. If treatment targets are not met within 3 to 6 months, escalation to a biologic DMARD (bDMARD) or targeted synthetic DMARD (tsDMARD) (e.g., tofacitinib, upadacitinib) is advised. Among bDMARDs, TNF inhibitors such as infliximab are recommended, typically in combination with MTX but may be used as monotherapy when MTX is unsuitable. The guidelines also stress the importance of considering comorbidities and safety profiles when selecting and escalating therapy (Smolen et al., 2023).

Sarcoidosis

Sodhi and colleagues (2009) stated that CNS involvement is a severe manifestation of sarcoidosis that often requires aggressive immunosuppressive therapy. The most effective approach for refractory disease is unknown. These investigators reviewed the cases of 4 subjects who demonstrated active progression of neurosarcoidosis while under treatment with cyclophosphamide, and who were subsequently treated with infliximab. All 4 subjects demonstrated rapid and substantial reversal of their clinical course. Radiological findings were concordant with the clinical responses. There were no notable toxicities. Infliximab may be more effective than cyclophosphamide for refractory CNS sarcoidosis. The authors noted that a larger, prospective study is warranted.

Drent and colleagues (2014) noted that in severe refractory sarcoidosis cases not responding to conventional immunosuppressive treatment, the 3rd-line TNF-α inhibitors IFX and ADA might be an alternative. However, appropriate studies to guide the clinician are lacking. These investigators established practical recommendations for the use of TNF-α inhibitors in the management of refractory sarcoidosis patients. Based on a literature search and the opinion of sarcoidosis experts worldwide, the recommendations were established. Studies conducted in sarcoidosis

were supplemented with data obtained from relevant studies in other inflammatory diseases. A Delphi method of polling, using an online survey addressing 12 clinical questions, was performed amongst 20 of the world's leading sarcoidologists to examine consensus in case of inadequate data to determine an objective answer. Of the 256 papers found, 101 were included; RCTs on the use of TNF-α inhibitors in sarcoidosis were limited; 95 % (19 of 20) of the sarcoidologists contacted, completed the questionnaire (Europe 68 %, North America 32 %). A total of 9 recommendations were formulated concerning general aspects of TNF-α inhibitor use; specific sarcoidosis related items, including indications, starting and maintenance dosage, interval of treatment, treatment duration, and discontinuation regimen of IFX and ADA, were addressed. The authors concluded that based on earlier studies and consensus amongst world's leading sarcoidologists, practical recommendations for the use of TNF-α inhibitors in sarcoidosis were established. These recommendations, with emphasis on indications, dosage and discontinuation regimens, had been developed to support the clinician in the management of refractory sarcoidosis patients.

Schimmelpennink and co-workers (2018) stated that IFX is effective 3rd-line therapy in severe sarcoidosis. The originator product IFX, Remicade, is expensive, limiting universal access. Recently, a less expensive biosimilar of IFX, Inflectra, has become available, but the efficacy and tolerability has not been studied in sarcoidosis. In this retrospective, cohort study, a total of 29 patients treated with the IFX biosimilar Inflectra, were analyzed. Patients received Inflectra intravenously monthly at a dose of 5 mg/kg. These researchers measured trough levels before every infusion. Before and after 6 months of induction therapy pulmonary function and disease activity were evaluated using standardized uptake value (SUV) of the 18F-fluorodeoxyglucose by positron emission tomography (18F-FDG PET), soluble interleukin-2 receptor (sIL-2R), angiotensin converting enzyme (ACE) and health-related quality of life (HRQOL). In patients with pulmonary sarcoidosis as main treatment indication ($n = 15$) the predicted FVC improved with 8.1 % ($p < 0.05$). Furthermore, in the whole group HRQoL improved significantly ($p < 0.001$), whereas SUVmax and sIL-2R significantly reduced ($p < 0.001$ and $p = 0.001$, respectively). Hospitalization due to infections occurred in 4 patients. None of the patients discontinued Inflectra due to side-effects. Furthermore, all patients had detectable trough levels indicating

development of neutralizing antibodies. The authors concluded that IFX biosimilar Inflectra appeared effective in the treatment of refractory sarcoidosis with a comparable safety profile to the reference product Remicade. These investigators stated that Inflectra can be considered as an alternative and less expensive option for patients with refractory sarcoidosis.

Adler and associates (2019) stated that although anti-TNFs have been recommended as 3rd-line therapy for sarcoidosis, an up-to-date systematic synthesis of their safety and efficacy is lacking. These investigators reviewed the literature to characterize the safety and efficacy of anti-TNFs in sarcoidosis. All countries and treatment settings were included. The authors searched Medline, Embase, CINAHL, Web of Science, ClinicalTrials.gov, Cochrane Library, and Google Scholar from inception to November 27, 2017. Studies of 5 or more cases of sarcoidosis treated with anti-TNFs were included. Descriptive statistics were performed. A total of 65 studies (including 5 RCTs) were identified, comprising 1,525 patients. For pulmonary sarcoidosis, 1 RCT found IFX significantly improved vital capacity versus placebo; a 2nd detected no difference. In non-randomized studies, IFX improved pulmonary function in 79 % of patients. For cutaneous sarcoidosis, compared to placebo, ADA showed greater Physician Global Assessment (PGA) response and significantly reduced target lesion area, and IFX significantly decreased Sarcoidosis Area and Severity Index (SASI) induration and erythema scores. In non-randomized studies of cutaneous, ocular, neurologic, and multi-system sarcoidosis, IFX improved 89 %, 69 %, 77 %, and 71 % of cases, respectively; ADA improved 77 % of ocular sarcoidosis cases; IFX displayed a steroid-sparing effect; 50 % of patients relapsed after discontinuation of IFX, ADA, ETA, or certolizumab pegol. In RCTs, compared to placebo, anti-TNFs had comparable overall and serious AEs and slightly more serious infections. The authors concluded that available evidence suggested the safety and efficacy of IFX in pulmonary, cutaneous, ocular, neurologic, and multi-system sarcoidosis, and ADA in cutaneous and ocular sarcoidosis.

Scleritis

An UpToDate review on “Treatment of scleritis” (Stone and Dana, 2014) states that “Case reports and small, uncontrolled case series suggest that the tumor necrosis factor (TNF) alpha inhibitor infliximab may be partially effective in the treatment of scleritis that is resistant to treatment with other agents. Doses in the range of 3 to 5 mg/kg administered every four to eight weeks have been employed for this purpose. Additional investigation of infliximab for this indication is required”.

Sjogren's Syndrome

Dry eye affects many individuals worldwide. Significant proportion of patients with dry eye has underlying Sjogren's syndrome (SS), a progressive autoimmune condition. Akpek et al (2011) reported treatment options for patients with dry eye secondary to SS. A search strategy was developed to identify prospective, interventional studies of treatments for SS-associated dry eye from electronic databases. Eligible references were restricted to English-language articles published after 1975. These sources were augmented by hand searches of reference lists from accessed articles. Study selection, data extraction, and grading of evidence were completed independently by 4 or more review authors. The searches identified 3,559 references as of August 10, 2010. After duplicate review of the titles and abstracts, 245 full-text papers were assessed, 62 of which were relevant for inclusion in the review. In the current literature on SS-associated dry eye, there is a paucity of rigorous clinical trials to support therapy recommendations. The review noted that prospective trials using infliximab and etanercept failed to demonstrate any objective improvement in the sicca component of SS syndrome.

Still's Syndrome

Kraetsch and colleagues (2001) examined the efficacy of infliximab in the treatment of patients with severe and active adult onset Still's syndrome (AOSD) despite conventional immunosuppressive therapy. A total of 6 patients with the diagnosis of AOSD according to the Yamagushi criteria of 1992 were treated with infliximab. All patients had severe disease with high clinical and serological activity. Patients were treated initially with high dose steroids or more intensive immunosuppressive therapy. Two

patients had a history of multiple disease modifying antirheumatic drug (DMARD) treatments. One patient had a history of 3 years of AOSD with fever, chills, pleural and pericardial effusions, and hepatosplenomegaly. Despite these treatments, the patient developed increasing serological signs of inflammation and arthritis of both hips and peripheral joints. Another patient had a history of 5 years of AOSD with oligoarthritis, myalgias, and recurrent fever despite multiple DMARD treatment, including cyclophosphamide pulse therapy. These patients with AOSD presented with massive polyarthralgias, polyarthritis, splenomegaly or hepatomegaly, the typical rash, sore throat, weight loss, serositis, continuing fever, leucocytosis, and raised C reactive protein (CRP), ESR, and ferritin levels. Four patients with early onset of the disease, fulfilling the diagnostic criteria for AOSD and a clinical and serological high disease activity, were included in this pilot study without any further DMARD treatment apart from the initial steroid treatment. Reduction of established treatment, mainly with steroids, caused a relapse of the disease in all these patients. Patients then received 3 to 5 mg/kg infliximab on weeks 0, 2, and 6, continuing with intervals of 6 to 8 weeks depending on the patient's individual disease activity. In all patients, fever, arthralgias, myalgias, hepatosplenomegaly, and the rash resolved after the first courses of treatment with infliximab. All serological variables (CRP, ESR, hyperferritinaemia) returned to normal. After 3 courses of infliximab infusions, splenomegaly could not be detected in any of these patients. After 3 courses of treatment with infliximab, splenomegaly could not be detected in any of the patients. Up to now, these patients have received infliximab infusion treatment for between 5 and 28 months. Throughout this period, all patients have continued to benefit from this treatment, with improvement in their clinical symptoms, joint counts, and serological disease activity. One of the patients had a moderate infusion reaction during the second treatment. The infusion was discontinued for 1 hour and then was resumed with no further problems. The authors concluded that the disease improved remarkably in all 6 patients with AOSD after treatment with infliximab, also in the early stage of AOSD. These preliminary data suggested the potential therapeutic benefit of anti-TNF-alpha treatment in AOSD.

In a prospective, non-comparative case-series study, Kokkinos et al (2004) examined the effects of infliximab on refractory AOSD. A total of 4 patients with severe and highly active AOSD, refractory to high doses of

corticosteroids (which had been combined with methotrexate in 3 of them) and methotrexate were treated with infliximab (initial dose 3 to 5 mg/kg, continuing at intervals depending on the patient's individual disease activity). Resolution of their symptoms, which was evident within few days after the first infusion, and a parallel rapid improvement of the acute inflammatory response indices were observed in all. Concomitant corticosteroid treatment was reduced after the first courses of treatment with infliximab, which was well-tolerated, and complete disease remission was sustained during a 5 to18 months follow-up period. The authors noted that although further studies are needed to confirm long-term efficacy and safety in larger numbers of patients, they suggested that administration of infliximab with observation for objective improvement is the treatment of choice in cases of AOSD refractory to conventional treatment.

Takayasu Arteritis

In a retrospective study, Molloy and co-workers (2008) assessed the efficacy of anti-TNF therapy to induce remission in patients with Takayasu arteritis (TAK) refractory to other immunosuppressive therapies. Patients (n = 25) were treated with infliximab (IFX) or etanercept (ETA) for up to 7 years; 21 with IFX (median 28 months; range of 2 to 84) and 9 with ETA (median 28 months; range of 4 to 82); 5 patients initially treated with ETA subsequently switched to IFX. Following anti-TNF therapy, remission was achieved and prednisone was discontinued in 15 patients (60 %) and successfully tapered below 10 mg/day in an additional 7 patients (28 %). Of 18 patients treated with other immunosuppressive agents concurrent with anti-TNF therapy, 9 (50 %) could taper or discontinue the additional agent. Major relapses occurred in 4 patients that initially achieved stable remission. Four patients suffered adverse events, including 1 with opportunistic infections and 1 with breast cancer. The authors concluded that in this group of patients with refractory TAK, anti-TNF therapy was associated with remission in a majority of patients, facilitating dose reduction or discontinuation of prednisone and other immunosuppressive therapy. These findings strengthen the rationale for the conducting of a randomized controlled trial of anti-TNF therapy in TAK.

Filocamo et al (2008) noted that 4 children with TAK were treated with TNF antagonists because of disease relapse during conventional therapy or as a first-line agent. Two patients went into remission; in the other 2, the response was partial. The authors concluded that anti-TNF agents can have a role in the treatment of TAK; however, further controlled studies are needed.

Tubulointerstitial Nephritis and Uveitis Syndrome

An UpToDate review on "Tubulointerstitial nephritis and uveitis (TINU syndrome)" (Lee and Ashfaq, 2013) does not mention the use of infliximab as a therapeutic option.

Ulcerative Colitis

Conventional treatment options for patients with severe corticosteroid-refractory ulcerative colitis include intravenous cyclosporine, which is frequently limited by toxicity, or colectomy. Studies have shown that infliximab, improves clinical, endoscopic, and histologic outcomes in patients with severely active ulcerative colitis refractory to conventional therapy, allowing corticosteroid sparing and reducing the need for colectomy. This is consistent with scientific observations that suggest a central role for TNF in the inflammatory cascade. Two phase III randomized, placebo-controlled clinical trials have demonstrated efficacy of infliximab in inducing and maintaining clinical response and remission of refractory moderate to severe ulcerative colitis (Rutgeerts et al, 2005; Sandborn et al, 2005). In these clinical trials, subjects with moderate-to-severe ulcerative colitis that was refractory to at least one standard therapy were randomly assigned to infliximab in doses of 5 mg per kg or 10 mg per kg, or to placebo. In 1 clinical trial involving 364 subjects with moderate-to-severe ulcerative colitis (Sandborn et al, 2005), 62 % of subjects in the 10-mg group and 69 % of subjects in the 5-mg group had a clinical response at 8 weeks, compared to 37 % of subjects in the placebo group ($p < 0.001$ for both). At that time, 32 % of subjects in the 10 mg group and 39 % of subjects in the 5 mg group were in clinical remission, versus 15 % of subjects in the placebo group ($p = 0.002$ and $p < 0.001$, respectively). By 30 weeks, 51 % of the 10-mg group and 52 % of the 5-mg group achieved clinical response versus 30 % of placebo-

treated subjects (p = 0.002 and p < 0.001). At that time, 37 % of the 10-mg group and 34 % of the 5-mg group were in clinical remission, versus 16 % of the placebo group (p < 0.001 and p = 0.001).

In a 2nd clinical trial involving 364 subjects with moderate-to-severe ulcerative colitis (Sandborn et al, 2005), 69 % of subjects in the 10-mg group and 65 % of subjects in the 5-mg group had a clinical response at 8 weeks, compared to 29 % of subjects in the placebo group (p < 0.001 for both). At that time, 28 % of subject in the 10-mg group and 34 % of subjects in the 5-mg group were in clinical remission, versus 6 % of subjects in the placebo group (p < 0.001 for both). By 30 weeks, 60 % of subjects in the 10-mg group and 47 % of subjects in the 5-mg group had a clinical response, compared to 26 % of subjects receiving placebo (p < 0.001 for both). At that time, 36 % of subjects in the 10-mg group and 26 % of subjects in the 5-mg group were in clinical remission, compared to 11 % in the placebo group (p < 0.001 and p = 0.003).

On September 23, 2011, the FDA approved infliximab to treat moderately to severely active ulcerative colitis in children older than 6 years who have had inadequate response to conventional therapy.

Additional studies are needed to evaluate the optimal timing and duration of infliximab therapy, the utility of adjuvant medical treatments during infliximab therapy, and the long-term safety and comparative efficacy of the various treatments for ulcerative colitis to better define the role of infliximab in the treatment of this condition. Other TNF antagonists for ulcerative colitis in various phases of investigation include the monoclonal antibody CDP 571, the fusion peptide etanercept, the phosphodiesterase inhibitor oxpentifylline, and thalidomide.

Adedokun and colleagues (2014) analyzed data collected during the Active Ulcerative Colitis Trials (ACT-1 and ACT-2) to evaluate relationships between serum concentrations of infliximab and outcomes of adults with moderate-to-severe ulcerative colitis. These researchers compared serum concentrations of infliximab with outcomes of 728 patients with moderately-to-severely active ulcerative colitis who participated in ACT-1 or ACT-2; efficacy data were collected at weeks 8, 30, and 54 (for ACT-1 only). Relationships between serum concentration of infliximab and efficacy outcomes were assessed using trend, logistic

regression, and receiver operating characteristic curve analyses. They also evaluated factors that affected the relationship between exposure and response. Median serum concentrations of infliximab at weeks 8, 30, and/or 54 were significantly higher in patients with clinical response, mucosal healing, and/or clinical remission than in patients who did not meet these response criteria. There were statistically significant relationships between quartile of infliximab serum concentration and efficacy at these time points ($p < 0.01$). Infliximab therapy was effective for a smaller proportion of patients in the lowest quartile, and these patients had lower serum levels of albumin and a higher incidence of antibodies to infliximab than patients in other quartiles. Although the relationship between exposure to infliximab and response varied among patients, approximate serum concentrations of 41 μg/ml infliximab at week 8 of induction therapy and 3.7 μg/ml at steady-state during maintenance therapy produced optimal outcomes in patients. The authors concluded that serum concentrations of infliximab are associated with efficacy in patients with moderate-to-severe ulcerative colitis; however, complex factors determine the relationship between exposure to this drug and response. They stated that a prospective evaluation of the value of measuring serum concentrations of infliximab should be performed before these data can be included in patient management strategies.

Uveitis

Infliximab is under investigation as a treatment for uveitis. Current evidence is limited to case reports and uncontrolled case series. In the largest series of infliximab for uveitis published to date, Suhler and colleagues (2005) reported that at 10 weeks' follow-up, 18 of 23 patients were considered successfully treated. Those successfully treated for 1 year fell to 7 of 14 eligible patients, with 5 not completing the 1 year of treatment because of significant adverse effects. An editorial by Robert Nussenblatt of the National Eye Institute (Nussenblatt, 2005) that accompanied this report noted, however, that only 4 of 23 patients demonstrated an improvement in their visual acuity of 2 lines or better. "Perhaps the most striking part of the report is the litany of adverse effects associated with the administration of this medication" (Nussenblatt, 2005).

Commenting on the available literature on infliximab for uveitis, Nussenblatt (2005) stated: "More than 60 publications have appeared in the literature to date describing the effects of infliximab in the treatment of uveitis. This large number reflects a serious problem that needs to be addressed in this field. While small, often single, case reports are important in generating hypotheses, these studies seem to produce not hypotheses but merely more of the same case reports. Further, the reports use criteria that are not standardized, so comparing one study to another is difficult to do. This is the case for the present study as well."

Well-designed clinical studies are necessary for assessing the effectiveness and safety of infliximab in uveitis. It should also be noted that another TNF inhibitor, etanercept, showed apparently positive results in the treatment of uveitis in case reports and uncontrolled case series, but subsequently published controlled clinical trials demonstrated no significant benefit (Foster et al, 2003; Smith et al, 2005).

Kahn et al (2006) reported their experience in using infliximab for the treatment of childhood uveitis. A total of 17 children (3 males, 14 females) with chronic uveitis were administered high-dose infliximab (10 to 20 mg/kg/dose). Main outcome measure was the ability to eliminate all signs of intra-ocular inflammation. All 17 patients showed a dramatic, rapid response, with no observed inflammation in 13 patients after the second infusion, and 4 patients requiring 3 to 7 infusions to achieve disease quiescence. Additional immunosuppressives and topical glucocorticoids were tapered when patients achieved no intra-ocular inflammation. The authors noted that in this series, high-dose infliximab was a rapidly effective, well-tolerated therapeutic agent for the treatment of chronic, medically refractory, non-infectious uveitis. Moreover, they stated that larger, randomized, controlled studies will provide a better understanding of the dose, interval, and duration of treatment needed and will provide data on long-term safety.

A phase II clinical trial by Suhler and associates (2009) reported the 2-year follow-up data of patients with refractory uveitis treated with intravenous infliximab as part of a prospective clinical trial. Their 1-year data, published in 2005 (Suhler, 2005) reported reasonable initial success, but an unexpectedly high incidence of adverse events. Of their 23 patients, 7 developed serious adverse events, including 3 thromboses,

1 malignancy, 1 new onset of congestive heart failure, and 2 cases of drug-induced lupus. Of patients who received at least 3 injections of infliximab, 75 % also developed elevated antinuclear antibody titers, although the significance of this is unknown. The protocol initially permitted 1 year of therapy, but was extended to 2 years midway through the study. Longer-term follow-up data were also collected for patients who continued with infliximab therapy after study completion. Of the 31 patients who received infliximab in this study, approximately 75 % had an initial favorable response, and the drug was effective in select patients for 2 to 4 years. Of 23 patients who demonstrated initial success at 10 weeks, 15 completed 1 year in the study and 8 completed 2 years of therapy, 7 in the study and 1 outside the study protocol prior to its extension to 2 years. Three patients developed a drug-related lupus-like illness. Two developed fatal solid malignancies, which were considered by the authors to have an unclear relation to the drug. There were no further cases of congestive heart failure or venous thrombosis. Unfortunately, 2 of the 7 patients who completed 2 years of the study were ultimately lost to follow-up, and another 2 patients who completed the 1-year protocol were also lost to follow-up. It may be, in fact, that the risk of long-term adverse effects is even higher than reported in this article. The study by Suhler is difficult to interpret because of the small size, lack of comparison group, and substantial loss to follow-up.

In an accompanying editorial of the afore-mentioned article, Goldstein (2009) noted that Suhler et al (2009) are to be congratulated for adding to the literature on the use of TNF inhibitors in uveitis. The editorialist stated that, while the trial is small and is neither randomized nor controlled, the data are collected in a prospective fashion, with at least 2 years of follow-up. Although infliximab was not universally successful at controlling inflammation in this series, it was effective in selected patients, with 60 % of patients retained in the study per year. This series also demonstrated that while infliximab is well-tolerated in terms of immediate treatment-limiting adverse effects, significant late toxicity may occur. Furthermore, Goldstein noted that the rate of adverse events reported by Suhler et al (2009) is much higher than reported in retrospective uveitis series. Because this is a prospective trial, its results may be more clinically relevant than those from retrospective series and may offer the clinician a

more accurate estimate of the risks associated with the use of infliximab in patients with uveitis. Clinical trials of infliximab for uveitis are currently ongoing.

Mirshahi et al (2012) stated that TNF-α plays an important role both in inflammation and apoptosis. In the eye, TNF-α appears to have a role in the pathogenesis of inflammatory, edematous, neovascular and neurodegenerative disorders. Several TNF-blocking drugs have been developed and approved, and are in clinical use for inflammatory diseases such as rheumatoid arthritis, psoriasis and ankylosing spondylitis. Tumor necrosis factor-α blockers are widely used in ophthalmology as an off-label alternative to "traditional" immunosuppressive and immune-modulatory treatments in non-infectious uveitis. Preliminary studies suggested a positive effect of intravenously administered TNF-α blockers, mainly infliximab, for treating refractory diabetic macular edema and neovascular age-related macular degeneration. Unfortunately, much of the current data raises considerable safety concerns for intra-vitreal use of TNF-α inhibitors, in particular, intra-ocular inflammatory responses have been reported after intra-vitreal injection of infliximab. Results of dose-finding studies and humanized antibody or antibody fragments (e.g., adalimumab) are anticipated in the coming years; these will shed light on potential benefits and risks of local and systemic TNF-α blockers used for treatment of diseases of the retina and choroid.

Cordero-Coma et al (2013) established evidence-based recommendations regarding the use TNF-α agents for managing uveitis patients. Medline was searched via OVID (1950 - October Week 3, 2011) using a Cochrane highly sensitive search (phases 1 and 2). Additional literature searches were also conducted including the following databases: the Cochrane, LILACS and the TRIP Database. A total of 54 studies met all of the inclusion criteria and were included in this review. A different level of recommendation and evidence is assigned to each anti-TNF-α agent. The overall rate of reported side effects with anti-TNF-α agents for the treatment of uveitis that required discontinuation of therapy was 2.2 % (26/1,147 patients). The authors concluded that based on the evidence gathered, infliximab and adalimumab seem to be effective in the management of immune-mediated uveitis. Moreover, they stated that further randomized studies evaluating the efficacy of these agents are

warranted. It is the most common cause of inflammatory eye disease, with an estimated prevalence of 115 cases per 100,000 persons. Endogenous or associated with a systemic disease, non-infectious uveitis accounts for approximately 75 % of total cases comprising of a heterogeneous group of inflammatory conditions responsible for about 10 % of legal blindness in developed nations. Endogenous uveitides are thought to have an autoimmune component mediated by T lymphocytes specific to intraocular antigens that have failed to successfully pass basic processes designed to maintain self-tolerance.

A systematic evidence review of tumor necrosis factor inhibitors in childhood chronic uveitis found that most of the studies were case series; the only randomized controlled trial (RCT) (of Enbrel) found no significant benefit (Simonini et al, 2013). Most of the evidence relates to uveitis association with a rheumatologic condition that would qualify persons for anti-TNF treatment; among the 229 children included in this systematic evidence review, only about 60 had idiopathic uveitis not associated with another rheumatologic condition. Simonini et al (2013) summarized the evidence regarding the effectiveness of anti-TNFα treatments in childhood autoimmune chronic uveitis (ACU), refractory to previous DMARDs. A systematic search between January 2000 and October 2012 was conducted using Embase, Ovid Medline, Evidence Based Medicine Reviews-ACP Journal Club, Cochrane libraries, and EBM Reviews. Studies investigating the efficacy of anti-TNFα therapy, in children (less than or equal to 16 yrs), as the first biologic treatment for ACU, refractory to topical and/or systemic steroid therapy and at least 1 DMARD, were eligible for inclusion. The primary outcome measure was the improvement of intraocular inflammation, as defined by the SUN working group criteria. The authors determined a combined estimate of the proportion of children responding to anti-TNFα: etanercept (ETA), infliximab (INF), or ADA. The authors initially identified 989 articles, of which 148 were potentially eligible; 22 retrospective chart reviews, and 1 RCT, were deemed eligible, thus including 229 children (ADA n =31; ETA n = 54 and INF n = 144). On pooled analysis of observational studies, the proportion of responding children was 87 % (95 % CI: 75 to 98 %) for ADA, 72 % (64 to 79 %) for INF, and 33 % (95 % CI: 19 to 47 %) for ETA. There was no difference in the proportion of responders between ADA and INF ($\chi(2) = 3.06$, $p = 0.08$), although both showed superior efficacy compared to ETA (ADA versus ETA $\chi(2) = 20.9$, $p < 0.001$; INF vs ETA

$\chi(2) = 20.9$, $p < 0.001$). The authors concluded that, although randomized controlled trials are needed, the available evidence suggests that INF and ADA provide proven similar benefits in the treatment of childhood ACU, and they are both superior to ETA.

A review by Sanchez-Cano (2013) found that evidence for tumor necrosis factor inhibitors in noninfectious uveitis is primarily case reports and case series. The authors observed that TNF α plays a central role in both the inflammatory response and that of the immune system. Thus, its blockade with the so-called anti-TNF agents (infliximab, etanercept, adalimumab, certolizumab pegol, and golimumab) has turned into the most important tool in the management of a variety of disorders, such as rheumatoid arthritis, spondyloarthropathies, inflammatory bowel disease, and psoriasis. The authors said that, theoretically, some other autoimmune disorders may benefit from these agents. The authors reviewed off-label uses of anti-TNF blockers in three common conditions: Behçet's disease, sarcoidosis, and noninfectious uveitis. The authors concluded that, due to the insufficient number of adequate clinical trials and consequently to their lower prevalence compared to other immune disorders, this review is mainly based on case reports and case series.

An UpToDate review on “Uveitis: Treatment” (Rosenbaum, 2013) states that “The role of tumor necrosis factor-alpha (TNF) inhibitors such as infliximab, adalimumab, or etanercept in the management of patients with uveitis is uncertain. Further data from controlled trials are needed to clarify the indications and risks of TNF inhibition in the treatment of ocular inflammatory disease”.

A consensus panel organized by the American Uveitis Society (Levy-Clarke, et al., 2014) provided recommendations for the use of TNF-α biologic agents in patients with ocular inflammatory disorders. The authors performed a systematic review of literature to generate guidelines for use of these agents in ocular inflammatory conditions. Recommendations were generated using the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) group criteria. The panel found that numerous studies including controlled clinical trials have demonstrated that anti-TNF-α biologic agents (in particular infliximab and adalimumab) are effective in the treatment of severe ocular inflammatory disease. Based on these studies, the panel

made the following recommendations. Infliximab and adalimumab can be considered as first-line immunomodulatory agents for the treatment of ocular manifestations of Behçet's disease. Infliximab and adalimumab can be considered as second-line immunomodulatory agents for the treatment of uveitis associated with juvenile arthritis. Infliximab and adalimumab can be considered as potential second-line immunomodulatory agents for the treatment of severe ocular inflammatory conditions including posterior uveitis, panuveitis, severe uveitis associated with seronegative spondyloarthropathy, and scleritis in patients requiring immunomodulation in patients who have failed or who are not candidates for antimetabolite or calcineurin inhibitor immunomodulation. Infliximab and adalimumab can be considered in these patients in preference to etanercept, which seems to be associated with lower rates of treatment success.

Furthermore, an UpToDate review on “Uveitis: Treatment” (Rosenbaum, 2019) states that “Non-infectious uveitis -- The approach to therapy is generally not affected by the etiology of the uveitis, but there are important exceptions. As examples, Behcet syndrome is especially responsive to infliximab … The response of uveitis associated with Behcet syndrome to infliximab has been encouraging in observational studies. For example, in an open-label observational study of 43 patients, 33 patients responded to conventional therapy (oral glucocorticoids, cyclosporine, azathioprine, or methotrexate). Ten patients who failed to respond were subsequently treated with infliximab (5 mg/kg) every 2 weeks for a total of 6 doses. Patients treated with infliximab had fewer relapses and longer remissions than patients who were treated conventionally. Infliximab has been approved in Japan to treat Behcet syndrome, which is a relatively common cause of uveitis in that country. Several reports suggest that infliximab may be more effective than etanercept for the treatment of uveitis. Etanercept, adalimumab, certolizumab, or infliximab appear to reduce the risk of developing eye inflammation in patients with ankylosing spondylitis, a disease frequently associated with acute anterior uveitis”.

Appendix

Appendix A: Examples of Clinical Reasons to Avoid Pharmacologic Treatment with Methotrexate, Hydroxychloroquine, Cyclosporine, Acitretin, or Leflunomide

I. Clinical diagnosis of alcohol use disorder, alcoholic liver disease or other chronic liver disease
II. Drug interaction
III. Risk of treatment-related toxicity
IV. Pregnancy or currently planning pregnancy
V. Breastfeeding
VI. Significant comorbidity prohibits use of systemic agents (e.g., liver or kidney disease, blood dyscrasias, uncontrolled hypertension)
VII. Hypersensitivity
VIII. History of intolerance or adverse event

Appendix B: RA Classification and Disease Activity

RA Classification

Rheumatoid arthritis (RA) classification criteria include a scoring system for the following four domains: joint distribution (0-5), serology (0-3), symptom duration (0-1), and acute phase reactants (0-1). A score of 6 or more (out of possible 10) equals definite RA; however, a person might fulfill the criteria prospectively over time (cumulatively), or retrospectively if data on all four domains have been adequately recorded in the past. The criteria are meant to be used for persons with clinical synovitis in at least one joint. The classification are not diagnostic criteria, but they can inform the diagnosis. For more information see, American College of Rheumatology (ACR) - Endorsed Criteria (https://www.rheumatology.org/Practice-Quality/Clinical-Support/Criteria).

Source: 2010 ACR/EULAR

Disease Activity

Disease activity is categorized as low, moderate, or high as per validated scales. Moderate and high disease activity categories have been combined (Singh et al., 2016). Structured assessments of disease activity which include multiple composite measures have been developed for RA. Eleven measures have been identified by the American College of Rheumatology (ACR) as having utility in clinical practice to accurately reflect disease activity. The choice of measure is based upon clinician preference; some measures require both patient and clinician input, while others are based only upon patient-reported data (England et al., 2019; Moreland and Cannella, 2020).

To assist clinicians in the care of persons with RA, the ACR recommend eleven measures that fulfilled minimum standard for RA disease activity. These measures include: Clinical Disease Activity Disease Index (CDAI), Disease Activity Score (DAS), Disease Activity Score 28 Joints (DAS28-ESR/CRP), Patient Derived DAS28, Hospital Universitario La Princesa Index (HUPI), Multi-Biomarker Disease Activity Score (MBDA score, VECTRA DA), Rheumatoid Arthritis Disease Activity Index (RADAI), Rheumatoid Arthritis Disease Activity Index 5 (RADAI-5), Routine Assessment of Patient Index Data 3 (RAPID3), Routine Assessment of Patient Index Data 5 (RAPID5), and Simplified Disease Activity Index (SDAI). For more information see 2019 Updated ACR Recommended RA Disease Activity Measures (https://www.rheumatology.org/Portals/0/Files/2019-Update-ACR-Recommended-Rheumatoid-Arthritis-Disease-Activity-Measures.pdf).

Source: 2019 Update of the American College of Rheumatology Recommended RA Disease Activity Measures

Appendix C: Hurley Staging System for Hidradenitis Suppurativa

Table: Hurley Staging System for Hidradenitis Suppurativa

Stage	Characteristics
I	Solitary or multiple isolated abscess formation without scarring or sinus tracts. (A few minor sites with rare inflammation; may be mistaken for acne.)

II	Recurrent abscesses, single or multiple widely separated lesions, with sinus tract formation. (Frequent inflammation restrict movement and may require minor surgery such as incision and drainage.)
III	Diffuse or broad involvement across a regional area with multiple interconnected sinus tracts and abscesses. (Inflammation of sites to the size of golf balls, or sometimes baseballs; scarring develops, including subcutaneous tracts of infection – see fistula. Obviously, patients at this stage may be unable to function.)

Source: Wieczorek and Walecka (2018)

Appendix D: Brands of Targeted Immune Modulators and FDA-approved Indications

Table: Brands of Targeted Immune Modulators and FDA-approved Indications (not an all-inclusive list)

Brand Name	Generic Name	FDA Labeled Indications
Actemra	tocilizumab	Coronavirus Disease 2019 (COVID-19) in hospitalized patients Cytokine release syndrome (CRS) Giant cell arteritis Juvenile idiopathic arthritis Rheumatoid arthritis Systemic juvenile idiopathic arthritis Systemic sclerosis-associated interstitial lung disease (SSc-ILD)
Arcalyst	rilonacept	Cryopyrin-associated periodic syndromes Deficiency of interleukin-1 receptor antagonist (DIRA) Recurrent pericarditis

Cimzia	certolizumab	Ankylosing spondylitis or non-radiographic axial spondyloarthritis Crohn's disease Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis
Cosentyx	secukinumab	Ankylosing spondylitis or non-radiographic axial spondyloarthritis Enthesitis-related arthritis Hidradenitis suppurativa Plaque psoriasis Psoriatic arthritis
Enbrel	etanercept	Ankylosing spondylitis Juvenile idiopathic arthritis Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis
Entyvio	vedolizumab	Crohn's disease Ulcerative colitis
Humira (for Humira biosimilars, see CPB 0655 - Adalimumab (../600_699/0655.html))	adalimumab	Ankylosing spondylitis Crohn's disease Hidradenitis suppurativa Juvenile idiopathic arthritis Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis Uveitis
Ilaris	canakinumab	Adult-onset Still's disease Gout flares Periodic fever syndromes Systemic juvenile idiopathic arthritis
Ilumya	tildrakizumab-asmn	Plaque psoriasis

Kevzara	sarilumab	Juvenile idiopathic arthritis Polymyalgia rheumatica Rheumatoid arthritis
Kineret	anakinra	Cryopyrin-associated periodic syndromes Deficiency of interleukin-1 receptor antagonist (DIRA) Rheumatoid arthritis
Olumiant	baricitinib	Alopecia areata COVID-19 in hospitalized adults Rheumatoid arthritis
Orencia	abatacept	Acute graft versus host disease Juvenile idiopathic arthritis Psoriatic arthritis Rheumatoid arthritis
Otezla	apremilast	Oral ulcers associated with Behçet's Disease Plaque psoriasis Psoriatic arthritis
Remicade (for Remicade biosimilars, see CPB 0341 - Infliximab (../300_399/0341.html))	infliximab	Ankylosing spondylitis Crohn's disease Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis
Rinvoq	upadacitinib	Ankylosing spondylitis or non-radiographic axial spondyloarthritis Atopic dermatitis Crohn's disease Juvenile idiopathic arthritis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis

Rituxan (for Rituxan biosimilars, see CPB 0314 - Rituximab (../300_399/0314.html))	rituximab	Chronic lymphocytic leukemia Granulomatosis with polyangiitis Microscopic polyangiitis Pemphigus vulgaris Rheumatoid arthritis Various subtypes of non-Hodgkin's lymphoma
Siliq	brodalumab	Plaque psoriasis
Simponi	golimumab	Ankylosing spondylitis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis
Simponi Aria	golimumab intravenous	Ankylosing spondylitis Juvenile idiopathic arthritis Psoriatic arthritis Rheumatoid arthritis
Skyrizi	risankizumab-rzaa	Crohn's disease Plaque psoriasis Psoriatic arthritis Ulcerative colitis
Stelara	ustekinumab	Crohn's disease Plaque psoriasis Psoriatic arthritis Ulcerative colitis
Taltz	ixekinumab	Ankylosing spondylitis or non-radiographic axial spondyloarthritis Plaque psoriasis Psoriatic arthritis
Tremfya	guselkumab	Plaque psoriasis Psoriatic arthritis Ulcerative colitis
Tysabri	natalizumab	Crohn's disease Multiple sclerosis

Xeljanz	tofacitinib	Ankylosing Spondylitis Polyarticular Course Juvenile Idiopathic Arthritis Psoriatic arthritis Rheumatoid arthritis Ulcerative Colitis
Xeljanz XR	tofacitinib, extended release	Ankylosing Spondylitis Polyarticular Course Juvenile Idiopathic Arthritis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis

References

The above policy is based on the following references:

1. Abu El-Asrar AM, Abboud EB, Aldibhi H, Al-Arfaj A. Long-term safety and efficacy of infliximab therapy in refractory uveitis due to Behcet's disease. Int Ophthalmol. 2005;26(3):83-92.
2. Adams DR, Yankura JA, Fogelberg AC, Anderson BE. Treatment of hidradenitis suppurativa with etanercept injection. Arch Dermatol. 2010;146:501-504.
3. Adedokun OJ, Sandborn WJ, Feagan BG, et al. Association between serum concentration of infliximab and efficacy in adult patients with ulcerative colitis. Gastroenterology. 2014;147(6):1296-1307.
4. Adler BL, Wang CJ, Bui TL, et al. Anti-tumor necrosis factor agents in sarcoidosis: A systematic review of efficacy and safety. Semin Arthritis Rheum. 2019;48(6):1093-1104.
5. Aeberli D, Oertle S, Mauron H, et al. Inhibition of the TNF-pathway: Use of infliximab and etanercept as remission-inducing agents in cases of therapy-resistant chronic inflammatory disorders. Swiss Med Wkly. 2002;132(29-30):414-422.

6. Agarwal A, Andrews JM. Systematic review: IBD-associated pyoderma gangrenosum in the biologic era, the response to therapy. Aliment Pharmacol Ther. 2013;38(6):563-572.
7. Agarwal A. Retinal vasculitis. American Academy of Ophthalmology (AAO) EyeWiki. San Francisco, CA: AAO; November 17, 2014. Available at: http://eyewiki.aao.org/Retinal_Vasculitis. Accessed February 8, 2019.
8. Akobeng AK, Zachos M. Tumor necrosis factor-alpha antibody for induction of remission in Crohn's disease. Cochrane Database Syst Rev. 2003;(4):CD003574.
9. Akpek EK, Lindsley KB, Adyanthaya RS, Treatment of Sjögren's syndrome-associated dry eye an evidence-based review. Ophthalmology. 2011;118(7):1242-1252.
10. Aletaha D, Neogi T, Silman, et al. 2010 Rheumatoid arthritis classification criteria: an American College of Rheumatology/European League Against Rheumatism collaborative initiative. Arthritis Rheum. 2010;62(9):2569-81.
11. Alghafeer IS, Sigal LH. Rheumatic manifestations of gastrointestinal diseases. Bull Rheum Dis. 2002:51(2):1.
12. Alikhan A, Sayed C, Alavi A, et al. North American clinical management guidelines for hidradenitis suppurativa: A publication from the United States and Canadian Hidradenitis Suppurativa Foundations Part I: Diagnosis, evaluation, and the use of complementary and procedural management. J Am Acad Dermatol. 2019; 81(1): 76-90.
13. Alikhan A, Sayed C, Alavi A, et al. North American clinical management guidelines for hidradenitis suppurativa: A publication from the United States and Canadian Hidradenitis Suppurativa Foundations Part II: Topical, intralesional, and systemic medical management. J Am Acad Dermatol. 2019; 81(1): 91-101.
14. American College of Rheumatology (ACR). Criteria: ACR-endorced criteria: Rheumatoid arthritis classification. Atlanta, GA: ACR; 2020. Available at: https://www.rheumatology.org/Practice-Quality/Clinical-Support/Criteria. Accessed August 26, 2020.
15. Amgen. Avsola (infliximab-axxq) for injection, for intravenous use. Prescribing Information. Thousand Oaks, CA: Amgen; revised December 2019a, September 2025.

16. Amgen. FDA approves Amgen's Avsola (infliximab-axxq), for the same indications as Remicade (infliximab). Press Release. Thousand Oaks, CA: Amgen; December 6, 2019b.
17. Amgen. Study to assess if ABP710 is safe & effective in treating moderate to severe rheumatoid arthritis compared to infliximab. ClinicalTrials.gov Identifier: NCT02937701. Bethesda, MD: National Library of Medicine; updated August 28, 2019c.
18. Andres PG, Friedman LS. Epidemiology and the natural course of inflammatory bowel disease. Gastroenterol Clin N Am. 1999;28(2):255-281.
19. Angeles-Han ST, Ringold S, Beukelman T, et al. 2019 American College of Rheumatology/Arthritis Foundation guideline for the screening, monitoring, and treatment of juvenile idiopathic arthritis-associated uveitis. Arthritis Care Res. 2019; 71(6):703-716.
20. Antoni C, Dechant C, Hanns-Martin Lorenz PD, et al. Open-label study of infliximab treatment for psoriatic arthritis: Clinical and magnetic resonance imaging measurements of reduction of inflammation. Arthritis Rheum. 2002;47(5):506-512.
21. Antoni C, Kalden JR. Combination therapy of the chimeric monoclonal anti-tumor necrosis factor alpha antibody (infliximab) with methotrexate in patients with rheumatoid arthritis. Clin Exp Rheumatol. 1999;17(6 Suppl 18):S73-S77.
22. Antoni CE, Kavanaugh A, Kirkham B, et al. Sustained benefits of infliximab therapy for dermatologic and articular manifestations of psoriatic arthritis: Results from the infliximab multinational psoriatic arthritis controlled trial (IMPACT). Arthritis Rheum. 2005;52(4):1227-1236.
23. Arguelles-Arias F, Castro-Laria L, Lobaton T, et al. Characteristics and treatment of pyoderma gangrenosum in inflammatory bowel disease. Dig Dis Sci. 2013;58(10):2949-2954.
24. Armuzzi A, De Pascalis B, Lupascu A, et al. Infliximab in the treatment of steroid-dependent ulcerative colitis. Eur Rev Med Pharmacol Sci. 2004;8(5):231-233.
25. Arroyo JG. Age-related macular degeneration: Treatment and prevention. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed February 2013.
26. Artornsombudh P, Gevorgyan O, Payal A, et al. Infliximab treatment of patients with birdshot retinochoroidopathy.

Ophthalmology. 2013;120(3):588-592.
27. Assasi N, Blackhouse G, Xie F, et al. Anti-TNF-a drugs for refractory inflammatory bowel disease: Clinical- and cost-effectiveness analyses. Technology Report No. 120. Ottawa, ON: Canadian Agency for Drugs and Technologies in Health (CADTH); 2009.
28. Atkins MB. Overview of the treatment of renal cell carcinoma. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed February 2019.
29. Baert F, Noman M, Vermeire S, et al. Influence of immunogenicity on the long-term efficacy of infliximab in Crohn's disease. N Engl J Med. 2003;348(7):601-608.
30. Baeten D, Kruithof E, Van den Bosch F, et al. Immunomodulatory effects of anti-tumor necrosis factor alpha therapy on synovium in spondylarthropathy: Histologic findings in eight patients from an open-label pilot study. Arthritis Rheum. 2001;44(1):186-195.
31. Basoulis D, Fragiadaki K, Tentolouris N, et al. Anti-TNFα treatment for recalcitrant ulcerative necrobiosis lipoidica diabeticorum: A case report and review of the literature. Metabolism. 2016;65(4):569-573.
32. Baughman RP, Judson MA, Costabel U, et al. Randomized, double-blind, placebo-controlled trial of infliximab in patients with chronic pulmonary sarcoidosis. Chest. 2005;128(4):202S.
33. Baughman RP, Lower EE. Infliximab for refractory sarcoidosis. Sarcoidosis Vasc Diffuse Lung Dis. 2001;18:70-74.
34. Beck KE, Blansfield JA, Tran KQ, et al. Enterocolitis in patients with cancer after antibody blockade of cytotoxic T-lymphocyte-associated antigen 4. J Clin Oncol. 2006;24(15):2283-2289.
35. Behm BW, Bickston SJ. Tumor necrosis factor-alpha antibody for maintenace of remission in Crohn's disease. Cochrane Database Syst Rev. 2008;(1):CD006893.
36. Belloni B, Andres C, Ollert M, et al. Novel immunological approaches in the treatment of atopic eczema. Curr Opin Allergy Clin Immunol. 2008;8(5):423-427.
37. Bernstein CN, Blanchard JF, Kliewer E, Wajda A. Cancer risk in patients with inflammatory bowel disease: A population-based study. Cancer. 2001;91(4):854-862.
38. Beukelman T, Patkar NM, Saag KG, et al. 2011 American College of Rheumatology recommendations for the treatment of juvenile

idiopathic arthritis: initiation and safety monitoring of therapeutic agents for the treatment of arthritis and systemic features. Arthritis Care Res. 2011;63(4):465-482.
39. Bitoun S, Hässler S, Ternant D, et al. Response to biologic drugs in patients with rheumatoid arthritis and antidrug antibodies. JAMA Netw Open. 2023;6(7):e2323098.
40. Blam ME, Stein RB, Lichtenstein GR. Integrating anti-tumor necrosis factor therapy in inflammatory bowel disease: Current and future perspectives. Am J Gastroenterol. 2001;96(7):1977-1997.
41. Blumenauer B, Burls A, Cranney A, et al. Infliximab for the treatment of rheumatoid arthritis. Cochrane Database Syst Rev. 2002;(3):CD003785.
42. Bongartz T, Sutton AJ, Sweeting MJ, et al. Anti-TNF antibody therapy in rheumatoid arthritis and the risk of serious infections and malignancies: Systematic review and meta-analysis of rare harmful effects in randomized controlled trials. JAMA. 2006;295:2275-2285.
43. Borhani Haghighi A, Safari A, Nazarinia MA, et al. Infliximab for patients with neuro-Behcet's disease: Case series and literature review. Clin Rheumatol. 2011;30(7):1007-1012.
44. Boudreau R, Blackhouse G, Goeree R, Mierzwinski-Urban M. Adalimumab, alefacept, efalizumab, etanercept, and infliximab for severe psoriasis vulgaris in adults: Budget impact analysis and review of comparative clinical- and cost-effectiveness. Technology Report No. 97. Ottawa, ON: Canadian Agency for Drugs and Technologies in Health (CADTH); 2007.
45. Brady B, Siebert U, Sroczynski G, et al. Long-term clinical and cost-effectiveness of infliximab and etanercept for rheumatoid arthritis. Technology Overview No. 29. Ottawa, ON: Canadian Agency for Drugs and Technologies in Health (CADTH); 2007.
46. Brandt J, Haibel H, Reddig J, et al. Successful short term treatment of severe undifferentiated spondyloarthropathy with the anti-tumor necrosis factor-alpha monoclonal antibody infliximab. J Rheumatol. 2002;29(1):118-122.
47. Brandt J, Haibel H, Sieper J, et al. Infliximab treatment of severe ankylosing spondylitis: One-year followup. Arthritis Rheum. 2001;44(12):2936-2937.

48. Braun J, Brandt J, Listing J, et al. Treatment of active ankylosing spondylitis with infliximab: A randomized controlled multicentre trial. Lancet. 2002;359(9313):1187-1193.
49. Braun J, van den Berg R, Baraliakos X, et al. 2010 update of the ASAS/EULAR recommendations for the management of ankylosing spondylitis. Ann Rheum Dis 2011;70:896–904.
50. Breban M, Vignon E, Claudepierre P, et al. Efficacy of infliximab in refractory ankylosing spondylitis: Results of a six-month open-label study. Rheumatology (Oxford). 2002;41(11):1280-1285.
51. Brightling C, Berry M, Amrani Y. Targeting TNF-alpha: A novel therapeutic approach for asthma. J Allergy Clin Immunol. 2008;121(1):5-10; quiz 11-12.
52. Bristol-Myers Squibb Co. (BMS). Yervoy (ipilimumab) injection, for intravenous infusion. Prescribing Information. Princeton, NJ: BMS; revised September 2019.
53. Bristol-Myers Squibb Co. (BMS). Yervoy (Ipilimumab) Immune-mediated Adverse Reaction Management Guide. 731US11REMS00201. Princeton, NJ: BMS; March 2011.
54. Brooklyn TN, Dunnill MG, Shetty A, et al. Infliximab for the treatment of pyoderma gangrenosum: A randomised, double blind, placebo controlled trial. Gut. 2006;55(4):505-509.
55. Brown A, Spry C. Infliximab versus adalimumab for patients with moderate to severe ulcerative colitis: A clinical and cost effectiveness review. Health Technology Inquiry Service (HTIS). Ottawa, ON: Canadian Agency for Drugs and Technologies in Health (CADTH); March 3, 2008.
56. Canadian Agency for Drugs and Technologies in Health (CADTH). Infliximab (Remicade, Centocor, Inc.). Indication: Ulcerative colitis. CEDAC Final Recommendation. Common Drug Review. Ottawa, ON: CADTH; April 22, 2009.
57. Canadian Coordinating Office for Health Technology Assessment (CCOHTA). Infliximab (Remicade) for the treatment of ankylosing spondylitis. Emerging Drug List Issue 54. Ottawa, ON: CCOHTA; 2004.
58. Casteele NV, Herfarth H, Katz J, et al. American Gastroenterological Association Institute technical review on the role of therapeutic drug monitoring in the management of inflammatory bowel disease. Gastroenterology. 2017;153(3):835-857.

59. Cauza E, Spak M, Cauza K, et al. Treatment of psoriatic arthritis and psoriasis vulgaris with the tumor necrosis factor inhibitor infliximab. Rheumatol Int. 2002;22(6):227-232.
60. Centers for Disease Control and Prevention (CDC). Quality ID #176: Tuberculosis screening prior to first course of biologic and/or immune response modifier therapy. Atlanta, GA: CDC; 2024. Available at: https://qpp.cms.gov. Accessed September 13, 2024.
61. Centers for Disease Control and Prevention (CDC). Testing for tuberculosis [website]. Atlanta, GA: CDC; June 17, 2024. Available at: https://www.cdc.gov/tb/testing/index.html. Accessed September 26, 2025.
62. Centers for Disease Control and Prevention (CDC). Tuberculosis (TB): Clinical and laboratory diagnosis for tuberculosis [website]. Atlanta, GA: CDC; January 31, 2025. Available at: https://www.cdc.gov/tb/hcp/testing-diagnosis/clinical-and-laboratory-diagnosis.html. Accessed September 26, 2025.
63. Celltrion USA, Inc. Zymfentra (infliximab-dyyb) injection, for subcutaneous use. Prescribing Information. Jersey City, NJ: Celltrion USA; revised February 2024.
64. Centocor, Inc. Remicade (infliximab) for IV injection. STN: BL 103772/5138 - Pediatric Crohn's Final Draft Labeling. Malvern, PA: Centocor; revised May 17, 2006.
65. Chaudhari U, Romano P, Mulcahy LD, et al. Efficacy and safety of infliximab monotherapy for plaque-type psoriasis: A randomised trial. Lancet. 2001;357(9271):1842-1847.
66. Cheema RA, Al-Askar E, Cheema HR. Infliximab therapy for idiopathic retinal vasculitis, aneurysm, and neuroretinitis syndrome. J Ocul Pharmacol Ther. 2011;27(4):407-410.
67. Chen Y-F, Jobanputra P, Barton P, et al. A systematic review of the effectiveness of adalimumab, etanercept and infliximab for the treatment of rheumatoid arthritis in adults and an economic evaluation of their cost-effectiveness. Health Technol Assess. 2006;10(42):1-248.
68. Cheng R, Cooper A, Kench J, et al. Ipilimumab-induced toxicities and the gastroenterologist. J Gastroenterol Hepatol. 2015;30(4):657-666.
69. Chey WY, Hussain A, Ryan C, et al. Infliximab for refractory ulcerative colitis. Am J Gastroenterol. 2001;96(8):2373-2381.

70. Chey WY. Infliximab for patients with refractory ulcerative colitis. Inflamm Bowel Dis. 2001;7 Suppl 1:S30-S33.
71. Clark W, Raftery J, Song F, et al. Systematic review and economic evaluation of the effectiveness of infliximab for the treatment of Crohn's disease. Health Technol Assess. 2003;7(3):1-67.
72. Coates LC, Soriano ER, Corp N, et al. Group for Research and Assessment of Psoriasis and Psoriatic Arthritis (GRAPPA): Updated treatment recommendations for psoriatic arthritis 2021. Nat Rev Rheumatol. 2022;18(8):465-479.
73. Cooper C, Shafran S, Greenbloom S, et al. Single-dose infliximab in hepatitis C genotype 1 treatment-naive patients with high serum tumour necrosis factor-alpha does not influence the efficacy of pegylated interferon alpha-2b/ribavirin therapy. Can J Gastroenterol. 2014;28(1):35-40.
74. Cordero-Coma M, Yilmaz T, Onal S. Systematic review of anti-tumor necrosis factor-alpha therapy for treatment of immune-mediated uveitis. Ocul Immunol Inflamm. 2013;21(1):12-20.
75. Couriel DR, Saliba R, de Lima M, et al. A phase III study of infliximab and corticosteroids for the initial treatment of acute graft-versus-host disease. Biol Blood Marrow Transplant. 2009;15(12):1555-1562.
76. Coyle D, Judd M, Blumenauer B, et al. Infliximab and etanercept in patients with rheumatoid arthritis: A systematic review and economic evaluation. Technology Report Issue 64. Ottawa, ON: Canadian Coordinating Office for Health Technology Assessment (CCOHTA); 2006.
77. D'Haens G, Van Deventer S, Van Hogezand R et al. Endoscopic and histological healing with infliximab anti-tumor necrosis factor antibodies in Crohn's disease: A European multicenter trial. Gastroenterology. 1999;116:1029-1034.
78. Dastmalchi M, Grundtman C, Alexanderson H, et al. A high incidence of disease flares in an open pilot study of infliximab in patients with refractory inflammatory myopathies. Ann Rheum Dis. 2008;67(12):1670-1677.
79. De Keyser F, Baeten D, Van den Bosch F, et al. Gut inflammation and spondyloarthropathies. Curr Rheumatol Rep. 2002;4(6):525-532.
80. de Vries C. Effects of TNF-alpha antagonists in people with rheumatoid arthritis. STEER: Succint and Timely Evaluated

Evidence Reviews. Bazian, Ltd., eds. London, UK: Wessex Institute for Health Research and Development, University of Southampton; 2001;1(19).

81. Deutschmann A, Mache CJ, Bodo K, et al. Successful treatment of chronic recurrent multifocal osteomyelitis with tumor necrosis factor-alpha blockage. Pediatrics. 2005;116(5):1231-1233.
82. D'haens G, Van Deventer S, Van Hogezand R, et al. Endoscopic and histological healing with infliximab anti-tumor necrosis factor antibodies in Crohn's disease: A European multicenter trial. Gastroenterology. 1999;116(5):1029-1034.
83. D'Haens G. Anti-TNF treatment in Crohn's disease: toward tailored therapy? Am J Gastroenterol. 2010;105(5):1140-1141.
84. Diab M, Coloe JR, Magro C, et al. Treatment of recalcitrant generalized morphea with infliximab. Arch Dermatol. 2010;146(6):601-604.
85. Doherty CB, Rosen T. Evidence-based therapy for cutaneous sarcoidosis. Drugs. 2008;68(10):1361-13 83.
86. Doherty G, Bennett G, Patil S, et al. Interventions for prevention of post-operative recurrence of Crohn's disease. Cochrane Database Syst Rev. 2009;(4):CD006873.
87. Drent M, Cremers JP, Jansen TL, Baughman RP. Practical eminence and experience-based recommendations for use of TNF-α inhibitors in sarcoidosis. Sarcoidosis Vasc Diffuse Lung Dis. 2014;31(2):91-107.
88. Dretzke J, Edliln R, Round J, et al. A systematic review and economic evaluation of the use of tumour necrosis factor-alpha (TNF-a) inhibitors, adalimumab and infliximab, for Crohn's disease. Health Technol Assess. 2011;15(6):1-250.
89. Eleftheriou D, Melo M, Marks SD, et al. Biologic therapy in primary systemic vasculitis of the young. Rheumatology (Oxford). 2009;48(8):978-986.
90. Elmets C, Korman N, et al. Guidelines of care for the management and treatment of psoriasis with topical therapy and alternative medicine modalities for psoriasis severity measures. J Am Acad Dermatol. 2021;84:432-470
91. El-Shabrawi Y, Hermann J. Anti-tumor necrosis factor-alpha therapy with infliximab as an alternative to corticosteroids in the treatment of human leukocyte antigen B27-associated acute anterior uveitis. Ophthalmology. 2002;109(12):2342-2346.

92. Emery P. Infliximab: A new treatment for rheumatoid arthritis. Hosp Med. 2001;62(3):150-152.
93. England BR, Tiong BK, Bergman MJ, et al. 2019 update of the American College of Rheumatology recommended rheumatoid arthritis disease activity measures. Arthrit Care Res. 2019;71(12):1540-1555.
94. Esteve M, Mahadevan U, Sainz E, et al. Efficacy of anti-TNF therapies in refractory severe microscopic colitis. J Crohns Colitis. 2011;5(6):612-618.
95. Eun LY. Infliximab, is it really a new horizon for the treatment of Kawasaki disease? Korean Circ J. 2019;49(2):192-193.
96. Falappone PC, Iannone F, Scioscia C, et al. [The treatment of recurrent uveitis with TNF-alpha inhibitors.] Reumatismo. 2004 Jul-Sep;56(3):185-9.
97. Farrell RJ, Peppercorn MA. Overview of the medical management of mild to moderate Crohn disease in adults. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed January 2017.
98. Feldman SR, Gordon KB, Bala, M. Infliximab treatment results in significant improvement in the quality of life of patients with severe psoriasis: A double-blind placebo-controlled trial. Br J Dermatol. 2005;152:954-960.
99. Ferguson ID, Weiser P, Torok KS, et al. A case report of successful treatment of recalcitrant childhood localized scleroderma with infliximab and leflunomide. Open Rheumatol J. 2015;9:30-35.
100. Feuerstein J, Ho E, Shmidt E, et al. AGA Clinical Practice Guidelines on the medical management of moderate to severe luminal and perianal fistulizing Crohn's disease. Gastroenterology. 2021;160:2496-2508.
101. Feuerstein JD, Isaacs KL, Schneider Y, et al. AGA Clinical Practice Guidelines on the management of moderate to severe ulcerative colitis. Gastroenterology. 2020;158:1450.
102. Feuerstein JD, Nguyen GC, Kupfer SS, Falck-Ytter Y, Singh S; American Gastroenterological Association Institute Clinical Guidelines Committee. American Gastroenterological Association Institute Guideline on Therapeutic Drug Monitoring in Inflammatory Bowel Disease. Gastroenterology. 2017;153(3):827-834.

103. Filocamo G, Buoncompagni A, Viola S, et al. Treatment of Takayasu's arteritis with tumor necrosis factor antagonists. J Pediatr. 2008;153(3):432-434.
104. Flores D, Marquez J, Garza M, Espinoza LR. Reactive arthritis: Newer developments. Rheum Dis Clin North Am. 2003;29(1):37-59, vi.
105. Foster CS, Tufail F, Waheed NK, Chu D, Efficacy of etanercept in preventing relapse of uveitis controlled by methotrexate. Arch Ophthalmol. 2003;121(4):437-440.
106. Fraenkel L, Bathon JM, England BR, et al. 2021 American College of Rheumatology guideline for the treatment of rheumatoid arthritis. Arthrit Care Res. 2021;0:1-16.
107. Furst DE, Keystone E, Maini RN, Smolen JS. Recapitulation of the round-table discussion -- assessing the role of anti-tumor necrosis factor therapy in the treatment of rheumatoid arthritis. Rheumatology (Oxford). 1999;38(Suppl 2):50-53.
108. Galor A, Perez VL, Hammel JP, Lowder CY. Differential effectiveness of etanercept and infliximab in the treatment of ocular inflammation. Ophthalmology. 2006;113(12):2317-2323.
109. Garnett WR, Yunker N. Treatment of Crohn's disease with infliximab. Am J Health Syst Pharm. 2001;58(4):307-316.
110. Gartlehner G, Hansen RA, Jonas BL, et al. Biologics for the treatment of juvenile idiopathic arthritis: A systematic review and critical analysis of the evidence. Clin Rheumatol. 2008;27(1):67-76.
111. Gelfand JM, Bradshaw MJ, Stern BJ, et al. Infliximab for the treatment of CNS sarcoidosis: A multi-institutional series. Neurology. 2017;89(20):2092-2100.
112. George, C, Deroide, F, Rustin, M. Pyoderma gangrenosum – a guide to diagnosis and management. Clin Med. 2019;19(3): 224-8.
113. Gladman DD, Antoni C, P Mease, et al. Psoriatic arthritis: epidemiology, clinical features, course, and outcome. Ann Rheum Dis. 2005;64(Suppl II):ii14–ii17.
114. Goldstein DA. Uncovering the risks of immunosuppressive therapy in patients with uveitis. Arch Ophthalmol. 2009;127(6):799-800.
115. Gossec L, Smolen JS, Ramiro S, et al. European League Against Rheumatism (EULAR) recommendations for the management of

psoriatic arthritis with pharmacological therapies; 2015 update. Ann Rheum Dis. 2016;75(3):499-510.

116. Gottlieb A, Korman NJ, Gordon KB, et al. Guidelines of care for the management of psoriasis and psoriatic arthritis: Section 2. Psoriatic arthritis: Overview and guidelines of care for treatment with an emphasis on the biologics. J Am Acad Dermatol. 2008;58(5):851-864.
117. Gottlieb AB, Chaudhari U, Mulcahy LD, et al. Infliximab monotherapy provides rapid and sustained benefit for plaque-type psoriasis. J Am Acad Dermatol. 2003;48(6):829-835.
118. Gottlieb AB, Evans R, Li Shu, et al. REMICADE induction therapy for patients with severe plaque-type psoriasis: A randomized, double-blind, placebo-controlled trial. J Am Acad Dermatol. 2004;51(4):534-542.
119. Gottlieb AB, Masud S, Ramamurthi R, et al. Pharmacodynamic and pharmacokinetic response to anti-tumor necrosis factor-alpha monoclonal antibody (infliximab) treatment of moderate to severe psoriasis vulgaris. J Am Acad Dermatol. 2003;48(1):68-75.
120. Graber E. Treatment of acne vulgaris. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed February 2019.
121. Grant A, Gonzalez T, Montgomery MO, et al. Infliximab therapy for patients with moderate to severe hidradenitis suppurativa: A randomized, double-blind, placebo-controlled crossover trial. J Am Acad Dermatol. 2010;62:205-217.
122. Greenberg SM. Cerebral amyloid angiopathy. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed August 2022.
123. Gul A. Standard and novel therapeutic approaches to Behçet's disease. Drugs. 2007;67(14):2013-2022.
124. Han CL, Zhao SL. Intravenous immunoglobulin gamma (IVIG) versus IVIG plus infliximab in young children with Kawasaki disease. Med Sci Monit. 2018;24:7264-7270.
125. Hanauer S. Safety of infliximab in clinical trials. Aliment Pharmacol Ther. 1999;13(Suppl 4):16-22.
126. Hanauer SB. Review article: Safety of infliximab in clinical trials. Aliment Pharmacol Ther. 1999;13(Suppl 4):16-22, discussion 38.
127. Harriman G, Harper LK, Schaible TF. Summary of clinical trials in rheumatoid arthritis using infliximab, an anti-TNFalpha treatment. Ann Rheum Dis. 1999;58(Suppl 1):I61-I64.

128. Harrison ML, Obermueller E, Maisey NR, et al. Tumor necrosis factor alpha as a new target for renal cell carcinoma: Two sequential phase II trials of infliximab at standard and high dose. J Clin Oncol. 2007;25(29):4542-4549.
129. Haslund P, Lee RA, Jemec GB. Treatment of hidradenitis suppurativa with tumour necrosis factor-alpha inhibitors. Acta Derm Venereol. 2009;89(6):595-600.
130. Hatemi G, Christensen R, Bang D, et al. 2018 update of the EULAR recommendations for the management of Behçet's syndrome. Ann Rheum Dis. 2018;77(6):808-818.
131. Health Technology Board for Scotland (HTBS). The use of infliximab for Crohn's disease. Glasgow, Scotland: HTBS; 2002.
132. Health Technology Inquiry Service (HTIS). Infliximab for maintenance therapy for treatment of Crohn's disease: A review of the clinical effectiveness and guidelines. Ottawa, ON: Canadian Agency for Drugs and Technologies in Health (CADTH); May 18, 2010.
133. Heidelberger V, Ingen-Housz-Oro S, Marquet A, et al. Efficacy and tolerance of anti-tumor necrosis factor α agents in cutaneous sarcoidosis: A French study of 46 cases. JAMA Dermatol. 2017;153(7):681-685.
134. Heller T, James SP, Drachenberg C, et al. Treatment of severe esophageal Crohn's disease with infliximab. Inflamm Bowel Dis. 1999;5(4):279-282.
135. Heneghan MA. Autoimmune hepatitis variants: Definitions and treatment. UpToDate [online serial], Waltham, MA: UpToDate; reviewed February 2018.
136. Hengstman GJ, De Bleecker JL, Feist E, et al. Open-label trial of anti-TNF-alpha in dermato- and polymyositis treated concomitantly with methotrexate. Eur Neurol. 2008;59(3-4):159-163.
137. Herfarth H, Obermeier F, Andus T, et al. Improvement of arthritis and arthralgia after treatment with infliximab (Remicade) in a German prospective, open-label, multicenter trial in refractory Crohn's disease. Am J Gastroenterol. 2002;97(10):2688-2690.
138. Herndon TM, Ausin C, Brahme NN, et al. Safety outcomes when switching between biosimilars and reference biologics: A systematic review and meta-analysis. PLoS One. 2023;18(10):e0292231.

139. Hillock NT, Heard S, Kichenadasse G, et al. Infliximab for ipilimumab-induced colitis: A series of 13 patients. Asia Pac J Clin Oncol. 2017;13(5):e284-e290.
140. Hinshaw A, Cares K, Thomas R, El-Baba M. Efficacy of infliximab biosimilar for maintenance therapy in pediatric inflammatory bowel disease following infliximab originator. JPGN Rep. 2022;3(4):e256.
141. Hochberg MC, Tracy JK, Hawkins-Holt M, Flores RH. Comparison of the efficacy of the tumour necrosis factor alpha blocking agents adalimumab, etanercept, and infliximab when added to methotrexate in patients with active rheumatoid arthritis. Ann Rheum Dis. 2003;62(Suppl 2):13-16.
142. Hommes DW, Oldenburg B, van Bodegraven AA, et al.; Dutch Initiative on Crohn and Colitis (ICC). Guidelines for treatment with infliximab for Crohn's disease. Neth J Med. 2006;64(7):219-229.
143. Huang H, Xu S, Huang F, et al. A meta-analysis of efficacy and safety of infliximab for prevention of postoperative recurrence in patients with Crohn's disease. Biomed Res Int. 2018;2018:2615978.
144. Hur G, Song MS, Sohn S, et al. Infliximab treatment for intravenous immunoglobulin-resistant Kawasaki disease: A multicenter study in Korea. Korean Circ J. 2019;49(2):183-191.
145. Ianiro G, Cammarota G, Valerio L, et al. Microscopic colitis. World J Gastroenterol. 2012;18(43):6206-6215.
146. Iqbal M, Kolodney MS. Acne fulminans with synovitis-acne-pustulosis-hyperostosis-osteitis (SAPHO) syndrome treated with infliximab. J Am Acad Dermatol. 2005;52(5 Suppl 1):S118-S120.
147. Jang DI, Lee AH, Shin HY, et al. The role of tumor necrosis factor alpha (TNF-α) in autoimmune disease and current TNF-α inhibitors in therapeutics. Int J Mol Sci. 2021;22(5):2719.
148. Janssen Biotech, Inc. Infliximab for injection, for intravenous use. Prescribing Information. Horsham, PA: Janssen Biotech; revised February 2025.
149. Janssen Biotech, Inc.. Remicade (infliximab) lyophilized concentrate for injection, for intravenous use. Prescribing Information. Horsham, PA: Janssen Biotech; revised February 2025.

150. Jemec G. Hidradinitis suppurativa. N Engl J Med. 2012;366(2):158-164.
151. Jessop S, Whitelaw DA, Delamere FM. Drugs for discoid lupus erythematosus. Cochrane Database Syst Rev. 2009;(4):CD002954.
152. Jobanputra P, Barton P, Bryan S, Burls A. The effectiveness of infliximab and etanercept for the treatment of rheumatoid arthritis: A systematic review and economic evaluation. Health Technol Assess. 2002;6(21): 1-110.
153. Johnston RL, Lutzky J, Chodhry A, Barkin JS. Cytotoxic T-lymphocyte-associated antigen 4 antibody-induced colitis and its management with infliximab. Dig Dis Sci. 2009;54(11):2538-2540.
154. Jones G, Halbert J, Crotty M, Shanahan EM, et al. The effect of treatment on radiological progression in rheumatoid arthritis: A systematic review of randomized placebo-controlled trials. Rheumatology. 2003;42(1):6-13.
155. Jones RE, Moreland LW. Tumor necrosis factor inhibitors for rheumatoid arthritis. Bull Rheum Dis. 1999;48(3):1-4.
156. Jordan J, Konstantinou K, O'Dowd. Herniated lumbar disc. In: BMJ Clinical Evidence. London, UK: BMJ Publishing Group; July 2008.
157. Joseph A, Raj D, Dua HS, et al. Infliximab in the treatment of refractory posterior uveitis. Ophthalmology. 2003;110(7):1449-1453.
158. Kabbaha S, Milano A, Aldeyab MA, Thorlund K. Infliximab as a second-line therapy for children with refractory Kawasaki disease: A systematic review and meta-analysis of randomized controlled trials. Br J Clin Pharmacol. 2023;89(1):49-60.
159. Kahn P, Weiss M, Imundo LF, Levy DM. Favorable response to high-dose infliximab for refractory childhood uveitis. Ophthalmology. 2006;113(5):860-864.
160. Kaiser MJ, Sany J. Efficacy of infliximab (Remicade) in the treatment of spondyloarthropathies. Two case reports. Joint Bone Spine. 2001;68(6):525-527.
161. Karim F, Paridaens D, Westenberg LE, et al. Infliximab for IgG4-related orbital disease. Ophthal Plast Reconstr Surg. 2017;33(3S Suppl 1):S162-S165.
162. Kemper AR, Coeytaux R, Sanders GD, et al. Disease-modifying antirheumatic drugs (DMARDs) in children with juvenille idiopathic arthritis (JIA). Comparative Effectiveness Review No.

28. Rockville, MD: Agency for Healthcare Research and Quality; 2011.
163. Keystone EC. The utility of tumour necrosis factor blockade in orphan diseases. Ann Rheum Dis. 2004;63 Suppl 2:ii79-ii83.
164. Kimura Y. Systemic juvenile idiopathic arthritis: Treatment. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed December 2020.
165. Kobayashi K, Ueno F, Bito S, et al. Development of consensus statements for the diagnosis and management of intestinal Behçet's disease using a modified Delphi approach. J Gastroenterol. 2007;42(9):737-745.
166. Kobelt G, Sobocki P, Sieper J, Braun J. Comparison of the cost-effectiveness of infliximab in the treatment of ankylosing spondylitis in the United Kingdom based on two different clinical trials. Int J Technol Assess Health Care. 2007;23(3):368-375.
167. Kokkinos A, Iliopoulos A, Greka P, et al. Successful treatment of refractory adult-onset Still's disease with infliximab. A prospective, non-comparative series of four patients. Clin Rheumatol. 2004;23(1):45-49.
168. Korhonen T, Karppinen J, Paimela L, et al. The treatment of disc herniation-induced sciatica with infliximab: One-year follow-up results of FIRST II, a randomized controlled trial. Spine. 2006;31(24):2759-2766.
169. Kraetsch HG, Antoni C, Kalden JR, Manger B. Successful treatment of a small cohort of patients with adult onset of Still's disease with infliximab: First experiences. Ann Rheum Dis. 2001;60 Suppl 3:iii55-iii57.
170. Kremer JM. Rational use of new and existing disease-modifying agents in rheumatoid arthritis. Ann Intern Med. 2001;134(8):695-706.
171. Krogstad P. Hematogenous osteomyelitis in children: Management. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed January 2016.
172. Kruh JN, Yang P, Suelves AM, Foster CS. Infliximab for the treatment of refractory noninfectious uveitis: A study of 88 patients with long-term follow-up. Ophthalmology. 2014;121(1):358-364.
173. Kruithof E, Van den Bosch F, Baeten D, et al. Repeated infusions of infliximab, a chimeric anti-TNFalpha monoclonal antibody, in

patients with active spondyloarthropathy: One year follow up. Ann Rheum Dis 2002 Mar;61(3):207-122.

174. Lanthier N, Parc C, Scavennec R, et al. Infliximab in the treatment of posterior uveitis in Behcet's disease. Long term follow up in four patients. Presse Med. 2005;34(13):916-918.
175. Larkin JM, Ferguson TR, Pickering LM, et al. A phase I/II trial of sorafenib and infliximab in advanced renal cell carcinoma. Br J Cancer. 2010;103(8):1149-1153.
176. Lateef A, Petri M. Biologics in the treatment of systemic lupus erythematosus. Curr Opin Rheumatol. 2010;22(5):504-509.
177. Lawson MM, Thomas AG, Akobeng AK. Tumour necrosis factor alpha blocking agents for induction of remission in ulcerative colitis. Cochrane Database Syst Rev. 2006;(3):CD005112.
178. Lebwohl M. Psoriasis. Lancet. 2003;361(9364):1197-1204.
179. Lee G, Ashfaq A. Tubulointerstitial nephritis and uveitis (TINU syndrome). UpToDate [online serial]. Waltham, MA: UpToDate; reviewed February 2013.
180. Levy-Clarke G, Jabs DA, Read RW, et al. Expert panel recommendations for the use of anti-tumor necrosis factor biologic agents in patients with ocular inflammatory disorders. Ophthalmology. 2014;121(3):785-796.
181. Lichtenstein GR, Hanauer SB, Sandborn WJ, et al. Management of Crohn’s disease in adults. Am J Gastroenterol. 2009;104:465-483.
182. Lichtenstein GR, Loftus EV, Afzali A, et al. ACG Clinical Guideline: Management of Crohn's disease in adults. Am J Gastroenterol. 2025;120(6):1225-1264.
183. Lichtenstein GR, Loftus Jr EV, Isaacs KI, et al. ACG Clinical Guideline: Management of Crohn’s Disease in Adults. Am J Gastroenterol. 2018;113:481-517.
184. Lichtenstein GR. Treatment of Crohn disease in adults: Dosing and monitoring of tumor necrosis factor-alpha inhibitors. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed October 2022.
185. Lindstedt EW, Baarsma GS, Kuijpers RW, van Hagen PM. Anti-TNF-alpha therapy for sight threatening uveitis. Br J Ophthalmol. 2005;89(5):533-536.
186. Lipsky PE, Van Der Heijde D, St. Clair EW et al. Infliximab and methotrexate in the treatment of rheumatoid arthritis. N Engl J Med. 2000;343:1594-1602.

187. Lofberg R. Treatment of fistulas in Crohn's disease with infliximab. Gut. 1999;45(5):642-643.
188. Lord JD, Hackman RC, Moklebust A, et al. Refractory colitis following anti-CTLA4 antibody therapy: Analysis of mucosal FOXP3+ T cells. Dig Dis Sci. 2010;55(5):1396-1405.
189. Lu D, Song H, Shi G. Anti-TNF-α treatment for pelvic pain associated with endometriosis. Cochrane Database Syst Rev. 2010;(3):CD008088.
190. Maini R, St Clair EW, Breedveld F, et al. Infliximab (chimeric anti-tumor necrosis factor alpha monoclonal antibody) versus placebo in rheumatoid arthritis patients receiving concomitant methotrexate: A randomized phase III trial. ATTRACT Study Group. Lancet. 1999;354(9194):1932-1939.
191. Maini RN, Breedveld FC, Kalden JR, et al. Sustained improvement over two years in physical function, structural damage, and signs and symptoms among patients with rheumatoid arthritis treated with infliximab and methotrexate. Arthritis Rheum. 2004;50:1051-1065.
192. Maksymowych WP, Jhangri GS, Lambert RG, et al. Infliximab in ankylosing spondylitis: A prospective observational inception cohort analysis of efficacy and safety. J Rheumatol. 2002;29(5):959-965.
193. Malottki K, Barton P, Tsourapas A, et al. Adalimumab, etanercept, infliximab, rituximab and abatacept for the treatment of rheumatoid arthritis after the failure of a TNF inhibitor: A systematic review and economic evaluation. Health Technol Assess. 2011;15(14):1-278.
194. Mariette X, Ravaud P, Steinfeld S, et al. Inefficacy of infliximab in primary Sjogren's syndrome: Results of the randomized, controlled Trial of Remicade in Primary Sjogren's Syndrome (TRIPSS). Arthritis Rheum. 2004;50(4):1270-1276.
195. Marshall J, Blackhouse G, Goeree R, et al. Infliximab for the treatment of Crohn's disease: a systematic review and cost-utility analysis. Technology Report No. 24. Ottawa, ON: Canadian Coordinating Office for Health Technology Assessment (CCOHTA); 2002.
196. Marshall JK, Blackhouse G, Goeree R, et al. Clinical and economic assessment: Infliximab for the treatment of Crohn's disease. Technology Overview Issue 8. Ottawa, ON: Canadian

Coordinating Office for Health Technology Assessment (CCOHTA); 2002.
197. Marteau P. Inflammatory bowel disease. Endoscopy. 2000;32(2):131-137.
198. Marzano AV, Ishak RS, Saibeni S, et al. Autoinflammatory skin disorders in inflammatory bowel diseases, pyoderma gangrenosum and Sweet's syndrome: A comprehensive review and disease classification criteria. Clin Rev Allergy Immunol. 2013;45(2):202-210.
199. Merative L.P. Infliximab. In-Depth Answers. Merative Micromedex. Ann Arbor, MI: Merative; 2025. Available at: www.micromedexsolutions.com. Accessed August 15, 2025.
200. McLeod C,1 Bagust A, Boland A, et al. Adalimumab, etanercept and infliximab for the treatment of ankylosing spondylitis: A systematic review and economic evaluation. Health Technol Assess. 2007;11(28):1-158.
201. Meador R, Hsia E, Kitumnuaypong T, Schumacher HR. TNF involvement and anti-TNF therapy of reactive and unclassified arthritis. Clin Exp Rheumatol. 2002;20(6 Suppl 28):S130-S134.
202. Mease PJ. Cytokine blockers in psoriatic arthritis. Ann Rheum Dis. 2001;60 Suppl 3:iii37-40.
203. Mease PJ. Disease-modifying antirheumatic drug therapy for spondyloarthropathies: Advances in treatment. Curr Opin Rheumatol. 2003;15(3):205-212.
204. Mehta, NA, Emani-Naeini P. A review of systemic biologics and local immunosuppressive medications in uveitis. J Ophthalmic Vis Res. 2022;17(2):276-289.
205. Menter A, Cordero KM, Davis DM, et al. Joint AAD-NPF guidelines of care for the management and treatment of psoriasis in pediatric patients. J Am Acad Dermatol. 2020;82(1):161-201.
206. Menter A, Gelfand JM, Connor C, et al. Joint AAD-NPF guidelines of care for the management of psoriasis with systemic nonbiologic therapies. J Am Acad Dermatol. 2020;82(6): 1445-86.
207. Menter A, Gottlieb A, Feldman SR, et al Guidelines of care for the management of psoriasis and psoriatic arthritis: Section 1. Overview of psoriasis and guidelines of care for the treatment of psoriasis with biologics. J Am Acad Dermatol. 2008;58(5):826-850.
208. Menter A, Korman NJ, Elmets CA, et al. Guidelines of care for the management of psoriasis and psoriatic arthritis. Section 6:

Guidelines of care for the treatment of psoriasis and psoriatic arthritis: case-based presentations and evidence-based conclusions. J Am Acad Dermatol. 2011;65(1):137-174.
209. Menter A, Strober BE, Kaplan DH, et al. Joint AAD-NPF guidelines of care for the management and treatment of psoriasis with biologics. J Am Acad Dermatol. 2019;80(4):1029-1072.
210. Merck Sharpe & Dohme Corp. Renflexis (infliximab-abda) for injection, for intravenous use. Prescribing Information. Kenilworth, NJ: Merck; issued April 2017.
211. Merrill SP, Reynolds P, Kalra A, et al. Early administration of infliximab for severe ipilimumab-related diarrhea in a critically ill patient. Ann Pharmacother. 2014;48(6):806-810.
212. Michet CJ. Treatment of relapsing polychondritis. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed February 2013.
213. Miller I, Lynggaard CD, Lophaven S, et al. A double-blind placebo-controlled randomized trial of adalimumab in the treatment of hidradenitis suppurativa. Br J Dermatol. 2011;165:391-398.
214. Miller ML, Rudnicki SA. Treatment of recurrent and resistant dermatomyositis and polymyositis in adults. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed January 2012.
215. Mills SC, von Roon AC, Orchard TR, Tekkis PP. Crohn's disease. In: BMJ Clinical Evidence. London, UK: BMJ Publishing Group; December 2009.
216. Minor DR, Chin K, Kashani-Sabet M. Infliximab in the treatment of anti-CTLA4 antibody (ipilimumab) induced immune-related colitis. Cancer Biother Radiopharm. 2009;24(3):321-325.
217. Mirshahi A, Hoehn R, Lorenz K, et al. Anti-tumor necrosis factor alpha for retinal diseases: Current knowledge and future concepts. J Ophthalmic Vis Res. 2012;7(1):39-44.
218. Molloy ES, Langford CA, Clark TM, et al. Anti-tumour necrosis factor therapy in patients with refractory Takayasu arteritis: Long-term follow-up. Ann Rheum Dis. 2008 Nov;67(11):1567-1569.
219. Moravan M, Segal BM. Treatment of CNS sarcoidosis with infliximab and mycophenolate mofetil. Neurology. 2009;72(4):337-340.
220. Moreland LW, Cannella A. General principles of management of rheumatoid arthritis in adults. UpToDate [online serial].

Waltham, MA: UpToDate; reviewed January 2020.

221. Motllo C, Ferra C, Lopez L, et al. Salvage treatment with infliximab of steroid-resistant graft-versus-host disease in allogeneic transplanted patients. Med Clin (Barc).2011;137(3):115-118.
222. Moulton K, Farrah K. Infliximab for the treatment of nonfistulizing Crohn's disease: Guidelines and cost-effectiveness. Health Technology Inquiry Service (HTIS). Ottawa, ON: Canadian Agency for Drugs and Technologies in Health (CADTH); March 12, 2008.
223. Mouser JF, Hyams JS. Infliximab: A novel chimeric monoclonal antibody for the treatment of Crohn's disease. Clin Ther. 1999;21(6):932-942.
224. Naldi L, Rzany B. Psoriasis (chronic plaque). In: BMJ Clinical Evidence. London, UK: BMJ Publishing Group; August 2007.
225. National Comprehensive Cancer Network (NCCN). Infliximab. NCCN Drugs & Biologics Compendium. Plymouth Meeting, PA: NCCN; August 2025.
226. National Comprehensive Cancer Network (NCCN). Management of immunotherapy-related toxicities: Immune checkpoint inhibitor-related toxicities. NCCN Clinical Practice Guidelines in Oncology, Version 2.2023. Fort Washington, PA: NCCN; May 9, 2023.
227. National Comprehensive Cancer Network (NCCN). Kidney cancer. NCCN Clinical Practice Guidelines in Oncology, Version 1.2024. Fort Washington, PA: NCCN; June 21, 2023.
228. National Health Service (NHS), National Prescribing Centre. Infliximab for rheumatoid arthritis. New Medicines on the Market. UK Medicines Information (UKMi). Monograph No. 4/00/17. London, UK: NHS; December 2000.
229. National Health Service (NHS), National Prescribing Centre. Infliximab for psoriasis. New Medicines Profile. UK Medicines Information (UKMi). Monograph No. 4/00/17. London, UK: NHS; December 2005.
230. National Horizon Scanning Centre (NHSC). Infliximab (Remicade) for ulcerative colitis - horizon scanning review. Birmingham, UK: NHSC; 2005.
231. National Horizon Scanning Centre (NHSC). Infliximab (Remicade) for paediatric ulcerative colitis. Horizon Scanning

Review. Birmingham, UK: National Horizon Scanning Centre (NHSC); 2011.
232. National Horizon Scanning Centre (NHSC). Infliximab for ankylosing spondylitis - horizon scanning review. Birmingham, UK: NHSC; 2003.
233. National Horizon Scanning Centre (NHSC). Infliximab for psoriasis -- horizon scanning review. Birmingham, UK: NHSC; 2004.
234. National Institute for Clinical Excellence (NICE). Guidance on the use of etanercept and infliximab for the treatment of rheumatoid arthritis. Technology Appraisal Guidance 36. London, UK: NICE; 2002.
235. National Institute for Clinical Excellence (NICE). Guidance on the use of infliximab for Crohn's disease. Technology Appraisal Guidance 40. London, UK: NICE; 2002.
236. National Institute for Health and Clinical Excellence (NICE). Adalimumab, etanercept and infliximab for ankylosing spondylitis. Technology Appraisal Guidance No. 143. London, UK: National Institute for Health and Clinical Excellence (NICE); May 2008.
237. National Institute for Health and Clinical Excellence (NICE). Adalimumab, etanercept, infliximab, rituximab and abatacept for the treatment of rheumatoid arthritis after the failure of a TNF inhibitor. Technology Appraisal Guidance 195. London, UK: NICE; August 2010.
238. National Institute for Health and Clinical Excellence (NICE). Etanercept and infliximab for the treatment of psoriatic arthritis. Technology Appraisal Guidance 102. London, UK: NICE; 2006.
239. National Institute for Health and Clinical Excellence (NICE). Etanercept, infliximab and adalimumab for the treatment of psoriatic arthritis. Technology Appraisal Guidance 199. London, UK: NICE; August 2010.
240. National Institute for Health and Clinical Excellence (NICE). Infliximab for subacute manifestations of ulcerative colitis. Technology Appraisal Guidance No. 140. London, UK: National Institute for Health and Clinical Excellence (NICE); April 2008.
241. National Institute for Health and Clinical Excellence (NICE). Infliximab for acute manifestations of ulcerative colitis.

Technology Appraisal Guidance No. 163. London, UK: NICE; December 2008.

242. National Institute for Health and Clinical Excellence (NICE). Infliximab for the treatment of adults with psoriasis. Technology Appraisal Guidance No. 134. London, UK: National Institute for Health and Clinical Excellence (NICE); January 2008.
243. National Institute for Health and Clinical Excellence (NICE). Infliximab (review) and adalimumab for the treatment of Crohn's disease. Technology Appraisal Guidance 187. London, UK: NICE; May 2010.
244. Newby EA, Sawczenko A, Thomas AG, Wilson D. Interventions for growth failure in childhood Crohn's disease. Cochrane Database Syst Rev. 2005;(3):CD003873.
245. No authors listed. Infliximab (Remicade) for Crohn's disease. Med Lett Drugs Ther. 1999;41(1047):19-20.
246. Nussenblatt R. Treating intraocular inflammatory disease in the 21st century. Arch Ophthalmol. 2005;123(7):1000-1001.
247. Ochsenkuhn T, Sackmann M, Goke B. Infliximab for acute, not steroid-refractory ulcerative colitis: A randomized pilot study. Eur J Gastroenterol Hepatol. 2004;16(11):1167-1171.
248. Ogilvie AL, Antoni C, Dechant C, et al. Treatment of psoriatic arthritis with antitumour necrosis factor-alpha antibody clears skin lesions of psoriasis resistant to treatment with methotrexate. Br J Dermatol. 2001;144(3):587-589.
249. O'Halloran JA, Ko ER, Anstrom KJ, et al; ACTIV-1 IM Study Group Members. Abatacept, cenicriviroc, or infliximab for treatment of adults hospitalized with COVID-19 pneumonia: A randomized clinical trial. JAMA. 2023;330(4):328-339.
250. Oili KS, Niinisalo H, Korpilahde T, Virolainen J. Treatment of reactive arthritis with infliximab. Scand J Rheumatol. 2003;32(2):122-124.
251. Olivieri I, Padula A, Palazzi C. Pharmacological management of SAPHO syndrome. Expert Opin Investig Drugs. 2006;15(10):1229-1233.
252. Onel KB, Horton DB, Lovell DJ, et al. 2021 American College of Rheumatology guideline for the treatment of juvenile idiopathic arthritis: Therapeutic approaches for oligoarthritis, temporomandibular joint arthritis, and systemic juvenile idiopathic arthritis. Arthritis Rheumatol. 2022;74(4):553-569.

253. Ordas I, Mould DR, Feagan BG, Sandborn WJ. Anti-TNF monoclonal antibodies in inflammatory bowel disease: Pharmacokinetics-based dosing paradigms. Clin Pharmacol Ther. 2012;91(4):635-646.
254. Organon LLC. Renflexis (infliximab-abda) for injection, for intravenous use. Prescribing Information. Jersey City, NJ: Organon; revised December 2023.
255. Pagès C, Gornet JM, Monsel G, et al. Ipilimumab-induced acute severe colitis treated by infliximab. Melanoma Res. 2013;23(3):227-230.
256. Panaccione R, Fedorak RN, Aumais G, et al. Canadian Association of Gastroenterology Clinical Practice Guidelines: The use of infliximab in Crohn's disease. Can J Gastroenterol. 2004;18(8):503-508.
257. Panaccione R; Canadian Consensus Group on the Use of Infliximab in Crohn's Disease. Infliximab for the treatment of Crohn's disease; Review and indications for clinical use in Canada. Can J Gastroenterol 2001;15(6):371-375.
258. Peluso R, Lervolino S, Vitiello M, et al. Extra-articular manifestations in psoriatic arthritis patients. Clin Rheumatol. 2015;34(4):745-53.
259. Pfizer Inc. Inflectra (infliximab-dyyb) lyophilized concentrate for injection, for intravenous use. Prescribing Information. New York, NY: Pfizer; revised September 2025.
260. Phan GQ, Weber JS, Sondak VK. CTLA-4 blockade with monoclonal antibodies in patients with metastatic cancer: Surgical issues. Ann Surg Oncol. 2008;15(11):3014-3021.
261. Pichon Riviere A, Augustovski F, Alcaraz A, et al. Etanercept, infliximab and adalimumab for the treatment of rheumatoid arthritis [summary]. Report IRR No. 79. Buenos Aires, Argentina: Institute for Clinical Effectiveness and Health Policy (IECS); 2006.
262. Pidala J, Kim J, Field T, et al. Infliximab for managing steroid-refractory acute graft-versus-host disease. Biol Blood Marrow Transplant. 2009;15(9):1116-1121.
263. Pincus T, O'Dell JR, Kremer JM. Combination therapy with multiple disease-modifying antirheumatic drugs in rheumatoid arthritis: A preventive strategy. Ann Intern Med. 1999;131(10):768-774.

264. Pola S, Fahmy M, Evans E, et al. Successful use of infliximab in the treatment of corticosteroid dependent collagenous colitis. Am J Gastroenterol. 2013;108(5):857-858.
265. Present DH, Rutgeerts P, Targan S, et al. Infliximab for the treatment of fistulas in patients with Crohn's disease. N Engl J Med. 1999;340(18):1398-1405.
266. Present DH. Infliximab therapy for ulcerative colitis: Many unanswered questions. Am J Gastroenterol. 2001;96(8):2294-2296.
267. Present DH. Review article: The efficacy of infliximab in Crohn's disease -- healing of fistulae. Aliment Pharmacol Ther. 1999;13(Suppl 4):23-28, discussion 38.
268. Pritchard C, Nadarajah K. Tumour necrosis factor alpha inhibitor treatment for sarcoidosis refractory to conventional treatments: A report of five patients. Ann Rheum Dis. 2004;63:318-320.
269. Prometheus Laboratories. PredictrPK [website]. San Diego, CA: Prometheus Laboratories; 2024. Available at: https://prometheuslabs.com/about_predictrpk/. Accessed September 27, 2024.
270. Prystowsky S, Sanchez M. Management of cutaneous sarcoidosis. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed February 2019.
271. Rager A, Frey N, Goldstein SC, et al. Inflammatory cytokine inhibition with combination daclizumab and infliximab for steroid-refractory acute GVHD. Bone Marrow Transplant. 2011;46(3):430-435.
272. Rambhatla PV, Lim HW, Hamzavi I. A systematic review of treatments for hidradenitis suppurativa. Arch Dermatol. 2012;148(4):439-446.
273. Reich K, Nestle FO, Papp K, et al. Improvement in quality of life with infliximab induction and maintenance therapy in patients with moderate-to-severe psoriasis: A randomized controlled trial. Br J Dermatol. 2006;154(6):1161-1168.
274. Reich K, Nestle FO, Papp K, et al. Infliximab induction and maintenance therapy for moderate-to-severe psoriasis: A phase III, multicentre, double-blind trial. Lancet. 2005;366:1367-1374.
275. Rennard SI, Fogarty C, Kelsen S, et al. The safety and efficacy of infliximab in moderate to severe chronic obstructive pulmonary disease. Am J Respir Crit Care Med. 2007;175(9):926-934.

276. Ricart E, Panaccione R, Loftus EV, et al. Successful management of Crohn's disease of the ileoanal pouch with infliximab. Gastroenterology. 1999;117(2):429-432.
277. Ricart E, Sandborn WJ. Infliximab for the treatment of fistulas in patients with Crohn's disease. Gastroenterology. 1999;117(5):1247-1248.
278. Ringold S, Weiss PF, Beukelman T, et al. 2013 Update of the 2011 American College of Rheumatology Recommendations for the Treatment of Juvenile Idiopathic Arthritis: Recommendations for the Medical Therapy of Children With Systemic Juvenile Idiopathic Arthritis and Tuberculosis Screening Among Children Receiving Biologic Medications. Arthritis Rheum. 2013;65:2499-2512.
279. Rodgers M, Epstein D, Bojke L, et al. Etanercept, infliximab and adalimumab for the treatment of psoriatic arthritis: A systematic review and economic evaluation. Health Technol Assess. 2011;15(10):1-329.
280. Rojas-Feria M, Castro M, Suarez E, et al. Hepatobiliary manifestations in inflammatory bowel disease: The gut, the drugs and the liver. World J Gastroenterol. 2013;19(42):7327-40.
281. Rosenbaum JT. Uveitis: Treatment. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed January 2013.
282. Rosenbaum JT. Uveitis: Treatment. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed February 2019.
283. Ross WA, Couriel D. Colonic graft-versus-host disease. Curr Opin Gastroenterol. 2005;21(1):64-69.
284. Rovensky J, Sedlackova M. Relapsing polychondritis. Cas Lek Cesk. 2012;151(2):64-68.
285. Rubin DT, Ananthakrishnan AN, et al. 2019 ACG Clinical Guideline: Ulcerative colitis in adults. Am J Gastroentrol. 2019;114:384-413.
286. Rubin DT, Ananthakrishnan AN, Siegel CA, et al. ACG Clinical Guideline Update: Ulcerative colitis in adults. Am J Gastroenterol. 2025;120(6):1187-1224.
287. Ruperto N, Lovell DJ, Cuttica R, et al.; Paediatric Rheumatology INternational Trials Organisation; Pediatric Rheumatology Collaborative Study Group. A randomized, placebo-controlled trial of infliximab plus methotrexate for the treatment of

polyarticular-course juvenile rheumatoid arthritis. Arthritis Rheum. 2007;56(9):3096-3106.

288. Ruperto N, Lovell DJ, Cuttica R, et al.; Paediatric Rheumatology INternational Trials Organization (PRINTO); Pediatric Rheumatology Collaborative Study Group (PRCSG). Long-term efficacy and safety of infliximab plus methotrexate for the treatment of polyarticular-course juvenile rheumatoid arthritis: Findings from an open-label treatment extension. Ann Rheum Dis. 2010;69(4):718-722.
289. Rutgeerts P, Baert F. New strategies in the management of inflammatory bowel disease. Acta Clin Belg. 1999;54(5):274-280.
290. Rutgeerts P, D'Haens G, Targan S, et al. Efficacy and safety of retreatment with anti-tumor necrosis factor antibody (infliximab) to maintain remission in Crohn's disease. Gastroenterology. 1999;117(4):761-769.
291. Rutgeerts P, Feagan BGF, Olson A, et al. A randomized, placebo-controlled trial of infliximab therapy for active ulcerative colitis: ACT 1 Trial. Digestive Disease Week 2005, Chicago, IL, May 14-19, 2005. Presentation No. 689. Bethesda, MD: Digestive Disease Week; 2005.
292. Rutgeerts PJ, Targan SR. Introduction: Anti-TNF strategies in the treatment of Crohn's disease. Aliment Pharmacol Ther. 1999;13(Suppl 4):1.
293. Rutgeerts PJ. Review article: Efficacy of infliximab in Crohn's disease -- induction and maintenance of remission. Aliment Pharmacol Ther. 1999;13(Suppl 4):9-15, discussion 38.
294. Saag KG, Teng GG, Patkar NM, et al. American College of Rheumatology 2008 recommendations for the use of nonbiologic and biologic disease-modifying antirheumatic drugs in rheumatoid arthritis. Arthritis Rheum. 2008;59(6):762-784.
295. Samsung Bioepis Co., Ltd. Samsung Bioepis obtains first drug approval in the United States, as the U.S. Food and Drug Administration approves Renflexis (infliximab-abda) across all eligible indications. Press Release. Incheon, Korea; Samsung Bioepis; April 24, 2017.
296. Sánchez-Cano D, Callejas-Rubio JL, Ruiz-Villaverde R, et al. Off-label uses of anti-TNF therapy in three frequent disorders: Behçet's disease, sarcoidosis, and noninfectious uveitis. Mediators Inflamm. 2013;2013:286857.

297. Sandborn WJ, Hanauer SB. Antitumor necrosis factor therapy for inflammatory bowel disease: A review of agents, pharmacology, clinical results, and safety. Inflamm Bowel Dis. 1999;5(2):119-133.
298. Sandborn WJ, Hanauer SB. Infliximab in the treatment of Crohn's disease: A user's guide for clinicians. Am J Gastroenterol. 2002;97(12):2962-2972.
299. Sandborn WJ, Rachmilewitz D, Hanauer SB, et al. Infliximab induction and maintenance therapy for ulcerative colitis: The ACT 2 Trial. Digestive Disease Week 2005, Chicago, IL, May 14-19, 2005. Presentation No. 688. Bethesda, MD: Digestive Disease Week; 2005.
300. Sands BE, Anderson FH, Bernstein CN et al. A randomized controlled trial of infliximab maintenance therapy for fistulizing Crohn’s disease (ACCENT II). N Engl J Med. 2004;350:876-885.
301. Scheinfeld N. Off-label uses and side effects of infliximab. J Drugs Dermatol. 2004;3(3):273-284.
302. Schimmelpennink MC, Vorselaars ADM, van Beek FT, et al. Efficacy and safety of infliximab biosimilar Inflectra® in severe sarcoidosis. Respir Med. 2018;138S:S7-S13.
303. Schmidt E, Hunzelmann N, Zillikens D, et al. Rituximab in refractory autoimmune bullous diseases. Clin Exp Dermatol. 2006;31(4):503-508
304. Schmitt J, Schäkel K, Schmitt N, Meurer M. Systemic treatment of severe atopic eczema: A systematic review. Acta Derm Venereol. 2007;87(2):100-111.
305. Schopf RE, Aust H, Knop J. Treatment of psoriasis with the chimeric monoclonal antibody against tumor necrosis factor alpha, infliximab. J Am Acad Dermatol. 2002;46(6):886-891.
306. Scottish Intercollegiate Guidelines Network. Diagnosis and management of psoriasis and psoriatic arthritis in adults. SIGN Guideline 121. Edinburgh, Scotland: NHS Quality Improvement Scotland (NHS QIS); October 2010.
307. Serrano MS, Schmidt-Sommerfeld E, Kilbaugh TJ, et al. Use of infliximab in pediatric patients with inflammatory bowel disease. Ann Pharmacother. 2001;35(7-8):823-828.
308. Sharma SM, Ramanan AV, Riley P, Dick AD. Use of infliximab in juvenile onset rheumatological disease associated refractory uveitis: Efficacy in joint and ocular disease. Ann Rheum Dis. 2007;66(6):840-841.

309. Shen EH, Das KM. Current therapeutic recommendations: Infliximab for ulcerative colitis. J Clin Gastroenterol. 2004;38(9):741-745.
310. Shuja F, Chan CS, Rosen T. Biologic drugs for the treatment of hidradenitis suppurativa: An evidence-based review. Dermatol Clin. 2010;28(3):511-521, 523-524.
311. Simonini G, Druce K, Cimaz R, et al. Current evidence of anti-tumor necrosis factor α treatment efficacy in childhood chronic uveitis: A systematic review and meta-analysis approach of individual drugs. Arthritis Care Res (Hoboken). 2014l;66(7):1073-1084.
312. Singh JA, Christensen R, Wells GA, et al. Biologics for rheumatoid arthritis: An overview of Cochrane reviews. Cochrane Database Syst Rev, 2009;(4):CD007848.
313. Singh JA, Guyatt G, Ogdie A, et al. 2018 American College of Rheumatology/National Psoriasis Foundation guideline for the treatment of psoriatic arthritis. Arthritis Rheumatol. 2019;71(1):5-32.
314. Singh JA, Saag KG, Bridges SL Jr, et al. 2015 American College of Rheumatology guideline for the treatment of rheumatoid arthritis. Arthritis Rheum. 2016;68(1):1-26.
315. Slangen RM, van den Eertwegh AJ, van Bodegraven AA, de Boer NKh. Diarrhoea in a patient with metastatic melanoma: Ipilimumab ileocolitis treated with infliximab. World J Gastrointest Pharmacol Ther. 2013;4(3):80-82.
316. Smith CH, Anstey AV, Barker J, et al. British Association of Dermatologists guidelines for use of biological interventions in psoriasis 2005. Br J Dermatol. 2005;153:486-497.
317. Smith JA, Thompson DJ, Whitcup SM, et al. A randomized, placebo-controlled, double-masked clinical trial of etanercept for the treatment of uveitis associated with juvenile idiopathic arthritis. Arthritis Rheum. 2005;53(1):18-23.
318. Smolen JS, Aletaha D. Assessment of rheumatoid arthritis activity in clinical trials and clinical practice. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed November 2019.
319. Smolen JS, Landewé R, Billsma J, et al. EULAR recommendations for the management of rheumatoid arthritis with synthetic and biological disease-modifying antirheumatic drugs: 2016 update. Ann Rheum Dis. 2017;0:1-18.

320. Smolen JS, Landewé RBM, Bergstra SA, et al. EULAR recommendations for the management of rheumatoid arthritis with synthetic and biological disease-modifying antirheumatic drugs: 2022 update [published correction appears in Ann Rheum Dis. 2023;82(3):e76].
321. Sodhi M, Pearson K, White ES, Culver DA. Infliximab therapy rescues cyclophosphamide failure in severe central nervous system sarcoidosis. Respir Med. 2009;103(2):268-273.
322. St. Clair EW, Van Der Heijde DM, Smolen JS et al. Combination of infliximab and methotrexate therapy for early rheumatoid arthritis: A randomized, controlled trial. Arthritis Rheum. 2004;50(11):3432-3443.
323. Stein RB, Hanauer SB. Medical therapy for inflammatory bowel disease. Gastroenterol Clin N Am. 1999;28(2):297-321.
324. Stone JH, Dana R. Treatment of scleritis. UpToDate [online seriall]. Waltham, MA: UpToDate; reviewed January 2014.
325. Stone M, Salonen D, Lax M, et al. Clinical and imaging correlates of response to treatment with infliximab in patients with ankylosing spondylitis. J Rheumatol. 2001;28(7):1605-1614.
326. Suarez-Almazor M, Ortiz Z, Lopez-Olivo M, et al. Infliximab and etanercept in rheumatoid arthritis: Systematic review of long-term clinical effectiveness, safety, and cost-effectiveness. Technology Report No. 85. Ottawa, ON: Canadian Agency for Drugs and Technologies in Health (CADTH); 2007.
327. Suhler EB, Smith JR, Giles TR, et al. Infliximab therapy for refractory uveitis: 2-year results of a prospective trial. Arch Ophthalmol. 2009;127(6):819-822.
328. Suhler EB, Smith JR, Wertheim MS, et al. A prospective trial of infliximab therapy for refractory uveitis: Preliminary safety and efficacy outcomes. Arch Ophthalmol. 2005;123(7):903-912.
329. Sundel R. Refractory Kawasaki disease. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed February 2019.
330. Swedish Council on Technology Assessment in Health Care (SBU). Infliximab (Anti-TNF-alfa) in treating Crohn's disease - early assessment briefs (ALERT). Stockholm, Sweden: SBU; 2001.
331. Sznol M, Callahan MK, Yuan J, Wolchok J. Key issues in the management of gastrointestinal immune-related adverse events associated with ipilimumab administration. Commun Oncol. 2013;10(12):351-358.

332. Talley NJ, Abreu MT, Achkar J, et al. An evidence-based systematic review on medical therapies for inflammatory bowel disease. Am J Gastroenterol. 2011;106(Suppl 1):S2-S25.
333. Tolentino M, Dana R. Retinal vasculitis associated with primary ocular disorders. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed February 2019.
334. Torres J, Bonovas S, Doherty G, et al. ECCO guidelines on therapeutics in Crohn's disease: medical treatment. J Crohns Colitis. 2020;14(1):4-22.
335. Toruner M, Akpınar H, Akyuz F, et al. 2019 Expert opinion on biological treatment use in inflammatory bowel disease management. Turk J Gastroenterol. 2019;30(Suppl 4):S913-S944.
336. Toussirot E, Pertuiset E. TNFα blocking agents and sarcoidosis: An update. Rev Med Interne. 2010 31(12):828-837.
337. Tynjala P, Lindahl P, Honkanen V, et al. Infliximab and etanercept in the treatment of chronic uveitis associated with refractory juvenile idiopathic arthritis. Ann Rheum Dis. 2007;66(4):548-550.
338. UpToDate Inc. Infliximab injection. UpToDate Lexidrug [online]. Waltham, MA: UpToDate; updated September 18, 2025.
339. U.S. Food and Drug Administration (FDA). Biosimilar and interchangeable biologics: More treatment choices. Silver Spring, MD: FDA; August 17, 2023.
340. U.S. Food and Drug Administration (FDA). FDA roundup: October 3, 2023. News Release. Silver Spring, MD: FDA; October 3, 2023.
341. U.S. Food and Drug Administration (FDA). Procise IFX. Device classification under section 513(f) (2) (De novo). DEN210056. Silver Spring, MD: FDA; September 29, 2023.
342. U.S. Food and Drug Administration (FDA). FDA approves Remicade for children with Crohn's disease. FDA News. P06-71. Rockville, MD: FDA; May 19, 2006.
343. U.S. Food and Drug Administration (FDA). FDA approves Remicade to treat ulcerative colitis in children older than 6 years. FDA News. Rockville, MD: FDA; September 23, 2011.
344. U.S. Food and Drug Administration (FDA). Interchangeable biological products. Silver Spring, MD: FDA; undated. Available at https://www.fda.gov/. Accessed February 1, 2024.
345. Uhlving HH, Buchvald F, Heilmann CJ, et al. Bronchiolitis obliterans after allo-SCT: Clinical criteria and treatment options. Bone Marrow Transplant. 2012;47(8):1020-1029.

346. Ulbricht KU, Stoll M, Bierwirth J, et al. Successful tumor necrosis factor alpha blockade treatment in therapy-resistant sarcoidosis. Arthritis Rheum. 2003;48:3542-3543.
347. van Assche G, Rutgeerts P. Anti-TNF agents in Crohn's disease. Expert Opin Investig Drugs. 2000;9(1):103-111.
348. Van Assche G. Immunogenicity of anti-TNF antibodies. Has the veil been lifted? Gut. 2011;60(3):285-286.
349. Van den Bosch F, Kruithof E, Baeten D, et al. Effects of a loading dose regimen of three infusions of chimeric monoclonal antibody to tumour necrosis factor alpha (infliximab) in spondyloarthropathy: An open pilot study. Ann Rheum Dis. 2000;59(6):428-433.
350. Van den Bosch F, Kruithof E, Baeten D, et al. Randomized double-blind comparison of chimeric monoclonal antibody to tumor necrosis factor alpha (infliximab) versus placebo in active spondylarthropathy. Arthritis Rheum. 2002;46(3):755-765.
351. Van den Bosch F, Kruithof E, De Vos M, et al. Crohn's disease associated with spondyloarthropathy: Effect of TNF-alpha blockade with infliximab on articular symptoms. Lancet. 2000;356(9244):1821-1822.
352. van der Heijde D, Ramiro S, Landewe R, et al. 2016 Update of the international ASAS-EULAR management recommendations for axial spondyloarthritis. Ann Rheum Dis. 2017;0:1-14.
353. van Deventer SJ. Review article: Targeting TNF alpha as a key cytokine in the inflammatory processes of Crohn's disease -- the mechanisms of action of infliximab. Aliment Pharmacol Ther. 1999;13(Suppl 4):3-8, discussion 38.
354. van Rappard DC, Leenarts MF, Meijerink-van 't Oost L, Mekkes JR. Comparing treatment outcome of infliximab and adalimumab in patients with severe hidradenitis suppurativa. J Dermatolog Treat. 2012;23(4):284-289.
355. van Rheenen PF, Aloi M, Assa A, et al. The medical management of paediatric Crohn’s disease: An ECCO-ESPGHAN guideline update. J Crohns Colitis. 2021;15(2):171-194.
356. Van Sanford C, Obeidat AZ, Hagen M3, Zabeti A. A case of fatal invasive aspergillosis in a patient with neurosarcoidosis treated with infliximab. Int J Neurosci. 2019;129(6):619-622.

357. Velayos FS, Kahn JG, Sandborn WJ, Feagan BG. A test-based strategy is more cost effective than empiric dose escalation for patients with Crohn's disease who lose responsiveness to infliximab. Clin Gastroenterol Hepatol. 2013;11(6):654-666.
358. Walker-Bone K, Fallow S. Rheumatoid arthritis. In: BMJ Clinical Evidence. London, UK: BMJ Publishing Group; June 2005.
359. Wang SL, Ohrmund L, Hauenstein S, et al. Development and validation of a homogeneous mobility shift assay for the measurement of infliximab and antibodies-to-infliximab levels in patient serum. J Immunol Methods. 2012;382(1-2):177-188.
360. Wang Y, Abu-Sbeih H, Mao E, et al. Immune-checkpoint inhibitor-induced diarrhea and colitis in patients with advanced malignancies: Retrospective review at MD Anderson. J Immunother Cancer. 2018;6(1):37.
361. Ward MM, Deodhar A, Akl EA, et al. American College of Rheumatology/Spondylitis Association of America/Spondyloarthritis Research and Treatment Network 2015 recommendations for the treatment of ankylosing spondylitis and nonradiographic axial spondyloarthritis. Arthritis Rheum. 2016;68(2):282-98.
362. Ward MM, Deodhar A, Gensler LS, et al. 2019 Update of the American College of Rheumatology/Spondylitis Association of America/Spondyloarthritis Research and Treatment Network recommendations for the treatment of ankylosing spondylitis and nonradiographic axial spondyloarthritis. Arthritis Rheum. 2019;71(10):1599-1613.
363. Wieczorek M, Walecka I. Hidradenitis suppurativa - known and unknown disease. Reumatologia. 2018;56(6):337–339.
364. Woolacott N, Bravo Vergel Y, Hawkins N, et al. Etanercept and infliximab for the treatment of psoriatic arthritis: A systematic review and economic evaluation. Health Technol Assess. 2006;10(31):1-239.
365. Yates M, Watts RA, Bajema IM, et al. EULAR/ERA-EDTA recommendations for the management of ANCA-associated vasculitis. Ann Rheum Dis. 2016;75(9):1583-1594.
366. Yee AM, Pochapin MB. Treatment of complicated sarcoidosis with infliximab anti-tumor necrosis factor-alpha therapy. Ann Intern Med. 2001;135:27-31.

367. Yeh SW, Sami N, Ahmed RA. Treatment of pemphigus vulgaris: Current and emerging options. Am J Clin Dermatol. 2005;6(5):327-342.
368. Zochling J, Maxwell L, Beardmore J, Boonen A. TNF-alpha inhibitors for ankylosing spondylitis (Protocol for Cochrane Review). Cochrane Database Syst Rev. 2005;(3):CD005468.

Copyright Aetna Inc. All rights reserved. Clinical Policy Bulletins are developed by Aetna to assist in administering plan benefits and constitute neither offers of coverage nor medical advice. This Clinical Policy Bulletin contains only a partial, general description of plan or program benefits and does not constitute a contract. Aetna does not provide health care services and, therefore, cannot guarantee any results or outcomes. Participating providers are independent contractors in private practice and are neither employees nor agents of Aetna or its affiliates. Treating providers are solely responsible for medical advice and treatment of members. This Clinical Policy Bulletin may be updated and therefore is subject to change.

Copyright © 2001-2026 Aetna Inc.

Language services can be provided by calling the number on your member ID card. For additional language assistance: Español | ____ | Tiếng Việt | 한국어 | Tagalog | Русский | العربية | Kreyòl | Français | Polski | Português | Italiano | Deutsch | 日本語 | فارسی | Other Languages... | (http://www.aetna.com/individuals-families/contact-aetna/information-in-other-languages.html)